

Faster Linear Programming with $\sqrt{\text{rank}}$ Linear System Solves

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Abstract

Lee and Sidford [LS19] showed that the linear program $\min\{c^\top x : A^\top x = b, l \leq x \leq u\}$ over $x \in \mathbb{R}^m$, where $A \in \mathbb{R}^{m \times n}$, can be solved to accuracy ϵ using $O(\sqrt{n} \log^{13} m \cdot \log(mU/\epsilon))$ solves of linear systems in $A^\top \mathbf{D} A$ for positive diagonal matrices $\mathbf{D}$, where U bounds the magnitudes of the input and of the initial point. Theirs is the first such bound governed by $\text{rank}(A) = n$ rather than by the number of constraints m .

Given an interior point and a finite certified magnitude bound U , we give an algorithm that uses $O(\sqrt{n} \log^5 m \cdot \log(mU/\epsilon))$ solves of such linear systems, improving the Lee–Sidford bound by a factor of $\log^8 m$.

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1 Introduction

Linear programming has driven the development of algorithmic optimization for more than seven decades. Dantzig’s simplex method [Dan51] moves from vertex to vertex and remains a foundational practical algorithm. Its worst-case running time, however, is exponential: the Klee–Minty family forces standard pivot rules to visit exponentially many vertices [KM72]. Khachiyan’s ellipsoid method gave the first polynomial-time algorithm for linear programming [Kha79]. For square instances of dimension d and input bit length L , its classical analysis uses $O(d^4 L)$ arithmetic operations. Karmarkar’s projective method then initiated the modern theory of polynomial-time interior-point algorithms [Kar84], improving this bound to $O(d^{3.5} L)$ arithmetic operations on $O(L)$ -bit numbers.

The interior-point framework was subsequently simplified and generalized by Newton and path-following methods [Ren88, NN94]. Renegar obtained $\tilde{O}(\sqrt{r} \log(1/\epsilon))$ Newton iterations for an LP with r inequalities, and Vaidya’s volumetric-center method sharpened the arithmetic complexity to $O((rd^2 + r^{3/2}d)L)$ for r inequalities and d variables [Vai89]. These developments shifted attention from polynomiality alone to the cost of the linear algebra performed along a central path.

Modern work accelerates that linear algebra. Lee and Sidford obtained $\tilde{O}(\sqrt{\text{rank}(A)} \log(1/\epsilon))$ path-following iterations, each using only $\tilde{O}(1)$ linear-system solves [LS14, LS19]. For dense standard-form LPs with d variables, Cohen, Lee, and Song achieved $\tilde{O}(d^\omega + d^{2.5-\alpha/2} + d^{2+1/6})$ time, which is $\tilde{O}(d^\omega)$ for the current matrix-multiplication parameters [CLS19]. Here ω is the exponent of matrix multiplication; currently, $\omega < 2.371177$ [DEK⁺26]. Brand subsequently derandomized the Cohen–Lee–Song algorithm while preserving its asymptotic running time [Bra20]. For tall dense linear programs with n constraints and d variables, Brand, Lee, Sidford, and Song gave a randomized $\tilde{O}(nd + d^3)$ -time algorithm [BLSS20]. Later, that running time has been improved to $\tilde{O}(nd + d^{2.5})$ [BLL⁺21].

1.1 Our results

We study the linear program

$$\text{OPT} := \min_{\substack{x \in \mathbb{R}^m : A^\top x = b \\ l_i \leq x_i \leq u_i \text{ for all } i \in [m]}} c^\top x, \quad (1)$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^n$, $c \in \mathbb{R}^m$ and $l, u \in (\mathbb{R} \cup \{-\infty, +\infty\})^m$. Throughout we assume, as in [LS19], that A is nondegenerate, meaning that it has full column rank and no zero row; that $l_i < u_i$ and $(l_i, u_i) \neq \mathbb{R}$ for every $i \in [m]$; and that the feasible region $\Omega := \{x \in \mathbb{R}^m : A^\top x = b, l_i \leq x_i \leq u_i\}$ has nonempty interior $\Omega^\circ := \{x \in \mathbb{R}^m : A^\top x = b, l_i < x_i < u_i\}$. For the complexity statement, assume that Ω is bounded and that the input includes a finite certified magnitude bound U . Let $\mathcal{S}_0$ be the set of distances from x_0 to all finite coordinate endpoints,

$$\mathcal{S}_0 := \{x_{0,i} - l_i : l_i > -\infty\} \cup \{u_i - x_{0,i} : u_i < +\infty\}.$$

The supplied bound is required to satisfy

$$U \geq 4 \max\{1, \|c\|_\infty, \text{diam}_\infty(\Omega), \max \mathcal{S}_0, \max\{s^{-1} : s \in \mathcal{S}_0\}\}.$$

The theorem does not require the algorithm to compute this global bound. This normalization implies the standard interval-barrier bounds $\|\phi'(x_0)\|_\infty \leq U$ and $\sqrt{\phi''_i(x_i)} \geq 1/U$ for every $x \in \Omega^\circ$. As usual, logarithmic factors in asymptotic complexity bounds in this section are understood to be at least 1.

Lee and Sidford [LS19] solve Eq. (1) to accuracy ϵ using $\tilde{O}(\sqrt{n} \log(1/\epsilon))$ iterations, each of which solves a constant number of linear systems in $A^\top \mathbf{D} \mathbf{A}$ for a positive diagonal matrix $\mathbf{D}$. Theirs is the first method whose iteration count is governed by $\text{rank}(A) = n$ rather than by the number of constraints m , and this distinction is substantive: no known transformation puts Eq. (1) in standard form without increasing the rank. Written out, the bound they obtain carries thirteen logarithmic factors,

$$O(\sqrt{n} \log^{13} m \cdot \log(\frac{mU}{\epsilon}) \cdot \mathcal{T}_w) \text{ work.}$$

The corresponding parallel depth bound has the same thirteen logarithmic factors, with $\mathcal{T}_w$ replaced by $\mathcal{T}_d$, the depth of one linear-system solve.

We state our result as follows.

Theorem 1.1 (Linear programming, informal version of Theorem 8.16). *Given an interior point of Eq. (1) and a finite certified magnitude bound U satisfying the preceding display, there is an algorithm that outputs a feasible x with $c^\top x \leq \text{OPT} + \epsilon$ with constant probability using*

$$O(\sqrt{n} \log^5 m \cdot \log(\frac{mU}{\epsilon}) \cdot \mathcal{T}_w) \text{ work} \quad \text{and} \quad O(\sqrt{n} \log^{3.5} m \cdot \log(\frac{mU}{\epsilon}) \cdot \mathcal{T}_d) \text{ parallel depth,}$$

where U is defined above, and $\mathcal{T}_w$ and $\mathcal{T}_d$ are the work and depth needed to compute $(A^\top \mathbf{D} \mathbf{A})^{-1} \mathbf{q}$ for an input positive diagonal matrix $\mathbf{D}$ and vector $\mathbf{q}$.

Thus our result improves the Lee–Sidford bound [LS19] by a factor of $\log^8 m$.

2 Technique Overview

This section contrasts the source of the logarithmic overhead in the Lee–Sidford method with the mechanisms used to reduce it here. Section 2.1 reviews the two nested loops in their analysis and identifies how path following and repeated Lewis-weight computation combine to produce thirteen logarithmic factors. Section 2.2 then explains how sharper centrality geometry, defect-adaptive prediction, residual chasing, and diagonal self-transport reduce the resulting linear-system bound to five logarithmic factors, while certified checkpoints keep the confidence horizon independent of path length.

2.1 Why the Lee–Sidford method pays thirteen logarithms

Two nested loops. Write κ for the length of the path to be traversed. The Lee–Sidford method [LS19] uses $O(\sqrt{n} \log^5 m \cdot \kappa)$ centering steps. Each step asks for Lewis weights of a diagonal rescaling of A to relative accuracy R , and one such call costs $O(\log^8 m)$ systems in $A^\top \mathbf{D} \mathbf{A}$. Thus the thirteen logarithms split as $5 + 8$.

The outer loop. The path step is $\Theta(R/(\sqrt{n} \log^2 m))$. In the parameter choice made in the proof of Theorem 40 of [LS19], $K = \frac{1}{16c_k}$ and $R = \frac{K}{48c_k \log(36c_1 c_s c_k m)}$. Thus, writing $L_{\text{ch}} := \Theta(\log m)$ solely for exponent accounting, $R = \Theta((c_k^2 L_{\text{ch}})^{-1})$. Since their $c_k = \Theta(\log m)$, substitution in their displayed path-step formula gives the stated five-logarithm step count. This paragraph uses the Θ notation only to explain the ledger; the proof below does not import it as a lemma.

The inner loop. The damped fixed-point computation makes $\Theta(\log m)$ passes. Every pass estimates all leverage scores to relative accuracy $\Theta(R)$ using $O(R^{-2} \log m)$ systems. Hence one weight computation costs

$$\Theta(\log m) \cdot O(R^{-2} \log m) = O(\log^8 m).$$

The precision factor $\log(mU/\epsilon)$ is part of κ and is kept separate from these thirteen logarithms.

2.2 How the five-logarithm proof works

The proof in Sections 4–8 keeps the Lee–Sidford central path and changes the weight geometry, predictor, and score provider. The new score provider uses exact diagonal self-transport for the part of a query transition caused by rescaling its own row, and sketches only the remaining nonlocal response. Every sketch is drawn after its two query endpoints are fixed.

The new split. The first reduction is from five logarithmic factors to 2.5 in the number of centering calls. The path-composition bound is $N_{\text{step}} = O(c_k \sqrt{nq} R^{-1} \kappa)$. Here $c_k = \Theta(q)$ and $R = \Theta(\ell^{-1})$, so $N_{\text{step}} = O(\sqrt{nq}^{3/2} \ell \kappa)$; since $q \leq \ell$, the factors $q\sqrt{q}\ell$ contribute at most $1 + 1/2 + 1 = 2.5$ logarithms.

The second reduction is from eight logarithmic factors to 2.5 in the amortized work per centering call. Instead of recomputing an R -accurate Lewis fixed point through $\Theta(\ell)$ passes, hybrid score maintenance charges $O(E^{-2} L_{\text{conf}})$ systems per unit of effective drift, while one centering call creates only $O(E)$ effective drift. Its amortized transition cost is therefore $O(E^{-1} L_{\text{conf}}) = O(\sqrt{q} \ell^2) \leq O(\ell^{2.5})$. Thus the two new contributions multiply to $O(\sqrt{nq}^2 \ell^3 \kappa) \leq O(\sqrt{n} \ell^5 \kappa)$. The second reduction comes from amortizing score maintenance across transitions and restarts, not from accelerating the old fixed-point loop (Lemma 8.10 and Theorem 8.16).

2.2.1 Centering calls

Regularized Lewis weights and sharper centrality geometry. The regularization floor is placed inside the Lewis fixed point. Taking $q := \log(4m/n)$ and $p = 1 - \Theta(1/q)$ makes the sensitivity constant while the consistency is $c_k = \Theta(q)$. The orthogonal-projector and Lewis-Jacobian Pythagorean identities permit the mixed norm $N(z) := \|z\|_\infty + C\|z\|_w$ with $C = \Theta(\sqrt{q})$, together with the tracking tolerance $K = \Theta(q^{-1/2})$ (Lemmas 4.5 and 4.9).

Retaining and estimating the Jacobian defect. For a logarithmic barrier movement ψ , let B be the Lewis-weight Jacobian and define $d^2 := p^2 \|\psi\|_g^2 - \|B\psi\|_g^2$. This is the exact ideal-metric defect; maintained and surrogate metrics are used only through the comparison lemmas in Section 4. The retained-movement inequality gains a margin $\Theta(C^2 d^2 / \delta)$. The off-diagonal part of $B\psi$ factors through $Q = W^{-1/2}(P \circ P)W^{-1/2}$, and the two-projection estimator has variance $O(y^\top Q y) = O(d^2)$. Thus the same defect that creates centrality slack pays for the stochastic predictor (Lemmas 5.5, 5.6, and 5.9). The defect can also be certified with the required one-sided guarantee (Lemma 6.9).

A larger centrality radius. The centrality geometry uses $E = \Theta(q^{-1/2} \ell^{-1})$ and $R = \Theta(CE) = \Theta(\ell^{-1})$, where $\ell := \log(4m)$. The predictor radius is linear in the observed defect and in δ/c_k . Young’s inequality charges the linear defect against the quadratic retained-movement margin, so the centrality radius is larger than the observation scale by the factor C . The adaptive residual

game then tracks the remaining stochastic error without changing the deterministic movement cap. For the principal score transition, the scalar-free row-log displacement is

$$d^\circ = \psi + \frac{\vartheta}{2}\hat{u}, \quad \|d^\circ\|_w = O(R/C) = O(E), \quad \|d^\circ\|_\infty = O(R).$$

Hence its weighted branch has effective drift $O(E)$. The old-query-to-surrogate transition and the constant number of same-point fixed-point refinements have ordinary centered drift $O(E)$. Consequently every centering call contributes $O(E)$ effective drift, and hence $\Omega = O(N_{\text{step}}E)$ (Lemma 8.10). The certified surrogate and the complete path-following phase preserve these radius and movement budgets (Algorithms 5 and 6, and Lemmas 7.1 and 7.5).

2.2.2 Inner loops

Diagonal self-transport. Consider two consecutive score queries D_- and D_+ . After removing the irrelevant common scalar, write their row-log displacement as d . For a positive companion $\tilde{w}$, define $\lambda_w := (\sum_{i=1}^m \tilde{w}_i d_i) / (\sum_{i=1}^m \tilde{w}_i)$ and $\rho_w := \|d - \lambda_w \mathbf{1}\|_{\tilde{w}}$. If only row i is multiplied by γ , its leverage score s changes exactly according to $\mathcal{T}_\gamma(s) := \gamma^2 s / (1 + (\gamma^2 - 1)s)$ (Lemma 8.2). The algorithm applies this scalar map to its incoming approximate score and uses one fresh shared sketch only for the response of all the other rows. The exact one-row identity makes the deterministic error multiplier at most $e^{O(\rho_w)}$, while the centered sketch innovation has conditional variance proxy $O(\rho_w^2/k)$. Thus a large coordinatewise row displacement incurs no additional asymptotic cost when its weighted centered energy is small (Lemma 8.3).

Hybrid effective drift and restarts. For transitions better controlled coordinatewise, define $\rho_\infty := \min_\lambda \|d - \lambda \mathbf{1}\|_\infty$. The provider uses the eligible branch with smaller effective drift, namely $\eta_t := \min\{C_\infty \rho_{\infty,t}, C_w \rho_{w,t}\}$. It uses $k_t = \Theta(\eta_t E^{-2} L_{\text{conf}})$ rows for that transition and restarts from a fresh absolute Gaussian sketch before the cumulative effective drift exceeds a small constant. After rescaling by the accumulated drift, the relative-error recurrence is contractive, so maximal Bernstein gives simultaneous raw relative accuracy (Algorithm 10 and Lemma 8.7).

Long paths and certified checkpoints. The score-maintenance restarts above control error inside one score-maintenance block; they do not by themselves make a confidence bound independent of the total path length. For that purpose the complete path is also divided into polynomial-length epochs. At an epoch boundary the algorithm uses exact score diagonals to recenter the path point and greedily reduce the Lewis-weight tracking potential, certifies the resulting state outward, and then starts new score and residual-game epochs. A failed certificate causes only the mathematical state of that epoch to be retried—resource counters are never rolled back. Because the normalization has fixed polynomial cost and epochs are chosen longer than that cost, checkpoints add only a constant factor to the ledger while allowing all confidence logarithms to remain $O(\log m)$ for arbitrarily long paths (Lemmas 7.6 and 8.1, and Algorithm 8). The two path phases and their final output guarantee are supplied by the outer reduction (Algorithm 9 and Lemma 6.5).

The resulting score cost is $O(M + E^{-2} L_{\text{conf}}(1 + \Omega))$, where $\Omega := \sum_{t=1}^M \eta_t$. Fresh randomness enters at every block base and every transition, so no algebraically recycled anchor or estimate is used.

2.2.3 Putting it all together

The ledger. There are $N_{\text{step}} = O(\sqrt{n}q^{3/2}\ell\kappa)$ centering steps. Since $E = \Theta(q^{-1/2}\ell^{-1})$ and $L_{\text{conf}} = O(\ell)$,

$$\Omega = O(N_{\text{step}}E) = O(\sqrt{n}q\kappa), \quad E^{-2}L_{\text{conf}} = O(q\ell^3).$$

Thus score maintenance costs $O(\sqrt{q}\ell^2)$ systems per step on average, plus $O(q\ell^3)$ for a fresh absolute base. Geometrically accurate fresh score sketches also supply the initial weight and score state. The base term is absorbed by the path term, and the total is

$$O(\sqrt{n}q^2\ell^3\kappa + \min\{q\ell^3, n \log \ell\}) \leq O(\sqrt{n}\ell^5\kappa)$$

systems (Theorem 8.16).

3 Preliminaries

Subsection 3.1 fixes the basic vector and matrix notation used throughout the paper. Subsection 3.2 then records the barrier, centrality, and path-following quantities needed in the algorithm and its analysis.

3.1 Basic notation

For a positive integer d , write $[d] := \{1, \dots, d\}$. All vectors are column vectors; e_i denotes the i -th standard basis vector, and $\mathbf{1}$ denotes the all-ones vector of the dimension determined by context. For $x \in \mathbb{R}^d$, write $\|x\|_1 := \sum_{i=1}^d |x_i|$, $\|x\|_2 := (\sum_{i=1}^d x_i^2)^{1/2}$, and $\|x\|_\infty := \max_{i \in [d]} |x_i|$. The Euclidean inner product is $\langle x, y \rangle := x^\top y$.

For a matrix M , the symbols $M^\top$, M^{-1} , and $M^\dagger$ denote its transpose, its inverse when it is nonsingular, and its Moore–Penrose pseudoinverse, respectively. A superscript $+$ is instead reserved for an updated quantity. The identity matrix I always has the dimension determined by context. For symmetric matrices, $M \succeq N$ means that $M - N$ is positive semidefinite. If $M \succeq 0$, then $M^{1/2}$ denotes its positive semidefinite square root; negative powers are used only when the relevant matrix is positive definite. Given vector norms $\|\cdot\|_a$ and $\|\cdot\|_b$, the induced operator norm is $\|M\|_{a \rightarrow b} := \sup_{x \neq 0} \|Mx\|_b / \|x\|_a$.

For a vector x , $\text{Diag}(x)$ is the diagonal matrix with diagonal x ; for a square matrix M , $\text{diag}(M)$ is its diagonal vector. We use $\odot$ and $\circ$ for the coordinatewise and Hadamard products, respectively. Vector inequalities are coordinatewise, as are vector products, quotients, powers, square roots, exponentials, and logarithms. For a scalar z , define $\cosh z := (e^z + e^{-z})/2$ and $\sinh z := (e^z - e^{-z})/2$; both functions act coordinatewise on vectors.

3.2 Centrality and Lee–Sidford path facts

For each coordinate interval (l_i, u_i) , let ϕ_i be the one-self-concordant interval barrier used by Lee–Sidford [LS19], define $\Phi''(x) := \text{Diag}(\phi_i''(x_i))$, and write $A_x := \Phi''(x)^{-1/2}A$. For a positive vector w , let $W := \text{Diag}(w)$,

$$\|y\|_w := (y^\top W y)^{1/2}, \quad N_w(y) := \|y\|_\infty + C\|y\|_w.$$

Definition 3.1 (Centrality and permitted systems). *For an active cost vector c_{path} , define*

$$f_t^{c_{\text{path}}}(x, w) := tc_{\text{path}}^\top x + \sum_{i=1}^m w_i \phi_i(x_i)$$

and

$$\delta_t^{c_{\text{path}}}(x, w) := \min_{\eta \in \mathbb{R}^n} N_w\left(\frac{tc_{\text{path}} + w \odot \phi'(x) - A\eta}{w \odot \sqrt{\phi''(x)}}\right).$$

When the cost is clear, the superscript is omitted. Define

$$P_{x,w} := I - W^{-1}A_x(A_x^\top W^{-1}A_x)^{-1}A_x^\top.$$

For

$$r_t^{c_{\text{path}}}(x, w) := \frac{tc_{\text{path}} + w \odot \phi'(x)}{w \odot \sqrt{\phi''(x)}}, \quad \bar{\delta}_t^{c_{\text{path}}}(x, w) := N_w(P_{x,w}r_t^{c_{\text{path}}}(x, w)),$$

the first centrality-equivalence inequality is unconditional. Whenever $4g(x)/5 \leq w \leq 5g(x)/4$, Lemma 4.5 supplies the uniform constant c_γ used below and gives

$$\delta_t^{c_{\text{path}}}(x, w) \leq \bar{\delta}_t^{c_{\text{path}}}(x, w) \leq c_\gamma \delta_t^{c_{\text{path}}}(x, w). \quad (2)$$

All later uses of the upper bound are under this tracking band. Thus $\bar{\delta}_t$ is a computable centrality proxy, obtained with the same permitted solve as the projected Newton residual, and $\bar{\delta}_t = 0$ if and only if $\delta_t = 0$. We also write $s_i(x) := \phi''_i(x_i)^{-1/2}$ for the positive barrier row scaling, so that $A_x = \text{Diag}(s(x))A$. The projected Newton step is

$$h_t^{c_{\text{path}}}(x, w) := -\Phi''(x)^{-1/2}P_{x,w}\frac{tc_{\text{path}} + w \odot \phi'(x)}{w \odot \sqrt{\phi''(x)}}.$$

A permitted system is a solve in $A^\top DA$ for a positive diagonal matrix D and one right-hand side.

We use the following Lee–Sidford facts with the preceding notation. Stating them here fixes the hypotheses under which they are invoked later.

Lemma 3.2 (Lee–Sidford path facts). *Suppose $\delta := \delta_t(x, w) \leq 1/10$ and $4g(x)/5 \leq w \leq 5g(x)/4$. If $x^+ = x + h_t(x, w)$, then x^+ remains in the barrier domain,*

$$\delta_t(x^+, w) \leq 4\delta^2, \quad N_w(\sqrt{\Phi''(x)}(x^+ - x)) \leq c_\gamma \delta.$$

If $w' > 0$ and $z := N_w(\log w' - \log w) \leq 1/10$, then

$$\delta_t(x, w') \leq (1 + 4z)(\delta_t(x, w) + z).$$

For $|\xi| < 1$ and $(1 + \xi)t > 0$,

$$\delta_{(1+\xi)t}(x, w) \leq (1 + |\xi|)\delta_t(x, w) + |\xi|(1 + C\sqrt{\|w\|_1}).$$

Finally, define

$$\text{OPT}(c_{\text{path}}) := \min\{c_{\text{path}}^\top x : x \in \Omega\}.$$

If $x_t(w)$ minimizes $f_t^{c_{\text{path}}}(\cdot, w)$ on the affine feasible set, then

$$c_{\text{path}}^\top x_t(w) - \text{OPT}(c_{\text{path}}) \leq \frac{\|w\|_1}{t}.$$

Proof. The Newton contraction is Lemma 15 of [LS19], the weight-change estimate is Lemma 17, and the $\xi \geq 0$ case of the t -change estimate is Lemma 14. For $\xi < 0$, evaluate the new residual at the old minimizing multiplier scaled by $1 + \xi$; the old residual is multiplied by $1 + \xi$ and the added cost term has mixed norm at most $|\xi|(1 + C\sqrt{\|w\|_1})$. This proves the displayed signed extension directly rather than attributing the negative case to Lemma 14. The proofs of these three Lee–Sidford estimates use only that N_w is a norm dominating $\|\cdot\|_\infty$ and that the centrality-equivalence constant is at most two. They therefore apply verbatim to the present coefficient C . The normalized-step estimate is their centrality-equivalence lemma. The last display is their weighted duality-gap lemma and follows directly by pairing the first-order optimality equation at $x_t(w)$ with an optimal feasible point. $\square$

4 Geometric foundations and the mixed-ball oracle

Section 4.1 establishes rowwise projector stability and the regularized Lewis-weight fixed point used by the score provider. Section 4.2 proves the projector and Jacobian defect inequalities that sharpen the mixed-norm geometry. Section 4.3 converts those inequalities into robust centrality and fixes the structural parameters. Finally, Section 4.4 gives the finite mixed-ball optimization oracle used by the residual-chasing step.

4.1 Projection-score perturbations and regularized Lewis fixed points

Overview. The argument uses a rowwise stability estimate for the orthogonal projector under a diagonal rescaling, the global fixed-point contraction, and a predicted warm start. Along the path, the hybrid score provider transports the diagonal response exactly, sketches only the nonlocal endpoint discrepancy, and restarts after constant cumulative effective drift. Every transition sketch is drawn only after both query endpoints are fixed.

This section exploits information unavailable within a single call to the approximate Lewis-weight routine `computeApwWeight`: consecutive centering steps request leverage scores for diagonal rescalings of the *same* matrix A that differ only by $e^{\pm O(R)}$. Recomputing an E -accurate estimate from scratch at every step does not use this continuity.

We first isolate the relative-drift estimate used by both transition branches of the hybrid provider. Its Gaussian increment primitive is recorded later in Lemma 7.3; no algebraically recycled anchor is used.

Throughout this section $A \in \mathbb{R}^{m \times n}$ is nondegenerate, $\mathbf{D}$ denotes a positive diagonal matrix, and

$$P(\mathbf{D}) := \mathbf{D}A(A^\top \mathbf{D}^2 A)^{-1} A^\top \mathbf{D}$$

is the orthogonal projection onto the range of $\mathbf{D}A$, so that $\sigma_i(\mathbf{D}A) = \|P(\mathbf{D})e_i\|_2^2$.

The first estimate is fundamental to the scheme: when two rescalings differ by a small *diagonal* factor, the corresponding columns of the two projections differ by a small *relative* amount. An absolute bound would not suffice, because σ_i can be as small as n/m .

Lemma 4.1 (Relative drift of projection columns). *Let $\mathbf{D} = \Gamma \mathbf{D}'$ with Γ positive diagonal and $\|\log \Gamma\|_\infty \leq \rho \leq 1/8$. Then for every $i \in [m]$,*

$$\|P(\mathbf{D})e_i - P(\mathbf{D}')e_i\|_2 \leq 10\rho\sqrt{\sigma_i(\mathbf{D}'A)}.$$

Moreover,

$$e^{-4\rho}\sigma_i(\mathbf{D}'A) \leq \sigma_i(\mathbf{D}A) \leq e^{4\rho}\sigma_i(\mathbf{D}'A).$$

Proof. Let $B \in \mathbb{R}^{m \times n}$ have orthonormal columns spanning the range of $\mathbf{D}'A$, so that $P(\mathbf{D}') = BB^\top$ and $\sigma_i(\mathbf{D}'A) = \|B^\top e_i\|_2^2$. The range of $\mathbf{D}A$ is the range of ΓB , and therefore

$$P(\mathbf{D}) = \Gamma B(B^\top \Gamma^2 B)^{-1} B^\top \Gamma.$$

Write $c := B^\top e_i$, so that $\|c\|_2^2 = \sigma_i(\mathbf{D}'A)$. Because Γ is diagonal we have $\Gamma e_i = \Gamma_{ii} e_i$, hence $B^\top \Gamma e_i = \Gamma_{ii} c$, and consequently

$$P(\mathbf{D})e_i - P(\mathbf{D}')e_i = (\Gamma_{ii} \Gamma B(B^\top \Gamma^2 B)^{-1} - B) c.$$

It suffices to bound the spectral norm of the matrix in parentheses. Define $F := (B^\top \Gamma^2 B)^{-1} - I$. From $\rho \leq 1/8$ we obtain $\|\Gamma - I\| \leq e^\rho - 1 \leq 1.14\rho$ and $|\Gamma_{ii} - 1| \leq 1.14\rho$, and also $\|B^\top \Gamma^2 B - I\| \leq e^{2\rho} - 1 \leq 2.6\rho \leq 0.33$, so that $\|F\| \leq 2.6\rho/(1 - 0.33) \leq 4\rho$. Using $\|B\| = 1$ and $\|\Gamma B\| \leq 1 + 1.14\rho$, the triangle inequality gives

$$\begin{aligned} \|\Gamma_{ii} \Gamma B(I + F) - B\| &\leq |\Gamma_{ii} - 1| \cdot \|\Gamma B(I + F)\| + \|\Gamma B\| \cdot \|F\| + \|\Gamma B - B\| \\ &\leq 1.14\rho(1 + 1.14\rho)(1 + 4\rho) + (1 + 1.14\rho) \cdot 4\rho + 1.14\rho \leq 10\rho, \end{aligned}$$

where the last step uses $\rho \leq 1/8$ and that each coefficient increases in ρ . For the leverage-score comparison, note that

$$e^{-2\rho}(\mathbf{D}'A)^\top \mathbf{D}'A \preceq (\mathbf{D}'A)^\top \Gamma^2 \mathbf{D}'A \preceq e^{2\rho}(\mathbf{D}'A)^\top \mathbf{D}'A,$$

so the inverse quadratic form in the i -th leverage score changes by a factor $e^{\pm 2\rho}$. The two copies of Γ_{ii} contribute another factor $e^{\pm 2\rho}$, giving the stated $e^{\pm 4\rho}$ comparison. $\square$

Remark 4.2. Diagonality of Γ is essential. For a general Γ with $\|\Gamma - I\| \leq \rho$, one obtains only $\|P(\mathbf{D})e_i - P(\mathbf{D}')e_i\|_2 = O(\rho)$, which is an absolute rather than a relative bound and provides no improvement for small leverage scores. The gain comes from $\Gamma e_i = \Gamma_{ii} e_i$, which allows the factor $\sqrt{\sigma_i}$ to be extracted from the perturbation.

Each centering step must issue only $O(1)$ score queries. The loop of `computeApxWeight` issues $T = \Theta(\log m)$ of them because its convergence is measured in a Euclidean norm and the conversion back to ℓ_∞ costs a factor of n . Replacing its inner loop by the fixed-point iteration below removes that loss, since the iteration contracts in ℓ_∞ directly. For $0 < p < 2$ define $\vartheta := 1 - \frac{2}{p}$ and, for a nondegenerate matrix B with rows $b_i^\top$ and $w \in \mathbb{R}_{>0}^m$, define $\Lambda_{B,i}(w) := b_i^\top (B^\top \mathbf{W}^\vartheta B)^{-1} b_i$, suppressing the subscript when the matrix is clear.

We work throughout with the *regularized* Lewis vector $w_p^v(B)$, the solution of $w = \sigma(\mathbf{W}^{\vartheta/2} B) + v$ for a fixed floor $v \geq 0$. Placing the regularizer inside the fixed point rather than adding it afterward is what bounds the dynamic range of the weights, and through it the sensitivity of the weight function; this is the content of Lemma 4.8, and it is the reason the exponents below are governed by $\log(4m/n)$ rather than by $\log(4m)$. The identity $\sigma_i(\mathbf{W}^{\vartheta/2} B) = w_i^\vartheta \Lambda_{B,i}(w)$ shows that a relative approximation of $\Lambda_B(w)$ is one leverage-score query, and rearranging the fixed-point equation as $w_i^{2/p} = \Lambda_{B,i}(w) + w_i^{-\vartheta} v_i$ exhibits $w_p^v(B)$ as the unique fixed point of

$$F_{B,i}(w) := (\Lambda_{B,i}(w) + w_i^{-\vartheta} v_i)^{p/2}.$$

Writing $\tau_i(w; B) := \sigma_i(\mathbf{W}^{\vartheta/2} B) + v_i$ for the *regularized* score, the same map takes the multiplicative form

$$F_{B,i}(w) = w_i^{1-p/2} \tau_i(w; B)^{p/2},$$

because $-\vartheta p/2 = 1 - p/2$. Two things are then immediate. The map sees the leverage scores only through τ , so only τ has to be estimated relatively, which is what Section 4.1 exploits; and

$$\log F_{B,i}(w) - \log w_i = \frac{p}{2}(\log \tau_i(w; B) - \log w_i),$$

so the residual of the fixed-point equation is directly observable from an estimate of τ . The last part of the lemma is stated for the starting error itself rather than for the movement of the target, because in the final construction the target moves by $\Theta(R)$ while the accuracy to be restored is E , smaller by an unbounded factor. Lemma 6.1 reconciles them.

Regularized Lewis fixed points. Before this fixed point can be used as a weight function, we need existence, uniqueness, and differentiability under row rescaling. The next lemma records these facts together with the Jacobian formula used throughout the proof.

Lemma 4.3 (Well-posed regularized Lewis weights). *Let $0 < p < 1$, let $\vartheta := 1 - 2/p < 0$, let $v > 0$ coordinatewise, and let $B \in \mathbb{R}^{m \times n}$ have full column rank and no zero row. Then $w = \sigma(W^{\vartheta/2}B) + v$ has a unique solution $w = w_p^v(B) > 0$. This solution depends smoothly on every positive diagonal row scaling of B . If $B = \text{Diag}(c)A$, $\Lambda := \text{Diag}(\sigma) - P \circ P$, and $a := -\vartheta = 2/p - 1$, then*

$$(W + a\Lambda) \, d \log w = 2\Lambda \, d \log c.$$

In particular, the derivative operator is invertible.

Proof. For $z = \log w$, define

$$\mathcal{J}_B(z) := \sum_{i=1}^m e^{z_i} - v^\top z - \frac{1}{\vartheta} \log \det(B^\top \text{Diag}(e^{\vartheta z})B).$$

The matrix in the determinant is positive definite. Differentiating gives

$$\nabla \mathcal{J}_B(z) = w - v - \sigma(W^{\vartheta/2}B), \quad \nabla^2 \mathcal{J}_B(z) = W + a\Lambda.$$

The projection identity $\Lambda = \text{Diag}(\sigma) - P \circ P \succeq 0$ and $W \succ 0$ show that the Hessian is positive definite. Moreover $\mathcal{J}_B(z) \rightarrow +\infty$ as $\|z\|_\infty \rightarrow \infty$: an unbounded positive coordinate is dominated by $\sum_{i=1}^m e^{z_i}$, while if the positive coordinates stay bounded and a coordinate tends to $-\infty$, the term $-v^\top z$ and the nonnegative logarithmic-determinant growth prevent escape. Thus $\mathcal{J}_B$ is coercive and strictly convex, so it has a unique minimizer, exactly the displayed fixed point.

The Hessian is nonsingular, hence the implicit-function theorem gives smooth dependence on positive row scalings. Finally, $d\sigma = 2\Lambda(d \log c + \vartheta \, d((\log w)/2))$. Differentiating $w - v = \sigma$ and rearranging gives the claimed equation. $\square$

The fixed-point iteration must remain contractive when its leverage estimates are approximate, and its target must move continuously under row rescaling. The following lemma gives both properties in the same logarithmic metric.

Lemma 4.4 (ℓ_∞ contraction and target stability). *Let $0 < p < 2$, let $v \geq 0$, and let $\widehat{F}_i = e^{\pm\delta} F_{A,i}(w)$ for every $i \in [m]$, which holds as soon as the quantity $\Lambda_{A,i}(w) + w_i^{-\vartheta} v_i = w_i^{-\vartheta} (\sigma_i(\mathbf{W}^{\vartheta/2}A) + v_i)$ is estimated to within a factor $e^{\pm 2\delta/p}$. Only that sum has to be relatively accurate: an estimate of σ_i with additive error $O(\delta)(\sigma_i + v_i)$ suffices, which is strictly weaker than a relative estimate of σ_i and is supplied by the hybrid score provider. Then*

$$\|\log \widehat{F} - \log w_p^v(A)\|_\infty \leq (1 - \frac{p}{2}) \|\log w - \log w_p^v(A)\|_\infty + \delta.$$

Moreover, if $A' = \mathbf{D}A$ for a positive diagonal matrix $\mathbf{D}$ satisfying $\|\log \mathbf{D}\|_\infty \leq r$, then

$$\|\log w_p^v(A') - \log w_p^v(A)\|_\infty \leq 4r.$$

Consequently, fix $p_0 > 0$. There is an integer $t_0 = O_{p_0}(1)$ with the following property. If $p \in [p_0, 1)$, the starting point satisfies

$$\|\log w^{(0)} - \log w_p^v(A')\|_\infty \leq E$$

for some $E > 0$, and every iteration $w^{(t+1)} = \widehat{F}_{A'}(w^{(t)})$ has logarithmic error $\delta \leq pE/8$, then

$$\|\log w^{(t_0)} - \log w_p^v(A')\|_\infty \leq E/2.$$

Proof. The comparison below applies to any two positive inputs, so the map $z \mapsto \log F_A(e^z)$ is a global contraction in ℓ_∞ with factor $1 - p/2 < 1$. Banach's theorem therefore supplies the unique positive fixed point $w_p^v(A)$. Write $w_* := w_p^v(A)$ and $w = e^u w_*$, so $\mathbf{W}^\vartheta = e^{\vartheta u} \mathbf{W}_*^\vartheta$ with $e^{\vartheta u}$ diagonal. Hence

$$e^{-|\vartheta||u|_\infty} A^\top \mathbf{W}_*^\vartheta A \preceq A^\top \mathbf{W}^\vartheta A \preceq e^{|\vartheta||u|_\infty} A^\top \mathbf{W}_*^\vartheta A,$$

and inverting and applying the resulting inequality to a_i gives

$$\Lambda_{A,i}(w) = e^{\pm|\vartheta||u|_\infty} \Lambda_{A,i}(w_*).$$

The second summand obeys the same estimate, since $w_i^{-\vartheta} v_i = e^{-\vartheta u_i} (w_*)_i^{-\vartheta} v_i$; as both summands are nonnegative, their sum is within $e^{\pm|\vartheta||u|_\infty}$ of its value $(w_*)_i^{2/p}$ at w_* . Raising to the power $p/2$ yields

$$F_{A,i}(w) = e^{\pm \frac{p}{2} |\vartheta||u|_\infty} (w_*)_i, \quad \frac{p}{2} |\vartheta| = 1 - \frac{p}{2},$$

and the error in $\widehat{F}$ contributes a further factor $e^{\pm\delta}$, which proves the first claim.

For target stability, note that for every $w > 0$

$$A'^\top \mathbf{W}^\vartheta A' = A^\top \mathbf{D} \mathbf{W}^\vartheta \mathbf{D} A$$

lies between $e^{-2r} A^\top \mathbf{W}^\vartheta A$ and $e^{2r} A^\top \mathbf{W}^\vartheta A$. The inverse quadratic form changes by $e^{\pm 2r}$, and the factor $\mathbf{D}_{ii}^2$ in the i -th row contributes another $e^{\pm 2r}$, so $\Lambda_{A',i}(w) = e^{\pm 4r} \Lambda_{A,i}(w)$. The second summand of F does not involve A at all, so the sum moves by at most $e^{\pm 4r}$ as well, and

$$\|\log F_{A'}(w) - \log F_A(w)\|_\infty \leq 2pr.$$

Define $w_* = w_p^v(A)$, $w'_* = w_p^v(A')$ and $d := \|\log w'_* - \log w_*\|_\infty$. Applying the first claim to the exact map $F_{A'}$ and then the preceding comparison gives $d \leq (1 - \frac{p}{2})d + 2pr$, so $d \leq 4r$. For the final claim define $\varrho := 1 - p/2 \leq 1 - p_0/2 < 1$. Unrolling the first claim, the inexact iteration satisfies

$$\|\log w^{(t)} - \log w_p^v(A')\|_\infty \leq \varrho^t E + \frac{\delta}{1 - \varrho} \leq \varrho^t E + \frac{E}{4},$$

where the first step follows from unrolling the inexact contraction recurrence, and the second step follows from $1 - \varrho = p/2$ and $\delta \leq pE/8$. Choosing t_0 with $\varrho^{t_0} \leq 1/4$ proves the result. $\square$

4.2 Projector and Jacobian defects

Section 4.1 changes the Lewis-weight computation while preserving the path geometry. This section changes the geometry while preserving score maintenance. The two ingredients are combined by the hybrid provider in Section 8.2. Throughout, define $\ell := \log(4m)$ and $q := \log \frac{4m}{n}$, and let $L_* := \ell$ denote the deterministic confidence scale. The checkpointed execution below keeps every stochastic horizon polynomial in m , independently of the total path length κ . $\|y\|_{w+\infty} := \|y\|_\infty + C\|y\|_w$ is the mixed centrality norm of [LS19], and K is the radius of the invariant $\|\log w - \log g(x)\|_\infty \leq K$ maintained by the chasing game. The parameters C, K, p, R, E will be chosen below. Here R is the centrality radius tolerated by the centering argument, whereas E is the score accuracy observed by the chasing game. Lemma 4.9 fixes C, K , and p ; Section 5.2 makes the final choices of R and E . Until then, R is only assumed to be at most r_0/c_k for a sufficiently small absolute constant $r_0 \leq 1/160$.

Three parameter choices in the analysis of [LS19] arise from bounds that appear linear but admit sharper estimates. The coefficient $C = 24\sqrt{c_s c_k}$ absorbs the $\Theta(1/C)$ error of two triangle inequalities, one for the weighted projector and one for the Lewis-weight Jacobian. The tolerance $K = 1/(16c_k)$ absorbs the $\Theta(K)$ error of converting the ideal norm into the maintained one. Finally, the consistency $c_k = \Theta(1/(1-p))$ grows to $\Theta(\log m)$ because the sensitivity of their weight function is $m^{\Theta(1-p)}$. The first two losses are quadratic under the refined analysis, and the third is an artifact of adding the regularizer after the Lewis fixed point instead of inside it. Taking

$$p := 1 - \frac{1}{4q}, \quad c_k := 8q, \quad C := 8\sqrt{2c_s c_k}, \quad K := \frac{1}{128\sqrt{c_k}}$$

therefore suffices. Enlarging K relaxes the accuracy E to which the weights must be maintained, a smaller C lengthens the admissible change in the path parameter, and the regularized weight function makes both run on q rather than on ℓ . Section 4.4 changes what the chasing game is asked to track rather than any of the three quantities above; the defect-adaptive construction in Section 5.2 subsequently separates R from E and supplies the radius used by the final theorem.

The first geometric improvement is a Pythagorean estimate for the projected Newton direction in the mixed norm. It replaces two independent triangle inequalities by one joint bound.

Lemma 4.5 (Projector defect). *Let $x \in \Omega^\circ$ and let w satisfy $\frac{4}{5}g(x) \leq w \leq \frac{5}{4}g(x)$. For every $C > 0$ and every $y \in \mathbb{R}^m$,*

$$\|P_{x,w}y\|_\infty + C\|P_{x,w}y\|_w \leq \|y\|_\infty + \sqrt{C^2 + 2c_s} \|y\|_w,$$

and consequently $c_\gamma \leq \sqrt{1 + 2c_s/C^2}$.

Proof. Write $P := P_{x,w}$, so that $I - P = W^{-1}A_x(A_x^\top W^{-1}A_x)^{-1}A_x^\top$. This matrix is idempotent, and $W(I - P) = A_x(A_x^\top W^{-1}A_x)^{-1}A_x^\top$ is symmetric, so $I - P$ is self-adjoint for the inner product $\langle u, y \rangle_w := u^\top W y$. Hence P is an orthogonal projection in that inner product, and with $s := \|Py\|_w$ and $t := \|(I - P)y\|_w$ we have $s^2 + t^2 = \|y\|_w^2$.

Under the hypothesis on w , [LS19] bound $\|I - P\|_{w \rightarrow \infty}^2 \leq 2c_s$. Applying that bound to $(I - P)y$ rather than to y , and using $(I - P)^2 = I - P$, gives $\|(I - P)y\|_\infty \leq \sqrt{2c_s} t$. The key observation is that the error may therefore be charged to t instead of to $\|y\|_w$. Hence $\|Py\|_\infty \leq \|y\|_\infty + \sqrt{2c_s} t$, and Cauchy–Schwarz in $\mathbb{R}^2$ gives $\sqrt{2c_s} t + Cs \leq \sqrt{2c_s + C^2} \sqrt{t^2 + s^2}$. The second claim follows from $\sqrt{C^2 + 2c_s} = C\sqrt{1 + 2c_s/C^2}$ together with $\sqrt{1 + 2c_s/C^2} \geq 1$. $\square$

The second geometric improvement retains the same Pythagorean slack inside the Lewis-weight Jacobian. The next lemma couples its weighted contraction with a coordinatewise approximation by a diagonal map.

Lemma 4.6 (Regularized Jacobian defects). *Let $0 < p < 1$, let $v > 0$ coordinatewise, let $c > 0$ be a row scaling and let $w := w_p^v(\text{Diag}(c)A)$, so that $w = \sigma(\mathbf{W}^{\vartheta/2}\text{Diag}(c)A) + v$ with $\vartheta = 1 - 2/p$. Let B denote the Jacobian of $\log w$ with respect to $\log c$ and define $b := Bh$. Then*

$$(a) \quad \|b - ph\|_w^2 + \|b\|_w^2 \leq p^2 \|h\|_w^2.$$

$$(b) \quad \text{There is a diagonal matrix } D \text{ with } 0 \preceq D \preceq pI \text{ such that } \|b - Dh\|_\infty^2 \leq \frac{4}{p^2}(p^2 \|h\|_w^2 - \|b\|_w^2).$$

$$(c) \quad \|b\|_{w+\infty} \leq p\sqrt{1 + 4/(p^2 C^2)} \|h\|_{w+\infty} \text{ for every } C > 0.$$

Proof. Proof of (a). Define $a := -\vartheta = \frac{2}{p} - 1 > 0$, let Σ and P be the leverage-score matrix and the orthogonal projection matrix of $\mathbf{W}^{\vartheta/2}\text{Diag}(c)A$, and let $\Lambda := \Sigma - P \circ P$. Lemma 4.3 makes the fixed point unique and smooth in $\log c$. Differentiating its equation and using $d\sigma = 2\Lambda \, d\log(\mathbf{W}^{\vartheta/2}\text{Diag}(c))$ therefore gives $(W + a\Lambda) \, d\log w = 2\Lambda \, d\log c$, so $B = (W + a\Lambda)^{-1}2\Lambda$. Passing to w -normalized coordinates, with

$$S := W^{-1/2}\Sigma W^{-1/2}, \quad Q := W^{-1/2}(P \circ P)W^{-1/2}, \quad L := S - Q,$$

we get $T := W^{1/2}BW^{-1/2} = 2(I + aL)^{-1}L$. Here $\Lambda \succeq 0$ gives $L \succeq 0$, and $\sigma_i \leq w_i$ gives $S \preceq I$, hence $0 \preceq Q \preceq S \preceq I$ and $0 \preceq L \preceq I$. The eigenvalues $2\lambda/(1 + a\lambda)$ of T increase in λ and equal p at $\lambda = 1$, so T is symmetric with $0 \preceq T \preceq pI$. Writing $x := W^{1/2}h$, so that $\|h\|_w = \|x\|_2$ and $\|b\|_w = \|Tx\|_2$, the first inequality is the scalar bound $(p - \lambda)^2 + \lambda^2 \leq p^2$ on $[0, p]$ applied in the eigenbasis of T .

Proof of (b). Define $D := 2(I + aS)^{-1}S$, diagonal because S is, with $0 \preceq D \preceq pI$ for the same reason as T . From $(I + aX)^{-1}X = a^{-1}(I - (I + aX)^{-1})$ we get $K := T - D = \frac{2}{a}((I + aS)^{-1} - (I + aL)^{-1})$, and the resolvent identity together with $S - L = Q$ gives

$$K = -2(I + aS)^{-1}Q(I + aL)^{-1},$$

which is symmetric, being the difference of two symmetric matrices. Factoring $p^2 - t^2$ at $t = 2\lambda/(1 + a\lambda)$ as $(p(1 + a\lambda) - 2\lambda)(p(1 + a\lambda) + 2\lambda)$ and using $pa = 2 - p$,

$$M := p^2I - T^2 = p^2(I - L)(I + \frac{4-p}{p}L)(I + aL)^{-2}.$$

Hence $KM^\dagger K = \frac{4}{p^2}(I + aS)^{-1}Q(I - L)^\dagger(I + \frac{4-p}{p}L)^{-1}Q(I + aS)^{-1}$, the powers of $I + aL$ cancelling because they commute with the other factors. The two middle factors are commuting positive semidefinite functions of L and the second is at most I , while $Q \preceq I - S + Q = I - L$ gives $Q(I - L)^\dagger Q \preceq Q$; therefore

$$KM^\dagger K \preceq \frac{4}{p^2}(I + aS)^{-1}Q(I + aS)^{-1}.$$

Taking the i -th diagonal entry, which is legitimate because $I + aS$ is diagonal, and using $Q_{ii} = \sigma_i^2/w_i \leq w_i$, we get

$$e_i^\top KM^\dagger K e_i \leq \frac{4}{p^2} w_i. \tag{3}$$

The pseudo-inverse is used legitimately here: $\ker M$ is the eigenspace of L at the eigenvalue 1, on which Q vanishes because $0 \preceq Q \preceq I - L$, so every column of K is orthogonal to $\ker M$. Since $b - Dh = W^{-1/2}Kx$, Cauchy-Schwarz in the seminorm of M gives

$$|(b - Dh)_i|^2 = \frac{|e_i^\top Kx|^2}{w_i} \leq \frac{e_i^\top KM^\dagger K e_i}{w_i} x^\top Mx \leq \frac{4}{p^2}(p^2 \|h\|_w^2 - \|b\|_w^2),$$

where the first step follows from $b - Dh = W^{-1/2}Kx$, the second step follows from Cauchy–Schwarz in the seminorm induced by M , and the third step follows from Eq. (3) and $x^\top Mx = p^2\|h\|_w^2 - \|b\|_w^2$. Taking the maximum over $i \in [m]$ gives the claimed ℓ_∞ bound.

Proof of (c). Define $r := \|h\|_w$, $s := \|b\|_w$ and $d := \sqrt{p^2r^2 - s^2}$. Then $\|b\|_\infty \leq \|Dh\|_\infty + \|b - Dh\|_\infty \leq p\|h\|_\infty + \frac{2}{p}d$, and Cauchy–Schwarz in $\mathbb{R}^2$ gives $\frac{2}{p}d + Cs \leq \sqrt{4/p^2 + C^2}\sqrt{d^2 + s^2} = pC\sqrt{1 + 4/(p^2C^2)}r$. As $\sqrt{1 + 4/(p^2C^2)} \geq 1$, the claim follows. $\square$

4.3 Robust centrality geometry

The centrality proof needs contraction in the norm defined by the maintained weights rather than by the inaccessible exact weights. The next estimate shows that this change of norm incurs only a quadratic loss.

Lemma 4.7 (Robust consistency). *Let $b = Bh$ be as in Lemma 4.6-(a), let $\tilde{w} > 0$ satisfy $\|\log \tilde{w} - \log w\|_\infty \leq H \leq 1/4$, and define $N(y) := \|y\|_\infty + C\|y\|_{\tilde{w}}$. Then, with $L_H := e^{3H/2} - 1 \leq 2H$,*

$$N(b) \leq p(1 + \frac{e^{H/2}}{2}(\frac{2}{pC} + L_H)^2)N(h).$$

Proof. Define $r := \|h\|_w$, $s := \|b\|_w$ and $d := \sqrt{p^2r^2 - s^2}$, which is real by Lemma 4.6-(a). Lemma 4.6-(a) gives $\|b - ph\|_w \leq d$, and Lemma 4.6-(b) gives $\|b\|_\infty \leq p\|h\|_\infty + \frac{2}{p}d$.

Let $F(y) := \|y\|_{\tilde{w}} - \|y\|_w$, a positively homogeneous function. For $y \neq 0$ define $\theta := \|y\|_w / \|y\|_{\tilde{w}}$, so that $\theta \in [e^{-H/2}, e^{H/2}]$. Then $\nabla F(y) = \|y\|_w^{-1}(\theta \tilde{W} - W)y$, so its norm dual to $\|\cdot\|_w$ is at most the operator norm of $\theta \tilde{W}W^{-1} - I$. The entries of $\tilde{W}W^{-1}$ lie in $[e^{-H}, e^H]$, so that norm is at most L_H , and therefore $|F(y) - F(y')| \leq L_H\|y - y'\|_w$.

Hence $C\|b\|_{\tilde{w}} - pC\|h\|_{\tilde{w}} = C(s - pr) + C(F(b) - F(ph)) \leq C(s - pr) + CL_Hd$, using homogeneity and $\|b - ph\|_w \leq d$. Adding the ℓ_∞ estimate gives $N(b) - pN(h) \leq (\frac{2}{p} + CL_H)d + C(s - pr)$. Write $d = pr u$ and $s = pr\sqrt{1 - u^2}$ with $u \in [0, 1]$; then $s - pr \leq -pru^2/2$, so the right-hand side is at most $pr((\frac{2}{p} + CL_H)u - \frac{C}{2}u^2) \leq pr(\frac{2}{p} + CL_H)^2/(2C)$. Finally $N(h) \geq C\|h\|_{\tilde{w}} \geq Ce^{-H/2}r$, and dividing proves the claim, since $(\frac{2}{p} + CL_H)^2/C^2 = (\frac{2}{pC} + L_H)^2$. $\square$

For the rest of the paper define

$$c_1 := \sup_{x \in \Omega^\circ} \|g(x)\|_1, \quad c_s := \sup_{x \in \Omega^\circ, i \in [m]} \frac{\sigma_i(G(x)^{-1/2}A_x)}{g_i(x)}.$$

The symbol c_γ denotes the centrality-equivalence constant in Eq. (2), and c_k is the explicit consistency scale fixed in Lemma 4.9.

To instantiate the abstract path-following framework, we need uniform bounds on both the total weight and its leverage sensitivity. Regularization makes both quantities absolute up to the natural factor n .

Lemma 4.8 (Size and sensitivity). *Let $p = 1 - \frac{1}{4q}$, $v := \frac{n}{m}\mathbf{1}$ and $g(x) := w_p^v(A_x)$. Then $c_1(g) \leq 2n$ and $c_s(g) \leq 2$.*

Proof. Summing the fixed-point equation gives $\|g(x)\|_1 = \sum_{i=1}^m \sigma_i(\mathbf{G}^{\vartheta/2}A_x) + \|v\|_1 = n + n = 2n$, which is the first claim. For the second, $\sigma_i \leq 1$ and $v_i = n/m$ give $g_i \in [n/m, 2]$, so the largest coordinate exceeds the smallest by a factor at most $2m/n$. Define $\gamma := \frac{1-p}{p} \geq 0$ and $M := \mathbf{G}^{\vartheta/2}A_x$,

so that $\mathbf{G}^{-1/2}A_x = \mathbf{G}^\gamma M$. For a positive diagonal D one has $M^\top D^2 M \succeq (\min_j d_j^2) M^\top M$ and hence $\sigma_i(DM) \leq (d_i^2 / \min_j d_j^2) \sigma_i(M)$; with $d_i = g_i^\gamma$ and $\sigma_i(M) \leq g_i$ this gives

$$c_s(g) = \max_{i \in [m]} \frac{\sigma_i(\mathbf{G}^{-1/2}A_x)}{g_i} \leq \left(\frac{2m}{n}\right)^{2\gamma} = \exp\left(\frac{2(1-p)}{p} \log \frac{2m}{n}\right) \leq \exp\left(\frac{1}{2p}\right) \leq 2,$$

where the last two steps use $1 - p = \frac{1}{4q}$ with $\log \frac{2m}{n} \leq q$, and $p \geq \frac{3}{4}$, which holds because $q \geq \log 4 > 1$. $\square$

For the executable algorithm fix the certified numerical bounds

$$\bar{c}_1 := 2n, \quad \bar{c}_s := 2, \quad \bar{c}_\gamma := 1 + \frac{1}{128c_k}.$$

The unbarred quantities remain the analytic suprema above. Every parameter and failure test in the pseudocode uses the displayed barred constants; no supremum over Ω° is evaluated by the algorithm.

This is the one place where the position of the regularizer matters. In [LS19] the floor is added after the Lewis fixed point, so the exponent $\frac{1}{2} - \frac{1}{p}$ acts on the *unregularized* Lewis weights, whose coordinates have no lower bound; the same computation then yields only $c_s = O(m^{\Theta(1-p)})$, and $c_s = O(1)$ forces $1 - p = O(1/\ell)$. Putting the floor inside the fixed point bounds the dynamic range by $2m/n$ and replaces ℓ by q throughout.

Since c_s is only required to be an upper bound on the sensitivity, we assume throughout that $c_s \in [1, 2]$.

The preceding estimates determine a compatible choice of the weight exponent, mixed-norm coefficient, and tracking radius. The next lemma verifies these parameters and the resulting contraction gap along every Newton segment.

Lemma 4.9 (Weight-function parameters). *With $p = 1 - \frac{1}{4q}$, $v = \frac{n}{m}\mathbf{1}$, $g(x) = w_p^v(A_x)$ and*

$$c_k := 8q, \quad C := 8\sqrt{2\bar{c}_s c_k}, \quad K := 1/(128\sqrt{c_k}),$$

one has $c_\gamma(g) \leq 1 + 1/(128c_k)$. Moreover, there is an absolute $r_0 > 0$ such that the following holds. If a Newton segment starts at

$$\delta := \delta_t(x, w) \leq R \leq r_0/c_k, \quad \|\log w - \log g(x)\|_\infty \leq K,$$

then, in the maintained norm, at every point of that segment,

$$\|G^{-1}J_g(x)(\Phi''(x))^{-1/2}\|_{w+\infty \rightarrow w+\infty} \leq 1 - \frac{15}{8c_k}.$$

Proof. Lemma 4.5 and $C^2 = 128\bar{c}_s c_k$ give $c_\gamma \leq \sqrt{1 + 1/(64c_k)} \leq 1 + 1/(128c_k)$.

We first verify the tracking band all along the segment without using the Jacobian estimate that is to be proved. Write $x_s := x + s\Delta$, where Δ is the projected Newton direction, and define $h := \sqrt{\Phi''(x)}\Delta$. Lemma 3.2 gives $N_w(h) \leq c_\gamma \delta$, hence $\|h\|_\infty \leq 2R$ after decreasing r_0 . Coordinatewise self-concordance gives

$$\left| \frac{d}{ds} \log s_i(x_s) \right| \leq \frac{|h_i|}{1 - s|h_i|}.$$

Consequently, for every $s \in [0, 1]$,

$$\|\log s(x_s) - \log s(x)\|_\infty \leq \frac{\|h\|_\infty}{1 - \|h\|_\infty} \leq 4\delta.$$

The target-stability part of Lemma 4.4 now gives

$$\|\log g(x_s) - \log g(x)\|_\infty \leq 16\delta \leq 16R.$$

Fix r_0 so that $16R \leq K$ uniformly for $R \leq r_0/c_k$. Thus $\|\log w - \log g(x_s)\|_\infty \leq 2K$ for the entire segment. This closes the former first-exit seam before the operator bound is invoked.

Apply Lemma 4.7 with $\tilde{w} := w$ and $H \leq 2K$. Then

$$L_H \leq 2H \leq \frac{1}{32\sqrt{c_k}} < \frac{0.032}{\sqrt{c_k}}. \quad (4)$$

Moreover, $p \geq 3/4$ and $\bar{c}_s \geq 1$ give

$$\frac{2}{pC} \leq \frac{1}{3\sqrt{2}\sqrt{c_k}} < \frac{0.236}{\sqrt{c_k}}. \quad (5)$$

Therefore

$$\frac{e^{H/2}}{2} \left(\frac{2}{pC} + L_H \right)^2 \leq \frac{e^{1/8} (0.236 + 0.032)^2}{2 c_k} < \frac{1}{8c_k},$$

where the first step follows from $H/2 \leq 1/8$, Eq. (5), and Eq. (4), and the second step follows from the numerical inequality $e^{1/8}(0.236 + 0.032)^2/2 < 1/8$. The chain rule through the barrier contributes the diagonal map $q_i = -\frac{1}{2}\phi_i'''(x_i)\phi_i''(x_i)^{-3/2}h_i$. Self-concordance gives $|q_i| \leq |h_i|$, so this map increases neither component of the maintained norm. Finally, $p = 1 - 2/c_k$, and hence

$$p(1 + \frac{1}{8c_k}) = 1 - \frac{15}{8c_k} - \frac{1}{4c_k^2} \leq 1 - \frac{15}{8c_k},$$

where the first step follows from $p = 1 - 2/c_k$, and the second step follows from $c_k > 0$. $\square$

4.4 The mixed-ball response oracle

The chasing game is played with an enlargement parameter: the ideal weights move inside a set U and the player is allowed to move inside $(1 + \epsilon_{\text{slack}})U$. The overshoot $\epsilon_{\text{slack}}|U|$ is charged against the centrality margin, which is only $\Theta(\delta/c_k)$, so [LS19] must take $\epsilon_{\text{slack}} = \Theta(1/c_k)$. Their chasing theorem then returns a tracking error $\frac{12E}{\epsilon_{\text{slack}}} \log \frac{12m\tau}{\epsilon_{\text{slack}}}$, and requiring this to be at most K forces $E = \Theta(K/(c_k L_{\text{ch}}))$. One whole factor c_k in E^{-1} is therefore the price of the enlargement.

That price is avoidable. Suppose that before moving the weights we can *predict* the movement $u_k := \log g(x_k) - \log g(x_{k-1})$ to within $\Theta(\delta/c_k)$ in the mixed norm. Chase the prediction error instead of the movement itself: the residual game has a movement set of radius $\Theta(\delta/c_k)$ to begin with, so a *constant* enlargement already costs only $\Theta(\delta/c_k)$, and the tracking error becomes $O(EL_{\text{ch}})$ rather than $O(Ec_k L_{\text{ch}})$. The stochastic residual game below proves the needed tracking statement; Section 5.2 supplies the final predictor.

Prediction buys a second saving, of a different kind, and it is the larger of the two. Once only the residual is chased, what the chasing theorem constrains is the accuracy E with which $g(x)$ is observed and the radius of the residual movement set; the centrality of the iterate enters neither

hypothesis. The final defect-adaptive construction exploits this separation by taking $R = \Theta(CE) = \Theta(1/\ell)$ while keeping $E = \Theta(q^{-1/2}/\ell)$.

The greedy player uses the following deterministic oracle. Recording its certified approximation guarantee closes one of the two non-system optimizations inside a centering step. All scalar-work and scalar-depth bounds for this oracle and the defect projection use the exact real-RAM model; the theorem makes no bit-complexity claim. In particular, logarithms, exponentials, linear-system solves, equality tests, and emptiness tests are exact real operations. The finite-precision certificates below show that an approximate scalar optimizer is never mistaken for an exact KKT point; they do not assert a Turing-model bit bound. A bit-complexity version would replace the zero and endpoint-equality branches by certified tolerance tests and is outside the present theorem.

Algorithm 1 Linear optimization over a mixed ball

```

1: procedure MIXEDBALLLINEARORACLE( $w, C, s, a, \epsilon_g$ ) ▷ Lemma 4.10
2:   if  $a = 0$  then
3:     return 0.
4:   end if
5:    $a \leftarrow a/\|a\|_\infty$ ,  $G \leftarrow \|a\|_1 + C^{-1}\|W^{-1/2}a\|_2$ , and  $H_{\text{lb}} \leftarrow s/(1 + C\sqrt{\max_{i \in [m]} w_i})$ .
6:    $J \leftarrow \lceil \log_2(2Gs/(\epsilon_g H_{\text{lb}})) \rceil$ ,  $t_- \leftarrow 0$ ,  $t_+ \leftarrow s$ , and  $\chi_{\text{best}} \leftarrow 0$ .
7:   for  $j = 1, \dots, J$  do
8:      $t \leftarrow (t_- + t_+)/2$ ,  $A \leftarrow (s - t)/C$ , and  $y_i \leftarrow -t \text{sign}(a_i)$ .
9:     if  $\|y\|_w \leq A$  then
10:       $\lambda \leftarrow 0$ ,  $\chi \leftarrow y$ , and  $\alpha_i \leftarrow |a_i|$  for every  $i$ .
11:     else
12:       $b_i \leftarrow |a_i|/(w_i t)$ ; sort the  $b_i$  in nonincreasing order.
13:      For each resulting prefix  $I$  with positive denominator, compute
          
$$\lambda_I \leftarrow \sqrt{\frac{\sum_{i \in [m] \setminus I} a_i^2/w_i}{A^2 - t^2 \sum_{i \in I} w_i}}$$

14:      Choose the prefix  $I$  satisfying  $b_i \geq \lambda_I$  on  $I$  and  $b_i \leq \lambda_I$  off  $I$ , and set  $\lambda \leftarrow \lambda_I$ .
15:       $\chi_i \leftarrow -\text{sign}(a_i) \min\{t, |a_i|/(\lambda w_i)\}$  and  $\alpha_i \leftarrow (|a_i| - \lambda w_i t)_+$ .
16:     end if
17:     If  $\langle a, \chi \rangle < \langle a, \chi_{\text{best}} \rangle$ , set  $\chi_{\text{best}} \leftarrow \chi$ .
18:      $g \leftarrow \lambda A/C - \sum_{i=1}^m \alpha_i$ .
19:     if  $g > 0$  then
20:        $t_+ \leftarrow t$ .
21:     else if  $g < 0$  then
22:        $t_- \leftarrow t$ .
23:     else
24:       return  $\chi$ .
25:     end if
26:   end for
27:   return  $\chi_{\text{best}}$ .
28: end procedure

```

The finite procedure above is useful only if it returns a feasible mixed-ball point with a certified near-optimal linear objective. The next lemma proves that guarantee and records its scalar work and depth.

Lemma 4.10 (Linear optimization over a mixed ball). *Let $w > 0$, $C, s > 0$, and $a \in \mathbb{R}^m$. For every $\epsilon_g \in (0, 1/10)$, the procedure MIXEDBALLLINEARORACLE (Algorithm 1) returns χ with $N_w(\chi) \leq s$ and*

$$\langle a, \chi \rangle \leq -(1 - \epsilon_g) s N_w^*(a).$$

If $0 < \underline{w} \leq w_i \leq \bar{w}$ for every i , define

$$L_w := \log(2 + (1 + C\sqrt{\bar{w}})(m + \sqrt{m/\underline{w}}/C)).$$

The algorithm uses no permitted systems and has $O(mJ \log m)$ scalar work and $O(J \log m)$ arithmetic depth, where $J = O(L_w + \log(1/\epsilon_g))$.

Proof. If $a = 0$, the claim is immediate. Scaling a to $\|a\|_\infty = 1$ changes neither the optimizer nor a relative objective guarantee. For $t \in [0, s]$, define $A(t) = (s - t)/C$ and let $\varphi(t)$ be the minimum of $\langle a, x \rangle$ subject to $|x_i| \leq t$ and $\|x\|_w \leq A(t)$. This is the partial minimum of a jointly convex problem, so φ is convex.

We first verify that every operation in the pseudocode is explicit. If $y_i = -t \text{sign}(a_i)$ lies in the w -ball, it is the exact fixed- t minimizer and the ball multiplier is $\lambda = 0$. Otherwise the ball is active and the KKT equations [KT51] give

$$x_i = -\text{sign}(a_i) \min\{t, |a_i|/(\lambda w_i)\}.$$

For a proposed clipped set I , the equality $\|x\|_w = A(t)$ gives exactly the displayed formula for λ . Sorting $b_i = |a_i|/(w_i t)$ partitions the positive line into at most $m + 1$ regions. Scanning the prefixes finds a region whose formula has positive denominator and satisfies its clipping inequalities; continuity and monotonicity of $\|x(\lambda)\|_w$ give existence, and all resulting vectors coincide at a tie. Thus the scan returns the exact fixed- t minimizer using $O(m \log m)$ scalar work and $O(\log m)$ depth.

At that KKT tuple define

$$\alpha_i = (|a_i| - \lambda w_i t)_+.$$

This is the multiplier of the active face with sign opposite to a_i ; the other face multiplier is zero. Differentiating only the moving constraints gives the exact subgradient

$$g(t) = \frac{\lambda A(t)}{C} - \sum_{i=1}^m \alpha_i \in \partial \varphi(t).$$

There is no multiplier ambiguity at a clipping tie because then $\alpha_i = 0$. Moreover $\sum_{i=1}^m \alpha_i \leq \|a\|_1$. Coordinatewise, $\lambda w_i |x_i| \leq |a_i|$, and hence

$$\lambda A(t) \leq \|W^{-1/2} a\|_2.$$

Consequently every returned subgradient has magnitude at most

$$G := \|a\|_1 + C^{-1} \|W^{-1/2} a\|_2.$$

The sign of an exact subgradient preserves an interval containing a minimizer of the convex function φ . After J iterations that interval has length at most $s 2^{-J}$. The midpoint evaluated at the last iteration is an endpoint of the retained interval, so it is within this distance of a minimizer. The subgradient bound gives

$$\langle a, \chi_{\text{best}} \rangle - \min_t \varphi(t) \leq G s 2^{-J} \leq \epsilon_g H_{\text{lb}}/2.$$

If $|a_j| = 1$, the one-coordinate vector with $|x_j| = s/(1 + C\sqrt{w_j})$ proves $-\min_t \varphi(t) \geq H_{\text{lb}}$. This yields the stated relative support guarantee.

Finally, $G \leq m + \sqrt{m/w}/C$, so the displayed definition of L_w bounds $\log(2Gs/H_{\text{lb}})$. Each of the J iterations performs one sort and one prefix scan, for total $O(mJ \log m)$ scalar work and $O(J \log m)$ arithmetic depth. In the application, $e^{-K}n/m \leq w_i \leq 2e^K$ and $C^2 = O(q)$, so $L_w = O(L_*)$. $\square$

We apply the mixed-ball oracle to the softmax potential using only an approximate observation of the residual. The following one-step estimate quantifies the resulting descent and the cost of the observation error.

Lemma 4.11 (Observed greedy softmax step). *Let $N(z) = \|z\|_\infty + C\|z\|_w$, $V = \{z : N(z) \leq s\}$, and $\Phi(d) = 2 \sum_{i=1}^m \cosh(\mu d_i)$. Suppose $b \in V/8$, $s \leq E/100$, $\mu E = 1/100$, and $\|q_{\text{obs}} - q\|_\infty \leq 5E$. Let χ be the output of Lemma 4.10 for radius $(1 + \epsilon_0)s$, gradient $\nabla\Phi(q_{\text{obs}})$, $\epsilon_0 = 1/10$, and a sufficiently small absolute ϵ_g . Then*

$$\Phi(q - b + \chi) \leq \Phi(q) - \frac{3}{4}sN^*(\nabla\Phi(q)) + C_{\text{sm}}\mu ms$$

for an absolute constant C_{sm} .

Proof. The elementary inequalities for $\phi'(t) = 2\mu \sinh(\mu t)$ imply, whenever $|r - t| \leq 5E$,

$$|\phi'(r)| \geq e^{-5\mu E}|\phi'(t)| - \mu, \quad |\phi'(r) - \phi'(t)| \leq (e^{5\mu E} - 1)|\phi'(t)| + \mu.$$

Because V is unconditional, these coordinatewise inequalities pass to its support function, with an additive loss at most μms . The oracle guarantee and $b \in V/8$ therefore give

$$\langle \nabla\Phi(q), -b + \chi \rangle \leq -((1 + \epsilon_0)(1 - \epsilon_g)e^{-5\mu E} - \frac{1}{8} - O(\mu E))sN^*(\nabla\Phi(q)) + O(\mu ms).$$

The coefficient in parentheses is larger than $4/5$ after fixing ϵ_g small. Also $N(-b + \chi) \leq (1/8 + 1 + \epsilon_0)s$ and $\| -b + \chi \|_\infty \leq (1/8 + 1 + \epsilon_0)s$. Taylor's theorem, $\phi'' = \mu^2\phi$, and

$$N(z \odot z) \leq \|z\|_\infty N(z), \quad \mu\phi(t) \leq |\phi'(t)| + 2\mu,$$

bound the quadratic remainder by $O(\mu s)sN^*(\nabla\Phi(q)) + O(\mu^2 ms^2)$. Since $\mu s \leq 10^{-4}$, this loss is smaller than the slack between $4/5$ and $3/4$, proving the claim. $\square$

5 Residual chasing and retained-defect geometry

Section 5.1 proves the stochastic chasing guarantee that keeps the maintained weights close to their moving target despite prediction error. Section 5.2 then identifies the Jacobian defect retained along a finite Newton segment and converts it into a large-radius split predictor. Finally, Section 5.3 gives a computable projection that turns the predictor and its defect certificate into a legal residual-game move.

5.1 Stochastic residual chasing

We combine the observed greedy step with predictable bias and martingale fluctuations in the residual motion. The next lemma is the tracking statement that will be invoked by every later centering call.

Lemma 5.1 (Stochastic residual chasing). *Let $s_k > 0$ be predictable, let $\hat{u}_k$ be a predicted ideal-weight increment, and write $e_k := u_k - \hat{u}_k$, $b_k := \mathbb{E}[e_k \mid \mathcal{F}_{k-1}]$ and $Z_k := e_k - b_k$. For the k th update write*

$$N_k(z) := \|z\|_\infty + C\|z\|_{w_{k-1}},$$

and let N_k^ be its dual norm. Thus every mixed norm in the k th hypotheses, movement set, and movement conclusion is evaluated at the predictable pre-step weight w_{k-1} . Define $\beta := 1 + C\sqrt{2c_1}$ and assume*

$$N_k(b_k) \leq \frac{s_k}{8}, \quad \|Z_k\|_\infty \leq \frac{s_k}{2} \text{ almost surely,} \quad \mathbb{E}[N_k(Z_k)^2 \mid \mathcal{F}_{k-1}] \leq \nu^2 s_k^2, \quad \nu := 1/5.$$

Play the game of [LS19] on the shifted pair $\bar{x}_k := \log w_k - \sum_{j=1}^k \hat{u}_j$, $\bar{y}_k := \log g(x_k) - \sum_{j=1}^k \hat{u}_j$, with movement sets $V_k := \{v : N_k(v) \leq s_k\}$, enlargement $\epsilon_0 := 1/10$, softmax parameter $\mu := 1/(100E)$, and observations of $\bar{y}_k$ carrying ℓ_∞ error at most E . The player uses Lemma 4.10 with a fixed sufficiently small ϵ_g ; its constant relative support loss is included in the numerical slack below. If $s_k \leq E/100$ for every k and

$$E \leq \frac{K}{400 L_{\text{ch}}},$$

let $A_{\text{ch}} \geq 3$ be a sufficiently large absolute constant, define

$$M_{\text{ch}} := A_{\text{ch}} m \beta, \quad d_0 := \log w_0 - \log g(x_0), \quad \Phi(d) := 2 \sum_{i=1}^m \cosh(\mu d_i),$$

and assume

$$\Phi(d_0) \leq M_{\text{ch}}.$$

Choose $A_0 \geq 40A_{\text{ch}}$ and require $L_{\text{ch}} \geq \log(A_0 N m \beta)$. Then with probability at least $19/20$ the stronger invariant

$$\|\log w_k - \log g(x_k)\|_\infty \leq K/4$$

holds simultaneously for all $k \leq N$. If in addition

$$N_k(\hat{u}_k) + 1.1s_k \leq (1 - \frac{1}{c_k})\delta_k,$$

then, deterministically,

$$N_k(\log w_k - \log w_{k-1}) \leq (1 - \frac{1}{c_k})\delta_k.$$

The same conclusions hold for an externally stopped process. Namely, let τ_{ext} be such that the decision to execute step k , equivalently $\mathbf{1}_{\{k < \tau_{\text{ext}}\}}$, is measurable in the pre-noise sigma-field $\mathcal{F}_{k-1}$, and freeze the process from τ_{ext} onward. Then the tracking and movement conclusions hold through $N \wedge (\tau_{\text{ext}} - 1)$; on $\{\tau_{\text{ext}} > N\}$ they give the full-horizon conclusions above.

Proof. The shifted target moves by $\bar{y}_k - \bar{y}_{k-1} = e_k$ and the player by $\bar{x}_k - \bar{x}_{k-1} = \chi_k \in (1 + \epsilon_0)V_k$, so $d_k := \bar{x}_k - \bar{y}_k = \log w_k - \log g(x_k)$ satisfies $d_k = d_{k-1} - e_k + \chi_k$. Stop at the first index τ_K for which $\|d_k\|_\infty > K$, and at the optional external stop τ_{ext} from the statement, freezing the process after the first of these times. For every executed step $k < \tau_K \wedge \tau_{\text{ext}}$, the pre-step tracking invariant gives

$$\|w_{k-1}\|_1 \leq e^K \|g(x_{k-1})\|_1 \leq 2c_1.$$

Thus, for the stopped process, each predictable V_k is symmetric and convex with ℓ_∞ radius at most $s_k \leq E/100$, and $\|v\|_{w_{k-1}} \leq \|v\|_\infty \sqrt{\|w_{k-1}\|_1}$ gives $N_k(v) \leq \beta \|v\|_\infty$, so V_k contains an ℓ_∞ ball of

radius s_k/β . Write $\Phi(d) := 2 \sum_{i=1}^m \cosh(\mu d_i)$ and, for the current step, abbreviate $N := N_k$ and $N^* := N_k^*$, so that the support function of V_k is $h_k(a) = s_k N^*(a)$.

The centered part. The mixed norm is absolute, and coordinatewise $\|z \odot z\|_\infty = \|z\|_\infty^2$ while $\|z \odot z\|_{w_{k-1}} \leq \|z\|_\infty \|z\|_{w_{k-1}}$. Thus, with $\alpha := \|z\|_\infty$ and $\gamma := C\|z\|_{w_{k-1}}$,

$$N(z \odot z) \leq \alpha^2 + \alpha\gamma = \alpha(\alpha + \gamma) \leq N(z)^2.$$

Let x be $\mathcal{F}_{k-1}$ -measurable. Taylor's theorem with $\phi(t) := e^{\mu t} + e^{-\mu t}$, $\phi'' = \mu^2 \phi$ and $\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] = 0$ gives

$$\mathbb{E}[\Phi(x - Z_k) \mid \mathcal{F}_{k-1}] \leq \Phi(x) + \frac{1}{2} \mu^2 e^{\mu \|Z_k\|_\infty} \mathbb{E}[\langle \phi(x), Z_k \odot Z_k \rangle \mid \mathcal{F}_{k-1}],$$

ϕ acting coordinatewise. Since $\cosh \leq |\sinh| + 1$ we have $\mu\phi(t) \leq |\phi'(t)| + 2\mu$, whence

$$\mu \langle \phi(x), z \odot z \rangle \leq \langle |\nabla \Phi(x)|, z \odot z \rangle + 2\mu \|z \odot z\|_1 \leq N^*(\nabla \Phi(x)) N(z)^2 + 2\mu m \|z\|_\infty^2,$$

by Hölder for the pair N, N^* and the displayed quadratic bound. With $\mu \|Z_k\|_\infty \leq \mu s_k/2 \leq 10^{-4}$ and $\mathbb{E}[N(Z_k)^2 \mid \mathcal{F}_{k-1}] \leq \nu^2 s_k^2$ this gives

$$\mathbb{E}[\Phi(x - Z_k) \mid \mathcal{F}_{k-1}] \leq \Phi(x) + \nu^2 \mu s_k h_k(\nabla \Phi(x)) + 2\nu^2 \mu^2 m s_k^2.$$

The key feature of the construction is that a mean-zero error enters the potential only through its *second* moment, and only against the same support function h_k that the player's move decreases.

The bias and the player. We make the conditioning order explicit because the player's greedy move is chosen after Z_k is revealed. Condition first on $\mathcal{F}_{k-1}$ and then on a realization of Z_k , and define $q_k := d_{k-1} - Z_k$. The observed residual used by the greedy move differs from q_k by at most $E + \|b_k\|_\infty \leq E + s_k/8 < 5E$. Lemma 4.11, applied with $b_k \in V_k/8$, gives

$$\Phi(q_k - b_k + \chi_k) \leq \Phi(q_k) - \frac{3}{4} h_k(\nabla \Phi(q_k)) + C_0 \mu m s_k.$$

This is the required deterministic one-step estimate; in particular, the factor 1/8 from the bias body is present in the descent coefficient.

It remains to compare the post-noise gradient in this display with the pre-noise gradient in the centered Taylor estimate. The coordinate inequalities for $\phi'(t) = 2\mu \sinh(\mu t)$ imply that whenever $\|x - y\|_\infty \leq a \leq 1/(5\mu)$ and V_k is symmetric and contained in an ℓ_∞ ball of radius s_k ,

$$h_k(\nabla \Phi(x)) \geq e^{-\mu a} h_k(\nabla \Phi(y)) - \mu m s_k.$$

Indeed, coordinatewise $|\phi'(x_i)| \geq e^{-\mu a} |\phi'(y_i)| - \mu$, and symmetry of V_k allows the maximizing vector to be chosen with the signs of the gradient; summing the additive loss against that vector costs at most $\mu m s_k$. Taking $x = q_k$, $y = d_{k-1}$ and $a = \|Z_k\|_\infty \leq s_k/2$ therefore gives, for every realization of Z_k ,

$$h_k(\nabla \Phi(q_k)) \geq e^{-\mu s_k/2} h_k(\nabla \Phi(d_{k-1})) - \mu m s_k.$$

This is the comparison needed before averaging; no independence between the greedy move and Z_k is being asserted.

Take conditional expectations in the deterministic player inequality and combine it with the centered estimate above. Since $\mu s_k \leq 10^{-4}$, $2\nu^2 \mu^2 m s_k^2 = O(\nu^2 \mu m s_k)$, and the fixed choice $\nu = 1/5$ leaves ample slack below the 3/4 descent coefficient, the result is

$$\mathbb{E}[\Phi(d_k) \mid \mathcal{F}_{k-1}] \leq \Phi(d_{k-1}) - \frac{1}{2} h_k(\nabla \Phi(d_{k-1})) + O(\mu m s_k).$$

From one step to all steps. Since $V_k \supseteq \{v : \|v\|_\infty \leq s_k/\beta\}$ we have $h_k(a) \geq (s_k/\beta)\|a\|_1$, and $|\phi'(t)| \geq \mu(\phi(t) - 2)$ gives $\|\nabla\Phi(d)\|_1 \geq \mu(\Phi(d) - 2m)$. Choose A_{ch} so that

$$\Phi(d_{k-1}) \geq M_{\text{ch}} \quad \implies \quad \mathbb{E}[\Phi(d_k) \mid \mathcal{F}_{k-1}] \leq \Phi(d_{k-1}),$$

so Φ is a nonnegative supermartingale as long as it stays above M_{ch} . By hypothesis $\Phi(d_0) \leq M_{\text{ch}}$. One step moves d by $\|e_k\|_\infty + \|\chi_k\|_\infty = O(s_k)$ in ℓ_∞ , hence changes Φ by a factor at most $e^{O(\mu s_k)} \leq 2$, so every excursion above M_{ch} begins at a value at most $2M_{\text{ch}}$.

Define the excursion stopping times as follows. Define $\tau_0 := 0$ and, successively,

$$\sigma_j := \inf\{k > \tau_{j-1} : \Phi(d_k) > M_{\text{ch}}\}, \quad \tau_j := \inf\{k \geq \sigma_j : \Phi(d_k) \leq M_{\text{ch}}\} \wedge (N+1),$$

with the convention that an empty infimum is $N+1$. Conditional on $\mathcal{F}_{\sigma_j}$, the process $\Phi(d_{(\sigma_j+r) \wedge \tau_j})$ is a nonnegative supermartingale: before τ_j its state is above M_{ch} , exactly where the one-step drift inequality applies, and optional stopping handles the terminal step. Moreover $\Phi(d_{\sigma_j}) \leq 2M_{\text{ch}}$. Doob's maximal inequality therefore gives

$$\Pr\left[\max_{\sigma_j \leq k \leq \tau_j-1} \Phi(d_k) > 40NM_{\text{ch}} \mid \mathcal{F}_{\sigma_j}\right] \leq \frac{1}{20N}.$$

There are at most N nonempty excursions. A conditional union bound shows that

$$\Pr\left[\max_{0 \leq k \leq N} \Phi(d_k) > 40NM_{\text{ch}}\right] \leq \frac{1}{20}.$$

On the complementary event,

$$\|d_k\|_\infty \leq 100E \log(40NA_{\text{ch}}m\beta) \leq 100EL_{\text{ch}} \leq \frac{K}{4}.$$

Thus τ_K cannot occur before τ_{ext} or before the end of the horizon. Removing the tracking stop proves the externally stopped claim; on $\{\tau_{\text{ext}} > N\}$ it proves the original claim.

The movement. This part is deterministic, independently of the estimator outputs. Since $N_k(\chi_k) \leq (1 + \epsilon_0)s_k = 1.1s_k$, the additional hypothesis gives

$$N_k(\log w_k - \log w_{k-1}) = N_k(\widehat{u}_k + \chi_k) \leq (1 - \frac{1}{c_k})\delta_k,$$

where the first step follows from the definition of the shifted player and the update $\bar{x}_k - \bar{x}_{k-1} = \chi_k$, and the second step follows from the triangle inequality, $N_k(\chi_k) \leq (1 + \epsilon_0)s_k = 1.1s_k$, and the additional hypothesis of the lemma. $\square$

Because the predictor is a Jacobian–vector product, we must control the variation of the Jacobian along a centering step. The resolvent representation below provides the appropriate form.

Lemma 5.2 (Resolvent form). *With $\Lambda := \Sigma - P \circ P$ and $\mathcal{L} := W^{-1}\Lambda$, the Jacobian from Lemma 4.6-(a) is*

$$B = 2(I + a\mathcal{L})^{-1}\mathcal{L} = \frac{2}{a}(I - (I + a\mathcal{L})^{-1}),$$

and $(I + a\mathcal{L})^{-1}$ is entrywise nonnegative with all row sums equal to 1, and is self-adjoint and positive definite for $\langle \cdot, \cdot \rangle_w$. In particular

$$\|(I + a\mathcal{L})^{-1}\|_{\infty \rightarrow \infty} = 1 \quad \text{and} \quad \|(I + a\mathcal{L})^{-1}\|_{w \rightarrow w} \leq 1.$$

Proof. Lemma 4.3 justifies the derivative used here, and the proof of Lemma 4.6-(a) gives $B = (W + a\Lambda)^{-1}2\Lambda$. Hence $B = (I + aW^{-1}\Lambda)^{-1}W^{-1}2\Lambda = 2(I + a\mathcal{L})^{-1}\mathcal{L}$, and the identity $(I + aX)^{-1}X = a^{-1}(I - (I + aX)^{-1})$ gives the second form. Now $\Lambda_{ii} = \sigma_i - \sigma_i^2$ and $\Lambda_{ij} = -P_{ij}^2$ for $j \neq i$, so $I + a\mathcal{L}$ has positive diagonal and nonpositive off-diagonal entries; and $\sum_{j=1}^m P_{ij}^2 = P_{ii} = \sigma_i$ gives $\Lambda\mathbf{1} = 0$, so all its row sums are exactly 1. It is therefore strictly diagonally dominant with nonpositive off-diagonals, hence a nonsingular M -matrix, so its inverse is entrywise nonnegative; and $(I + a\mathcal{L})\mathbf{1} = \mathbf{1}$ gives $(I + a\mathcal{L})^{-1}\mathbf{1} = \mathbf{1}$. A nonnegative matrix with unit row sums has ℓ_∞ operator norm 1. Finally $W\mathcal{L} = \Lambda$ is symmetric, so $\mathcal{L}$ is w -self-adjoint, and $\langle y, \mathcal{L}y \rangle_w = y^\top \Lambda y \geq 0$ since $\Lambda \succeq 0$; hence the w -spectrum of $I + a\mathcal{L}$ lies in $[1, \infty)$. Both properties hold for *every* positive w , not only at the Lewis fixed point: $\Lambda\mathbf{1} = 0$ and the symmetry of Λ are identities of the projection of $\mathbf{W}^{\vartheta/2}\text{Diag}(c)A$ whatever w is, because Σ is by definition the diagonal of that projection. They therefore survive the substitution of an approximate weight vector for the exact one; the resulting stability estimate is quantified below. $\square$

The resolvent representation converts changes of the Lewis-weight fixed point into a perturbation problem for a stochastic M -matrix. The next lemma uses that structure to control the Jacobian simultaneously in both maintained norms.

Lemma 5.3 (Jacobian stability). *For two row scalings $c, c' > 0$, let $w = w_p^v(\text{Diag}(c)A)$ and $w' = w_p^v(\text{Diag}(c')A)$ be the corresponding exact regularized Lewis vectors, and let B, B' be the Jacobians of their logarithms with respect to the logarithmic row scalings. Suppose $\|\log(c'/c)\|_\infty \leq \rho$ and $\|\log(w'/w)\|_\infty \leq \rho \leq 1/16$. Then*

$$\|B' - B\|_{w+\infty \rightarrow w+\infty} = O(\rho).$$

Proof. By Lemma 5.2 and the resolvent identity,

$$B' - B = 2(I + a\mathcal{L}')^{-1}(\mathcal{L}' - \mathcal{L})(I + a\mathcal{L})^{-1},$$

with no conjugation by powers of W . The two outer factors are contractions in ℓ_∞ and, up to a factor $e^\rho \leq 2$ coming from w' versus w , in $\|\cdot\|_w$. Hence $\|B' - B\|_{\infty \rightarrow \infty} \leq 4\|\Delta\mathcal{L}\|_{\infty \rightarrow \infty}$ and $\|B' - B\|_{w \rightarrow w} \leq 4\|\Delta\mathcal{L}\|_{w \rightarrow w}$, and since an operator bounded by α in both $\|\cdot\|_\infty$ and $\|\cdot\|_w$ is bounded by α in $\|\cdot\|_{w+\infty}$, it suffices to bound $\Delta\mathcal{L} := \mathcal{L}' - \mathcal{L}$ in those two norms.

The ℓ_∞ norm. The projection is that of $\mathbf{W}^{\vartheta/2}\text{Diag}(c)A$, so the two hypotheses combine into a single drift of its rescaling,

$$\rho_* := (1 + \frac{|\vartheta|}{2})\rho \leq 2\rho \leq \frac{1}{8},$$

using $|\vartheta| = \frac{2}{p} - 1 \leq 2$ for $p \geq \frac{2}{3}$; it is ρ_* , not ρ , with which Lemma 4.1 must be applied. Row by row,

$$\sum_{j=1}^m |\Delta\mathcal{L}_{ij}| \leq \frac{1}{w'_i} \sum_{j=1}^m |\Lambda'_{ij} - \Lambda_{ij}| + \left| \frac{w_i}{w'_i} - 1 \right| \frac{1}{w_i} \sum_{j=1}^m |\Lambda_{ij}|.$$

The exact identity $\sum_{j=1}^m |\Lambda_{ij}| = 2\sigma_i(1 - \sigma_i) \leq 2w_i$ makes the second term at most 3ρ . For the first, Lemma 4.1 gives $\|(P' - P)e_i\|_2 \leq 10\rho_*\sqrt{\sigma_i}$ and $\sigma'_i = e^{\pm 4\rho_*}\sigma_i$, whence $\|P'e_i\|_2 + \|Pe_i\|_2 \leq (1 + e^{2\rho_*})\sqrt{\sigma_i} \leq 2.3\sqrt{\sigma_i}$ and

$$\sum_{j=1}^m |P'^2_{ij} - P^2_{ij}| \leq \|(P' - P)e_i\|_2 (\|P'e_i\|_2 + \|Pe_i\|_2) \leq 23\rho_*\sigma_i \leq 46\rho\sigma_i;$$

the diagonal contributes $|\sigma'_i - \sigma_i| + |\sigma'^2_i - \sigma^2_i| \leq (e^{4\rho_*} - 1)(2 + e^{4\rho_*})\sigma_i \leq 38\rho\sigma_i$, using $\sigma_i \leq 1$. Dividing by $w'_i \geq e^{-\rho}w_i \geq e^{-\rho}\sigma_i$ leaves at most 90ρ , so $\|\Delta\mathcal{L}\|_{\infty \rightarrow \infty} \leq 93\rho$.

The w norm. Here reversibility does all the work, and no spectral perturbation argument is needed. Since $W\mathcal{L} = \Lambda$ is symmetric, $w_i\mathcal{L}_{ij} = w_j\mathcal{L}_{ji}$, and likewise for the primed quantities. Hence

$$\frac{w_i}{w_j} \Delta\mathcal{L}_{ij} = \frac{w_i w'_j}{w'_i w_j} \mathcal{L}'_{ji} - \mathcal{L}_{ji}, \quad \frac{w_i w'_j}{w'_i w_j} = e^{\pm 2\rho},$$

so the weighted column sums of $\Delta\mathcal{L}$ are controlled by its row sums. Writing $\alpha_r := \|\Delta\mathcal{L}\|_{\infty \rightarrow \infty}$ and α_c for that weighted column sum,

$$\alpha_c \leq \alpha_r + (e^{2\rho} - 1) \|\mathcal{L}'\|_{\infty \rightarrow \infty} \leq 93\rho + 5\rho \leq 98\rho,$$

using $\|\mathcal{L}\|_{\infty \rightarrow \infty} = \max_{i \in [m]} 2\sigma_i(1 - \sigma_i)/w_i \leq 2$. The Schur test in $\ell_2(w)$ states that $\|X\|_{w \rightarrow w}^2$ is at most the product of the two, and follows from Cauchy–Schwarz applied to $\sum_{i=1}^m w_i (\sum_{j=1}^m X_{ij} d_j)^2$; it gives $\|\Delta\mathcal{L}\|_{w \rightarrow w} \leq \sqrt{93 \cdot 98} \rho \leq 96\rho$. Collecting, $\|B' - B\|_{w+\infty \rightarrow w+\infty} \leq 4 \max\{93\rho, 96\rho\} \leq 384\rho$. $\square$

The predictor is evaluated over an entire Newton segment rather than at one endpoint. The next lemma integrates the Jacobian stability estimate and bounds the error of replacing the varying Jacobian by its initial value.

Lemma 5.4 (Finite-segment Jacobian error). *Under the Newton-segment hypotheses of Lemma 4.9, define*

$$x_s := x + s\Delta, \quad \psi_s := \frac{d}{ds} \log s(x_s), \quad \psi := \int_0^1 \psi_s \, ds,$$

and let B_s be the Jacobian of $\log g(x_s)$ with respect to the logarithmic row scaling. With $B := B_0$ and $u := \log g(x_1) - \log g(x_0)$, one has

$$N_w(\psi) \leq (1 + 8R)c_\gamma \delta, \quad N_w(u - B\psi) \leq a_1 \delta^2$$

for an absolute constant a_1 .

Proof. Keep the notation $h = \sqrt{\Phi''(x)}\Delta$ from the proof of Lemma 4.9. Coordinatewise self-concordance and Lemma 3.2 give, for every $s \in [0, 1]$,

$$|\psi_{s,i}| \leq \frac{|h_i|}{1 - s|h_i|}, \quad N_w(\psi_s) \leq \frac{N_w(h)}{1 - \|h\|_\infty} \leq (1 + 8R)c_\gamma \delta.$$

Integration proves the first claim and also $\|\log s(x_s) - \log s(x)\|_\infty \leq 3\delta$ after decreasing r_0 . Target stability then gives $\|\log g(x_s) - \log g(x)\|_\infty \leq 12\delta$. Apply Lemma 5.3 to the endpoints 0 and s . The exact $g(x)$ -mixed norm and the maintained w -mixed norm differ by at most the absolute factor $e^{K/2}$, so, uniformly in s ,

$$\|B_s - B_0\|_{w+\infty \rightarrow w+\infty} \leq A_J \delta$$

for an absolute A_J . The differentiability conclusion of Lemma 4.3 gives

$$u = \int_0^1 B_s \psi_s \, ds.$$

Therefore

$$N_w(u - B_0\psi) \leq \int_0^1 \|B_s - B_0\|_{w+\infty \rightarrow w+\infty} N_w(\psi_s) \, ds \leq A_J(1 + 8R)c_\gamma \delta^2 \leq a_1 \delta^2,$$

where the first step follows from the integral representations of u and ψ , the triangle inequality, and the definition of the operator norm, the second step follows from the uniform bounds on $B_s - B_0$ and ψ_s above, and the third step follows from $R \leq r_0/c_k$ and $c_\gamma \leq 1 + 1/(128c_k)$ by choosing a sufficiently large absolute constant a_1 . $\square$

5.2 Large-radius defect geometry

This subsection combines the sharper geometry and stochastic residual game above. The objective is to use the negative term discarded in Lemma 4.7 to pay for the part of the Jacobian–vector product that is costly to estimate. Throughout this subsection, $N(z) := \|z\|_\infty + C\|z\|_w$, where w is the maintained weight at the beginning of the centering step. All conditional expectations are with respect to the algorithmic filtration immediately before the fresh sketches named in the statement are drawn.

Let g be the exact regularized Lewis vector at the beginning of a centering step and let B be the Jacobian of $\log g$ with respect to the logarithmic row scaling. For a direction h , define

$$r := \|h\|_g, \quad b := Bh, \quad s := \|b\|_g, \quad d(h)^2 := p^2 r^2 - s^2.$$

Lemma 4.6-(a) makes $d(h)$ real.

The basic contraction estimate discards the slack measured by $d(h)$, but the predictor must retain that slack to pay for its approximation error. The next lemma makes the negative defect term explicit.

Lemma 5.5 (Retained Jacobian defect). *Suppose $\|\log w - \log g\|_\infty \leq H \leq 1/4$. Then*

$$N(Bh) \leq p(1 + e^{H/2}(\frac{2}{pC} + L_H)^2)N(h) - \frac{C^2}{4pe^{H/2}} \frac{d(h)^2}{N(h)},$$

where $L_H = e^{3H/2} - 1$. The last quotient is interpreted as zero when $h = 0$.

Proof. The proof of Lemma 4.7, before its last maximization, gives

$$N(Bh) - pN(h) \leq A_H d + C(s - pr), \quad A_H := \frac{2}{p} + CL_H,$$

where $d = d(h)$. If $r > 0$, then

$$s - pr = -\frac{d^2}{pr + s} \leq -\frac{d^2}{2pr}.$$

Young's inequality gives

$$A_H d \leq \frac{A_H^2 pr}{C} + \frac{Cd^2}{4pr},$$

and hence

$$N(Bh) \leq pN(h) + \frac{A_H^2 pr}{C} - \frac{Cd^2}{4pr}.$$

Since $N(h) \geq C\|h\|_w \geq Ce^{-H/2}r$, we have $pr \leq pe^{H/2}N(h)/C$. Applying this bound to both terms in the last display proves the claim. The case $r = 0$ is immediate. $\square$

We enlarge the centrality radius while retaining logarithmic cost per step. Keep

$$E = \frac{K}{A_E L_{\text{ch}}} = \Theta(q^{-1/2} \ell^{-1})$$

for a sufficiently large absolute A_E , and set

$$R := c_R C E = \Theta(\ell^{-1})$$

for a sufficiently small absolute $c_R > 0$. These choices give

$$R = O(1/c_k), \quad E = O(1/c_k), \quad R^2 = O(E).$$

The last relation follows from $C^2 E = O(\sqrt{q}/\ell) = O(1)$.

The next lemma extends the retained-movement estimate to a centrality radius R that may be larger than the score accuracy E . It uses only the three scale inequalities above, not the stronger relation $R = \Theta(E)$.

Lemma 5.6 (Large-radius retained movement). *There are absolute constants $r_0, \gamma > 0$ such that, with the parameters of Lemma 4.9, the following holds. Suppose $E \leq r_0/c_k$, $R \leq r_0/c_k$, and $\delta := \delta_t(x, w) \leq R$. For*

$$\psi = \log s(x^+) - \log s(x), \quad u = \log g(x^+) - \log g(x), \quad d = d(\psi),$$

one has

$$N(u) \leq (1 - \frac{4}{3c_k})\delta - \gamma \frac{C^2 d^2}{\delta}, \quad d^2 \leq \frac{5\delta^2}{C^2}.$$

Proof. Lemma 4.9 keeps the entire Newton segment in the band $H \leq 2K$. Its numerical estimate gives

$$e^{H/2}(\frac{2}{pC} + L_H)^2 < \frac{1}{4c_k}.$$

Since $p = 1 - 2/c_k$, Lemma 5.5, applied directly at the base point to $h = \psi$, therefore gives

$$N(B\psi) \leq (1 - \frac{7}{4c_k})N(\psi) - \gamma_1 \frac{C^2 d^2}{N(\psi)}$$

for an absolute $\gamma_1 > 0$. This derivation uses only the segment band and the parameter inequalities, not any separate small-radius specialization. Lemma 5.4 gives

$$N(\psi) \leq (1 + 8R)c_\gamma \delta, \quad N(u - B\psi) \leq a_1 \delta^2$$

for an absolute a_1 . Since $c_\gamma \leq 1 + 1/(128c_k)$ and $R, \delta \leq r_0/c_k$, the total positive error in substituting these two displays is at most

$$(1/(128c_k) + 8r_0/c_k + O(r_0^2/c_k^2))\delta + a_1 r_0 \delta / c_k.$$

Choose r_0 so that this is at most $\delta/(6c_k)$. Also $N(\psi) \leq 2\delta$, after the same choice, so the negative term is at most $-(\gamma_1/2)C^2 d^2/\delta$. Decreasing its absolute coefficient gives the first claim. Finally metric equivalence gives

$$C\|\psi\|_g \leq e^K N(\psi) \leq 2e^K \delta,$$

and therefore $d^2 \leq p^2 \|\psi\|_g^2 \leq 5\delta^2/C^2$. $\square$

The retained defect is algorithmically useful only if the Jacobian action can be estimated at a cost proportional to that defect. The following lemma separates the weighted and coordinatewise pieces needed for this purpose.

Lemma 5.7 (Large-radius defect split predictor). *Under the hypotheses of Lemma 5.6, define $\Delta := \bar{\delta}_t(x, w)$. There is a coarse sigma-field $\mathcal{G}_k$ and estimates p^∞, p^w such that, except with conditional probability ζ ,*

$$\|p^\infty - u\|_\infty \leq \epsilon_1(\bar{d} + \frac{\Delta}{c_k}),$$

and, conditionally on $\mathcal{G}_k$,

$$\mathbb{E}[\|p^w - u\|_w^2 \mid \mathcal{G}_k] \leq \epsilon_1^2 \left(\frac{\bar{d}^2}{c_k} + \frac{\Delta^2}{C^2 c_k^2} \right).$$

The first estimate and the certificate cost $O(\log(m/\zeta))$ systems, the second costs $O(c_k)$ systems, and ϵ_1 may be made an arbitrarily small absolute constant. Algorithm 4 implements these estimates with the displayed failure budget; all block sizes and copy counts are fixed before their randomness is drawn.

Proof. Construct the certified surrogate of Lemma 7.1 at the fixed query. Algorithm 4 uses the explicit failure split and copy counts stated there. Its Jacobian has the form $W^{-1/2}(D + K)W^{1/2}$. Add the exactly computable diagonal term D to two independent estimators of K from Lemma 6.7: a coordinate chain and, drawn only after the coarse sigma-field is fixed, a weighted chain. This estimator construction and the certificate of Lemma 6.9 are algebraic statements about the fixed surrogate and have no small-radius hypothesis.

Write $J^+ := (W^+)^{-1/2}(D^+ + K^+)(W^+)^{1/2}$ for the surrogate Jacobian and define

$$e_{\text{det}} := J^+ \psi - u.$$

This is precisely the sum of the surrogate-operator perturbation and the finite-segment remainder. Lemmas 7.1 and 5.4, together with the constant equivalence of W^+ , g , and the maintained w , give

$$N(e_{\text{det}}) \leq a_2(E\delta + \delta^2)$$

for an absolute a_2 . Since $E, R \leq r_0/c_k$,

$$N(e_{\text{det}}) \leq 2a_2 r_0 \delta / c_k.$$

Choose $r_0 \leq \epsilon_1/(8a_2)$. Since $\delta \leq \Delta$, this gives

$$\|e_{\text{det}}\|_\infty \leq \frac{\epsilon_1}{4} \frac{\Delta}{c_k}, \quad \|e_{\text{det}}\|_w \leq \frac{\epsilon_1}{4C} \frac{\Delta}{c_k}.$$

Draw the coordinate chain and the certificate first. If $\xi^\infty := (W^+)^{-1/2}(\widehat{Kx}_\infty - K^+x^+)$ denotes the physical-coordinate stochastic error, then the normalized coordinate conclusion of Lemma 6.7 is exactly

$$\|\xi^\infty\|_\infty \leq \epsilon_\infty d_+.$$

The explicit construction in Lemma 6.9 gives $d_+^2 \leq \bar{d}^2$. Choose its absolute block size so that $\epsilon_\infty \leq \epsilon_1/2$. Hence, on the coarse event,

$$\|p^\infty - u\|_\infty \leq \|\xi^\infty\|_\infty + \|e_{\text{det}}\|_\infty \leq \frac{\epsilon_1}{2} \bar{d} + \frac{\epsilon_1}{4} \frac{\Delta}{c_k},$$

where the first step follows from $p^\infty - u = \xi^\infty + e_{\text{det}}$ and the triangle inequality, and the second step follows from $\|\xi^\infty\|_\infty \leq \epsilon_\infty d_+$, $d_+ \leq \bar{d}$, $\epsilon_\infty \leq \epsilon_1/2$, and the deterministic bound on $\|e_{\text{det}}\|_\infty$ above. This is stronger than the first displayed conclusion.

Let $\mathcal{G}_k$ be generated by the past, the fixed surrogate-score sketch, the coordinate chain, and the certificate. Draw the weighted chain only afterward. Set $\xi^w := (W^+)^{-1/2}(\widehat{Kx}_w - K^+x^+)$.

Conditional on $\mathcal{G}_k$, we can show

$$\mathbb{E}[\|\xi^w\|_w^2 \mid \mathcal{G}_k] \leq c_{\text{eq}} \frac{a_4 c_w^2 d_+^2}{c_k} \leq \frac{\epsilon_1^2}{8} \frac{\bar{d}^2}{c_k},$$

where the first step follows from the conditional Euclidean guarantee of Lemma 6.7 and the norm equivalence between W^+ and w , and the second step follows from $d_+^2 \leq \bar{d}^2$ and the choice $c_{\text{eq}} a_4 c_w^2 \leq \epsilon_1^2/8$ of the absolute accuracy multiplier c_w in Algorithm 4. The conditional statement is legitimate because S_w and every query operator are $\mathcal{G}_k$ -measurable before this independent chain is drawn.

We can show

$$\mathbb{E}[\|p^w - u\|_w^2 \mid \mathcal{G}_k] \leq \frac{\epsilon_1^2 \bar{d}^2}{4 c_k} + \frac{\epsilon_1^2 \Delta^2}{8 C^2 c_k^2} \leq \epsilon_1^2 \left(\frac{\bar{d}^2}{c_k} + \frac{\Delta^2}{C^2 c_k^2} \right),$$

where the first step follows from $p^w - u = \xi^w + e_{\text{det}}$, Young's inequality in the form $(a+b)^2 \leq 2a^2 + 2b^2$, the conditional bound on ξ^w , and the deterministic bound on $\|e_{\text{det}}\|_w$ above, and the second step follows from $1/4 \leq 1$ and $1/8 \leq 1$. The coordinate chain and certificate use $O(\log(m/\zeta))$ systems, and the independent weighted chain uses $O(c_k)$ systems. This proves every assertion directly at the large radius. $\square$

5.3 Projection onto the defect body

The key point is to use the computable proxy $\Delta_k := \bar{\delta}_{t_k}(x_k, w_k)$ and to measure the residual with a radius linear in the certified defect:

$$s_k := \eta_0 \left(\bar{d}_k + \frac{\Delta_k}{c_k} \right),$$

where $\eta_0 > 0$ is a sufficiently small absolute constant. Set

$$r_k := \left(1 - \frac{59}{48c_k} \right) \Delta_k - B \frac{C^2 \bar{d}_k^2}{\Delta_k}.$$

The branch $\Delta_k = 0$ is handled before these quantities are formed.

For a fixed t below, write $\text{FIXEDSLICE}(t, w, \bar{p}, l, h, r, C) = (x, g)$ for the exact active-set calculation in the proof of Lemma 5.8: it minimizes the weighted quadratic on the slice $|x_i| \leq t$, $l_i \leq x_i \leq h_i$, and $\|x\|_w \leq (r - t)/C$, and returns the displayed KKT subgradient g . At ties it uses the strict moving-face indicators $\mathbf{1}_{\{l_i < x_i = t < h_i\}}$ and $\mathbf{1}_{\{l_i < x_i = -t < h_i\}}$. To make the auxiliary routine explicit, define $I_i(t) := [\max\{l_i, -t\}, \min\{h_i, t\}]$ and $A(t) := (r - t)/C$. If $y_i := \text{clip}(\bar{p}_i, I_i(t))$ satisfies $\|y\|_w \leq A(t)$, set $x := y$ and $\lambda := 0$. Otherwise, sort the values at which $\bar{p}_i/(1 + \lambda)$ meets an endpoint of $I_i(t)$. On a region with free set F and clipped endpoint values b_i , set

$$1 + \lambda := \left(\frac{\sum_{i \in F} w_i \bar{p}_i^2}{A(t)^2 - \sum_{i \notin F} w_i b_i^2} \right)^{1/2}, \quad x_i := \begin{cases} \bar{p}_i/(1 + \lambda), & i \in F, \\ b_i, & i \notin F. \end{cases}$$

The scan selects the region satisfying its clipping inequalities. For either branch, define

$$\alpha_i^+ := w_i(\bar{p}_i - (1 + \lambda)t) \mathbf{1}_{\{l_i < x_i = t < h_i\}}, \quad \alpha_i^- := w_i(-\bar{p}_i - (1 + \lambda)t) \mathbf{1}_{\{l_i < x_i = -t < h_i\}},$$

and return $g := \lambda A(t)/C - \sum_{i=1}^m (\alpha_i^+ + \alpha_i^-)$.

Algorithm 2 Approximate projection onto the defect body

```
1: procedure DEFECTPROJECTION( $w, C, r, s, p^\infty, p^w$ ) ▷ Lemma 5.8
2:    $\bar{p} \leftarrow \min\{1, r/(C\|p^w\|_w)\}p^w$ , with multiplier 1 when  $p^w = 0$ .
3:    $l_i \leftarrow p_i^\infty - s/64$ ,  $h_i \leftarrow p_i^\infty + s/64$ , and  $z_i \leftarrow \text{clip}(0, [l_i, h_i])$  for every  $i$ .
4:   if  $\|z\|_\infty + C\|z\|_w > r$  then
5:     return FAIL.
6:   end if
7:    $\tau_P \leftarrow \epsilon_P s / C$ ,  $W_1 \leftarrow \sum_{i=1}^m w_i$ , and  $\Lambda_P \leftarrow 32(r/(C\tau_P))^2$ .
8:    $G_P \leftarrow \Lambda_P r / C^2 + \sqrt{W_1}\|\bar{p}\|_w + (1 + \Lambda_P)rW_1$ .
9:    $J \leftarrow \lceil \log_2(2 + 8G_P r / \tau_P^2) \rceil$ ,  $t_- \leftarrow \|z\|_\infty$ ,  $t_+ \leftarrow r - C\|z\|_w$ , and  $t_{\text{edge}} \leftarrow r - C\sqrt{\|z\|_w^2 + \tau_P^2/16}$ .
10:  if  $t_{\text{edge}} \leq t_-$  then
11:    return  $z$ .
12:  end if
13:   $(x_{\text{edge}}, g_{\text{edge}}) \leftarrow \text{FIXEDSLICE}(t_{\text{edge}}, w, \bar{p}, l, h, r, C)$ .
14:  if  $g_{\text{edge}} < 0$  then
15:    return  $z$ .
16:  else if  $g_{\text{edge}} = 0$  then
17:    return  $x_{\text{edge}}$ .
18:  end if
19:   $t_+ \leftarrow t_{\text{edge}}$  and  $v_{\text{best}} \leftarrow x_{\text{edge}}$ .
20:  for  $j = 1, \dots, J$  do
21:     $t \leftarrow (t_- + t_+)/2$  and  $(x, g) \leftarrow \text{FIXEDSLICE}(t, w, \bar{p}, l, h, r, C)$ .
22:    Retain  $x$  if it improves  $\frac{1}{2}\|x - \bar{p}\|_w^2$ .
23:    if  $g > 0$  then
24:       $t_+ \leftarrow t$ .
25:    else if  $g < 0$  then
26:       $t_- \leftarrow t$ .
27:    else
28:      return  $x$ .
29:    end if
30:  end for
31:  return  $v_{\text{best}}$ .
32: end procedure
```

The split predictor produces two approximate descriptions of the same motion, which must be reconciled without an implicit continuous optimization oracle. The next lemma certifies the finite projection procedure above.

Lemma 5.8 (Computable defect projection). *Let $w > 0$, $C, r, s > 0$, and $p^\infty, p^w \in \mathbb{R}^m$. Define*

$$\mathcal{C}(p^\infty; r, s) := \{v : \|v\|_\infty + C\|v\|_w \leq r, \|v - p^\infty\|_\infty \leq s/64\}.$$

Fix an absolute $\epsilon_P > 0$. If this set is nonempty, Algorithm 2 returns a feasible $\tilde{v}$ such that

$$\|\tilde{v} - \Pi_{\mathcal{C}}^w(\bar{p})\|_w \leq \tau_P := \epsilon_P s / C, \quad \bar{p} := \min\{1, r/(C\|p^w\|_w)\}p^w,$$

where $\Pi_{\mathcal{C}}^w$ is exact metric projection in $\|\cdot\|_w$. When $p^w = 0$, the minimum factor is interpreted as 1. If there is a scale $\Delta > 0$ such that $r \leq \Delta$ and $s \geq \eta_0 \Delta / c_k$, the routine uses no linear systems.

In the tracked-weight regime used below, where $\sum_{i=1}^m w_i \leq 2e^K n$, $C^2 = O(q)$, and $q \leq L_*$, it has

$$O(mJ \log m) \text{ scalar work}, \quad O(J \log m) \text{ arithmetic depth}, \quad J = O(L_* + \log(2 + c_k/(\eta_0 \epsilon_P))).$$

It also detects an empty set, in which case the calling algorithm returns FAIL.

Proof. Write $l_i = p_i^\infty - s/64$, $h_i = p_i^\infty + s/64$, and $z_i = \text{clip}(0, [l_i, h_i])$. The vector z has the smallest absolute value in every coordinate among all vectors in the box. Hence the set is nonempty exactly when

$$\|z\|_\infty + C\|z\|_w \leq r.$$

For $t \in [\|z\|_\infty, r - C\|z\|_w]$, define

$$I_i(t) := [\max\{l_i, -t\}, \min\{h_i, t\}], \quad A(t) := (r - t)/C,$$

and let $\varphi(t)$ be the minimum of $\frac{1}{2}\|x - \bar{p}\|_w^2$ over $x_i \in I_i(t)$ and $\|x\|_w \leq A(t)$. This is the partial minimum of a jointly convex problem, so φ is convex.

We first make the fixed- t computation explicit. If $y_i = \text{clip}(\bar{p}_i, I_i(t))$ satisfies $\|y\|_w \leq A(t)$, it is the exact minimizer and $\lambda = 0$. Otherwise the ball is active and

$$x_i(\lambda, t) = \text{clip}(\bar{p}_i/(1 + \lambda), I_i(t)).$$

The at most $2m$ nonnegative values at which $\bar{p}_i/(1 + \lambda)$ meets an interval endpoint partition $\lambda \geq 0$ into regions. In a region, the clipped coordinates are fixed endpoints b_i and the others form a free set F ; the equation $\|x\|_w = A(t)$ has exactly the formula displayed before the algorithm. Scanning the sorted regions finds the one satisfying its clipping inequalities. This is an exact active-set computation, so its running time does not depend on the magnitude of λ .

At its KKT tuple, assign a multiplier at a tie between a moving face and a fixed box face to the fixed face. Thus a moving upper face is charged only when $l_i < x_i = t < h_i$, and a moving lower face only when $l_i < x_i = -t < h_i$. The two formulas for $\alpha_i^\pm$ displayed before the algorithm are then the exact multipliers of the moving faces, including every degenerate tie, and

$$g(t) = \frac{\lambda A(t)}{C} - \sum_{i=1}^m (\alpha_i^+ + \alpha_i^-) \in \partial\varphi(t).$$

Thus every non-endpoint branch uses an exact subgradient; there are no undefined multiplier intervals or approximate lower certificates.

It remains to justify the quantitative edge and iteration rules. For any x in the original coordinate box, $x_i z_i \geq z_i^2$, and hence

$$\|x - z\|_w^2 \leq \|x\|_w^2 - \|z\|_w^2.$$

Consequently, if $A(t)^2 - \|z\|_w^2 \leq \tau_P^2/16$, every feasible point at every parameter $u \geq t$ is within $\tau_P/4$ of z .

The algorithm uses this fact without mixing two incompatible certificates. If $t_{\text{edge}} \leq t_-$, the preceding estimate applies to the whole parameter interval and returning z is valid. Otherwise compute the exact subgradient g_{edge} of the convex function φ at t_{edge} . If $g_{\text{edge}} < 0$, every minimizer of φ lies to the right of t_{edge} and is therefore within $\tau_P/4$ of z ; if $g_{\text{edge}} = 0$, the fixed-slice point is the exact projection. Only when $g_{\text{edge}} > 0$ does the algorithm retain the left interval. That interval contains a minimizer and every subsequent cut is certified by an exact subgradient. In particular, a point certified only by the right edge is never later overwritten by an objective comparison.

On every other branch, nonexpansiveness of coordinate clipping gives

$$\|x(\lambda, t) - z\|_w \leq \|\bar{p}\|_w / (1 + \lambda).$$

Since $\|\bar{p}\|_w \leq r/C$ and $A(t) \leq r/C$,

$$A(t)^2 - \|z\|_w^2 \leq 2A(t)\|x(\lambda, t) - z\|_w \leq \frac{2(r/C)^2}{1 + \lambda}.$$

On the retained interval the reverse edge inequality therefore implies $1 + \lambda \leq \Lambda_P$. With $W_1 = \sum_{i=1}^m w_i$, the explicit moving-face formulas give

$$\sum_{i=1}^m (\alpha_i^+ + \alpha_i^-) \leq \sqrt{W_1} \|\bar{p}\|_w + (1 + \lambda)tW_1.$$

Since $t \leq r$, every exact subgradient used outside the endpoint branch has magnitude at most G_P .

Exact-subgradient bisection on the left interval preserves an interval containing a minimizer. After J iterations the retained interval has length at most $r2^{-J}$, and the best evaluated exact fixed- t point has objective gap at most $2G_P r 2^{-J} \leq \tau_P^2/4$. The projection variational inequality then gives $\|\tilde{v} - \Pi_{\mathcal{C}}^w(\bar{p})\|_w \leq \tau_P$. Every returned point is feasible by construction.

Each outer iteration sorts at most $2m$ breakpoints and scans them once. Moreover $r/s \leq c_k/\eta_0$ under the stated scale hypothesis, $\tau_P = \epsilon_P s/C$, and in the application $\sum_{i=1}^m w_i \leq 2e^K n$, $C^2 = O(q)$, and $q \leq L_*$. Taking logarithms in the explicit formula for J gives the stated bound. Hence the scalar work and depth are $O(mJ \log m)$ and $O(J \log m)$, respectively. These costs are charged by the global scalar counter introduced below; they are not attributed to one linear-system solve. $\square$

We apply the computable projection to the two predictor estimates and the defect certificate. The following lemma verifies that the resulting motion fits the residual game while preserving the centrality budget.

Lemma 5.9 (Defect projection). *There are choices of the absolute constants in this construction for which, on the coarse event of Lemma 5.7, with $\delta_k := \delta_{t_k}(x_k, w_k)$ and $\Delta_k := \bar{\delta}_{t_k}(x_k, w_k)$,*

$$r_k \geq N(u_k), \quad r_k + 1.1s_k \leq (1 - \frac{1}{c_k})\delta_k, \quad s_k \leq E/100.$$

Let

$$\mathcal{C}_k := \{v : N(v) \leq r_k, \|v - p^\infty\|_\infty \leq s_k/64\}$$

and let $\hat{u}_k$ be the feasible output of Lemma 5.8, applied with scale Δ_k (indeed, $r_k \leq \Delta_k$ and $s_k \geq \eta_0 \Delta_k / c_k$), with $\tau_P = \epsilon_P s_k / C$. Then, with

$$e_k := u_k - \hat{u}_k, \quad b_k := \mathbb{E}[e_k \mid \mathcal{G}_k], \quad Z_k := e_k - b_k,$$

one has

$$N(b_k) \leq s_k/8, \quad \|Z_k\|_\infty \leq s_k/2, \quad \mathbb{E}[N(Z_k)^2 \mid \mathcal{G}_k] \leq \nu^2 s_k^2,$$

with $\nu = 1/5$, as required by Lemma 5.1.

Proof. The one-sided certificate gives

$$d_k^2 \leq \bar{d}_k^2 \leq A_d(d_k^2 + \frac{\Delta_k^2}{C^2 c_k}).$$

Choose B so that $BA_d \leq \min\{1/12, \gamma/2\}$. The certificate then gives

$$B \frac{C^2 \bar{d}_k^2}{\Delta_k} \leq \frac{\gamma}{2} \frac{C^2 d_k^2}{\Delta_k} + \frac{\Delta_k}{12c_k}.$$

Lemma 5.6, together with $\delta_k \leq \Delta_k$, gives

$$N(u_k) \leq (1 - \frac{4}{3c_k})\Delta_k - \gamma \frac{C^2 d_k^2}{\Delta_k},$$

so the definition with coefficient $59/48$ gives $r_k \geq N(u_k)$.

Define $a_0 := 1.1\eta_0$. Young's inequality gives

$$a_0 \bar{d}_k \leq B \frac{C^2 \bar{d}_k^2}{\Delta_k} + \frac{a_0^2}{4BC^2} \Delta_k.$$

Since $C^2 = 128\bar{c}_s c_k \geq 128c_k$, choosing η_0 so that

$$a_0 + \frac{a_0^2}{512B} \leq \frac{1}{6}$$

yields

$$r_k + 1.1s_k \leq (1 - \frac{17}{16c_k})\Delta_k.$$

The explicit bound $\bar{c}_\gamma = 1 + 1/(128c_k)$ gives

$$(1 - \frac{17}{16c_k})\bar{c}_\gamma \leq 1 - \frac{1}{c_k}.$$

Therefore the stronger, directly testable inequality

$$r_k + 1.1s_k \leq (1 - \frac{1}{c_k}) \frac{\Delta_k}{\bar{c}_\gamma} \leq (1 - \frac{1}{c_k})\delta_k$$

holds, because Eq. (2) gives $\Delta_k \leq c_\gamma \delta_k \leq \bar{c}_\gamma \delta_k$. Thus the body is nonempty and the literal failure test in Algorithm 5 is not triggered. The certificate and $d_k^2 \leq 5\delta_k^2/C^2 \leq 5\Delta_k^2/C^2$ also imply $\bar{d}_k = O(\Delta_k/C)$. Since $\Delta_k \leq c_\gamma R = c_R c_\gamma C E$ and $C/c_k = O(1)$,

$$s_k \leq \eta_0 c_R E (O(1) + C/c_k).$$

Decreasing $c_R \eta_0$ proves $s_k \leq E/100$.

Next choose the coordinate sample constant so that

$$\|p^\infty - u_k\|_\infty \leq \epsilon_1 (\bar{d}_k + \Delta_k/c_k) \leq s_k/64.$$

Thus $u_k \in \mathcal{C}_k$. Let $\bar{p}^w$ be the radial preprocessing of Lemma 5.8, and let π_k be its exact metric projection onto $\mathcal{C}_k$. Because the radial ball contains u_k , the radial projection and the projection variational inequality give

$$\|\pi_k - u_k\|_{w_k} \leq \|\bar{p}^w - u_k\|_{w_k} \leq \|p^w - u_k\|_{w_k}.$$

The approximation guarantee and feasibility give

$$\|\hat{u}_k - \pi_k\|_{w_k} \leq \epsilon_P s_k / C, \quad \|\hat{u}_k - u_k\|_\infty \leq s_k / 32.$$

Moreover $C^2/c_k \leq 256$ and $s_k^2 \geq \eta_0^2(\bar{d}_k^2 + \delta_k^2/c_k^2)$, so Lemma 5.7 gives

$$\mathbb{E}[C^2 \|p^w - u_k\|_{w_k}^2 \mid \mathcal{G}_k] \leq 256 \epsilon_1^2 s_k^2 / \eta_0^2.$$

Define

$$\alpha_P := 16\epsilon_1/\eta_0 + \epsilon_P$$

and choose the two absolute accuracy constants so that $\alpha_P \leq 1/16$. Minkowski's inequality and the preceding displays give

$$C(\mathbb{E}[\|e_k\|_{w_k}^2 \mid \mathcal{G}_k])^{1/2} \leq \alpha_P s_k.$$

Conditional Jensen and the coordinate bound imply

$$N(b_k) \leq s_k/32 + \alpha_P s_k < s_k/8, \quad \|Z_k\|_\infty \leq s_k/16.$$

Finally conditional centering can only decrease the second moment, so

$$\mathbb{E}[N(Z_k)^2 \mid \mathcal{G}_k] \leq 2s_k^2/16^2 + 2\alpha_P^2 s_k^2 \leq s_k^2/64 < s_k^2/25.$$

This proves the claims with $\nu = 1/5$. □

6 Initialization, path composition, and defect certification

Section 6.1 constructs the predicted warm start and the fresh fixed-point initialization needed at restarts. Section 6.2 composes centering and path-parameter updates and converts the terminal barrier state into an LP solution. Section 6.3 develops the adaptive residual game and the randomized certificates used to measure the retained Jacobian defect.

6.1 Warm starts and fixed-point initialization

The first leverage-score query after a large-radius step has drift $O(R)$, so recomputing it to accuracy E would cost $(R/E)^2$. The next lemma avoids that loss by advancing the auxiliary weight with the projected predictor.

Lemma 6.1 (Large-radius warm start and query drift). *Suppose the maintained auxiliary weight is $E/2$ -accurate before a centering step and advance its logarithm by the projected increment $\hat{u}_k$ of Lemma 5.9. Then the advanced vector is within E of the new regularized Lewis vector. An absolute number of rounds from Lemma 4.4 restores $E/2$ accuracy. The first query transition has logarithmic row drift $O(R)$, and every subsequent query in that refinement has drift $O(E)$.*

Proof. The target moves by u_k , while $\|u_k - \hat{u}_k\|_\infty \leq s_k/32 \leq E/3200$. Thus the advanced vector has error at most $E/2 + E/3200 < E$. The final part of Lemma 4.4 restores $E/2$ accuracy in an absolute number of rounds. Across the centering step the barrier scaling changes by $e^{\pm O(R)}$, and $\|\hat{u}_k\|_\infty = O(R)$, proving the first drift bound. During refinement both the current and next auxiliary vectors are $O(E)$ from the same target, while the barrier point is fixed, proving the second. □

Fix absolute constants $A_q \geq 1$ and $c_{\text{anc}} > 0$, with A_q large enough to dominate the companion-error constants in Lemmas 6.1 and 7.1, and then c_{anc} sufficiently small for every score consumer below. The fixed-point routine uses either fresh score sketches or exact score diagonals. Its final argument is a mode flag `auto` or `exact`; an omitted flag means `auto`. Exact mode forces every

score query onto the deterministic exact branch, even when the Gaussian row count would be smaller. Its input also includes a certified initial logarithmic error, which makes the cold-start and predicted warm-start costs explicit. The outer solver uses the hybrid score routine of Algorithm 10 throughout. Its fresh-state handoff stores the companion of the actual final query, as made explicit below.

Algorithm 3 Fixed-point computation with hybrid scores

```
1: procedure FIXEDPOINT( $x, z^{(0)}, D, \epsilon_{\text{fp}}, \zeta, L_{\text{conf}}, \mathcal{S}, \text{mode}$ ) ▷ Lemma 6.2
2:    $D_x \leftarrow \Phi''(x)^{-1/2}$ .
3:    $\varrho \leftarrow 1 - p/2$ ,  $e_0 \leftarrow D$ , and  $j \leftarrow 0$ .
4:   while  $e_j > \epsilon_{\text{fp}}$  or  $j = 0$  do
5:     if  $e_j \geq 1$  then
6:        $\delta_j^{\text{map}} \leftarrow (1 - \varrho)/8$ .
7:     else
8:        $\delta_j^{\text{map}} \leftarrow (1 - \varrho)e_j/4$ .
9:     end if
10:     $e_{j+1} \leftarrow \varrho e_j + \delta_j^{\text{map}}$  and  $j \leftarrow j + 1$ .
11:  end while
12:   $J \leftarrow j$ .
13:  for  $j = 0, \dots, J - 1$  do
14:     $\theta_j \leftarrow \min\{1/8, \delta_j^{\text{map}}/(2p)\}$ .
15:    if  $j = J - 1$  then
16:       $\theta_j \leftarrow \min\{\theta_j, c_{\text{anc}}E/16\}$ .
17:    end if
18:     $k_j \leftarrow \lceil C_{\text{fp}}\theta_j^{-2} \log(2mJ/\zeta) \rceil$ .
19:  end for
20:  if  $\mathcal{S} = \emptyset$  then
21:     $S_{\text{sketch}} \leftarrow \sum_{j=0}^{J-1} k_j$ .
22:    If  $\text{mode} = \text{exact}$ , use exact score diagonals; otherwise use Gaussian mode if  $S_{\text{sketch}} \leq nJ$ 
    and exact score diagonals if not.
23:  end if
24:  for  $j = 0, \dots, J - 1$  do
25:     $D_j \leftarrow \text{Diag}(z^{(j)})^{\vartheta/2} D_x$  and  $M_j \leftarrow D_j A$ .
26:    if  $\mathcal{S} = \emptyset$  then
27:      if the global mode is Gaussian then
28:        Draw a fresh  $G_j \in \mathbb{R}^{k_j \times m}$  with independent  $N(0, 1/k_j)$  entries after  $D_j$  is fixed.
29:         $\tilde{\sigma}_i^{(j)} \leftarrow \|G_j P(D_j) e_i\|_2^2$  for every  $i \in [m]$ .
30:      else
31:         $\tilde{\sigma}^{(j)} \leftarrow \sigma(D_j)$  exactly.
32:      end if
33:    else
34:       $(\tilde{\sigma}^{(j)}, \mathcal{S}) \leftarrow \text{HYBRIDSCORE}(D_j, z^{(j)}, L_{\text{conf}}, \mathcal{S})$ . ▷ Algorithm 10
35:    end if
36:     $\tilde{\tau}^{(j)} \leftarrow \tilde{\sigma}^{(j)} + v$ .
37:    for  $i \in [m]$  do
38:       $z_i^{(j+1)} \leftarrow (z_i^{(j)})^{1-p/2} (\tilde{\tau}_i^{(j)})^{p/2}$ .
39:    end for
40:  end for
41:  if  $\mathcal{S} = \emptyset$  then
42:     $\mathcal{S} \leftarrow (D_{J-1}, z^{(J-1)}, \tilde{\sigma}^{(J-1)}, 0)$ . ▷ Lemma 8.11
43:  end if
44:  return  $(z^{(J)}, \mathcal{S})$ .
45: end procedure
```

The warm-start argument handles continuation between nearby queries, but a new epoch still needs an absolute fixed-point construction. The next lemma proves both modes of the preceding routine and its initialization cost.

Lemma 6.2 (Fixed-point computation and initialization). *Suppose $\|\log z^{(0)} - \log g(x)\|_\infty \leq D$ and $0 < \epsilon_{\text{fp}} \leq \min\{D, 1/4\}$. In fresh mode assume additionally $\epsilon_{\text{fp}} \leq E/4$. If $\mathcal{S} = \emptyset$, then with probability at least $1 - \zeta$, Algorithm 3 returns $(z^{(J)}, \mathcal{S})$ with $\|\log z^{(J)} - \log g(x)\|_\infty \leq \epsilon_{\text{fp}}$. The returned hybrid score state is $\mathcal{S} = (D_{J-1}, z^{(J-1)}, \tilde{\sigma}^{(J-1)}, 0)$ and satisfies the raw-relative guarantee of Lemma 8.11; in particular, the companion stored with D_{J-1} is $z^{(J-1)}$, not $z^{(J)}$. Here $J = O(\log(2 + D/\epsilon_{\text{fp}}))$, the permitted-system cost is*

$$O(\min\{(J + \epsilon_{\text{fp}}^{-2}) \log(m(1 + J)/\zeta), nJ\}),$$

and the linear-system depth is $O(J)$. In hybrid maintained mode, if $D \leq E$, $\epsilon_{\text{fp}} = E/2$, and every queried score obeys

$$|\tilde{\sigma}_i^{(j)} - \sigma_i(D_j)| \leq c_{\text{anc}} E \sigma_i(D_j),$$

then $J = O(1)$, the same output guarantee holds, and the query sequence contributes $O(E)$ effective drift and $O(1)$ transitions to the score process. In exact mode the fresh-mode conclusions hold deterministically, the exact branch costs $O(nJ)$ permitted systems, and the score vector stored in the returned state is exact.

Proof. Let e_j be a deterministic upper bound on the error before round j , with $e_0 = D$, and define $\varrho = 1 - p/2 < 1$. The pseudocode makes the schedule literal. In the coarse regime $e_j \geq 1$, it requests map error $(1 - \varrho)/8$. In the fine regime it requests map error

$$\delta_j^{\text{map}} = \frac{1 - \varrho}{4} e_j.$$

Lemma 4.4 then gives

$$e_{j+1} \leq \varrho e_j + \delta_j^{\text{map}} \leq (\varrho + \frac{1 - \varrho}{4}) e_j$$

in the fine regime, while the coarse regime also contracts after changing an absolute constant. The first index J with $e_J \leq \epsilon_{\text{fp}}$ is therefore deterministic and satisfies $J = O(\log(2 + D/\epsilon_{\text{fp}}))$. The last pre-update bound is $e_{J-1} \leq \epsilon_{\text{fp}}/(\varrho + (1 - \varrho)/4) = O(\epsilon_{\text{fp}})$ in the fine regime. Thus, in fresh mode, $e_{J-1} \leq A_q E$ for an absolute A_q .

At round j , condition on all preceding iterates and use the fresh matrix G_j . For every nonzero score,

$$\frac{\tilde{\sigma}_i^{(j)}}{\sigma_i(D_j)} \stackrel{d}{=} \frac{\chi_{k_j}^2}{k_j}.$$

For $0 < \theta_j \leq 1$, the standard chi-square bound and the definition of k_j give, after fixing one sufficiently large absolute C_{fp} ,

$$\Pr[\exists i : |\tilde{\sigma}_i^{(j)} - \sigma_i(D_j)| > \theta_j \sigma_i(D_j) \mid D_j] \leq \frac{\zeta}{J}.$$

Zero-score coordinates are returned exactly. A tower argument and a union bound over the adaptive rounds show that all these events hold with probability at least $1 - \zeta$. Since the floor is nonnegative, relative accuracy for the leverage scores gives the same or better additive accuracy for the regularized score. The algorithm's definition of θ_j gives

$$|\log(\tilde{\sigma}_i^{(j)} + v_i) - \log(\sigma_i(D_j) + v_i)| \leq 2\theta_j \leq \delta_j^{\text{map}}/p.$$

After multiplication by $p/2$, the logarithmic fixed-point-map error is at most $\delta_j^{\text{map}}/2$, and hence is certainly within the recurrence budget above. The explicit row count satisfies

$$k_j = O((\delta_j^{\text{map}})^{-2} \log(2mJ/\zeta)),$$

with the final raw-relative restriction changing only its absolute constant. The requested accuracies decrease geometrically in the fine regime, so those squared sample counts form a geometric series dominated by the final fine round. Each coarse round uses only $O(\log(2mJ/\zeta))$ rows, but there can be J such rounds. Thus the Gaussian branch uses

$$O((J + \epsilon_{\text{fp}}^{-2}) \log(2m(1 + J)/\zeta))$$

systems. Alternatively, compute the complete score vector in every round using $O(n)$ systems, for a total of $O(nJ)$. Concretely, the exact branch solves the n unit right-hand sides of $A^\top D_j^2 A$, forms $D_j A (A^\top D_j^2 A)^{-1}$ by n sparse matrix products, and takes the rowwise inner products with $D_j A$. It therefore uses $O(n)$ permitted systems and $O(n \text{nnz}(A) + mn)$ scalar operations per round, which are charged to the global scalar counter. Algorithm 3 selects the cheaper branch, and all systems inside one round are parallel. In the last round $\theta_{J-1} \leq c_{\text{anc}} E/16$. The same chi-square event used for that round therefore gives the raw-relative handoff, with no additional failure event or asymptotic cost, and the algorithm stores its actual query companion $z^{(J-1)}$.

In hybrid maintained mode, the displayed raw-relative guarantee makes every inexact-map error at most cE for an arbitrarily small absolute c . Starting from $D \leq E$, the same contraction gives error $E/2$ after an absolute number of rounds. Consecutive companions are all $O(E)$ -accurate to the same fixed point, so their row scalings have total effective drift $O(E)$. $\square$

6.2 Path composition and outer reduction

The preceding ingredients now supply one certified centering move and one controlled change of the path parameter. The next lemma composes them and shows that all centrality and weight-tracking invariants persist.

Lemma 6.3 (Centrality and path-update composition). *Suppose*

$$\delta := \delta_t(x, w) \leq R \leq \frac{1}{160c_k}, \quad \|\log w - \log g(x)\|_\infty \leq K.$$

Let x^+ be the Lee–Sidford Newton step at fixed t, w , and suppose

$$z := \|\log w^+ - \log w\|_{w+\infty} \leq (1 - \frac{1}{c_k})\delta, \quad \|\log w^+ - \log g(x^+)\|_\infty \leq K.$$

Then

$$\delta_t(x^+, w^+) \leq (1 - \frac{1}{4c_k})\delta.$$

Moreover, define

$$\beta := 1 + C\sqrt{2c_1}, \quad \alpha := \frac{R}{16c_k\beta}.$$

For every ξ with $|\xi| \leq \alpha$ and $1 + \xi > 0$,

$$\delta_{(1+\xi)t}(x^+, w^+) \leq R.$$

Proof. The parameter choice gives $K \leq 1/128$, hence $4g(x)/5 \leq w \leq 5g(x)/4$. Lemma 15 of [LS19] gives $\delta_t(x^+, w) \leq 4\delta^2$. Since $z \leq \delta \leq 1/10$, Lemma 17 of [LS19] gives

$$\delta_t(x^+, w^+) \leq (1 + 4z)(4\delta^2 + z) \leq \delta(1 + 4\delta)(1 - \frac{1}{c_k} + 4\delta).$$

Using $\delta \leq 1/(160c_k)$,

$$(1 + 4\delta)(1 - \frac{1}{c_k} + 4\delta) \leq (1 + \frac{1}{40c_k})(1 - \frac{39}{40c_k}) \leq 1 - \frac{1}{4c_k}.$$

The tracking invariant and the size bound on the ideal weight give $\|w^+\|_1 \leq e^K c_1 \leq 2c_1$. The proof of Lemma 14 of [LS19], also for $\xi < 0$, gives

$$\delta_{(1+\xi)t}(x^+, w^+) \leq (1 + |\xi|)\delta_t(x^+, w^+) + |\xi|\beta.$$

Indeed, evaluate the new residual at the old minimizing multiplier scaled by $1 + \xi$. Since $R \leq \beta$ and $|\xi| \leq \alpha$,

$$\delta_{(1+\xi)t}(x^+, w^+) \leq R - \frac{R}{4c_k} + 2\alpha\beta \leq R - \frac{R}{8c_k} \leq R.$$

□

To terminate the outer reduction, maintained centrality must imply proximity to the exact minimizer of the fixed-weight barrier objective. The next lemma provides this conversion at the final accuracy scale.

Lemma 6.4 (Fixed-weight terminal distance). *Fix a positive weight vector w and a path parameter t . Suppose $\|\log w - \log g(x)\|_\infty \leq K \leq 1/128$ and define $\eta := \delta_t(x, w)$ for the maintained mixed norm. If $\eta \leq 2^{-20}\ell^{-3}$, then the fixed-weight barrier objective has a unique minimizer $x_t(w)$ and*

$$\|\sqrt{\phi''(x_t(w))}(x - x_t(w))\|_\infty \leq 8\eta, \quad N_w(\sqrt{\phi''(x_t(w))}(x - x_t(w))) \leq 8c_\gamma\eta.$$

Moreover the corresponding target displacement satisfies

$$N_w(\log g(x) - \log g(x_t(w))) \leq 32c_\gamma\eta.$$

Proof. Starting from $x^{(0)} = x$, run exact projected Newton steps for the fixed objective with weight w , and write $\eta_j := \delta_t(x^{(j)}, w)$. The Newton and normalized-step parts of Lemma 3.2 give

$$\eta_{j+1} \leq 4\eta_j^2, \quad \|\sqrt{\phi''(x^{(j)})}(x^{(j+1)} - x^{(j)})\|_\infty \leq 2\eta_j.$$

Since $\eta \leq 2^{-20}$, induction gives $\eta_j \leq 2^{-j}\eta$ and hence $\sum_{j=0}^\infty \eta_j \leq 2\eta$.

We justify both applicability and convergence, rather than assuming an infinite Newton sequence exists. Suppose the estimates hold through step j . Coordinatewise self-concordant Hessian comparison along the preceding steps gives

$$\|\sqrt{\phi''(x)}(x^{(j)} - x)\|_\infty \leq 2e^{4\eta} \sum_{r=0}^{j-1} \eta_r \leq 4e^{4\eta}\eta < \frac{1}{2}.$$

Thus every finite iterate remains in the same compact interior Dikin neighborhood of x . The same comparison gives $\|\log s(x^{(j)}) - \log s(x)\|_\infty = O(\sum_{r=0}^{j-1} \eta_r)$, so target stability in Lemma 4.4 yields

$$\|\log g(x^{(j)}) - \log g(x)\|_\infty \leq 32\eta < K.$$

Consequently $\|\log w - \log g(x^{(j)})\|_\infty < K + 1/16 < \log(5/4)$, and the hypotheses of Lemma 3.2 hold at the next iterate. This closes the induction.

The sum of the step lengths in the fixed norm $\|\sqrt{\phi''(x)} \cdot\|_\infty$ is finite, so $(x^{(j)})$ is Cauchy. Let its limit be x_* . The strict Dikin bound above places x_* in the interior, and continuity there together with $\eta_j \rightarrow 0$ shows that the projected gradient of the fixed-weight objective vanishes at x_* . Its Hessian is the positive diagonal matrix $\text{Diag}(w \odot \phi''(x_*))$ restricted to the affine feasible space, and is positive definite on every nonzero feasible direction. Therefore the objective is strictly convex on that space, x_* is its unique minimizer, and $x_* = x_t(w)$.

Finally the total normalized displacement is below $1/2$, so self-concordant comparison between every intermediate Hessian and the final one costs less than a factor two. Summing the steps gives

$$\|\sqrt{\phi''(x_t(w))}(x - x_t(w))\|_\infty \leq 4 \sum_{j=0}^{\infty} \eta_j \leq 8\eta.$$

The normalized-step conclusion of Lemma 3.2 also gives $N_w(\sqrt{\phi''(x^{(j)})}(x^{(j+1)} - x^{(j)})) \leq c_\gamma \eta_j$. The same Hessian transport and summation, with the preceding slack in the constant, gives the second displayed bound. On the entire Newton sequence the preceding argument places the maintained weight in the $2K$ tracking band. The robust-consistency calculation in the proof of Lemma 4.9, applied pointwise on each Newton segment, bounds the target Jacobian by one in the maintained mixed norm. Integrating it, transporting the norm by the total logarithmic weight and slack drift, and summing $\sum_{j=0}^{\infty} \eta_j \leq 2\eta$ gives the last display (with the stated factor 32 leaving room for both transports). $\square$

The terminal-distance estimate lets us translate the final barrier state into an approximate solution of the original linear program. The following lemma performs this outer reduction and specifies the output guarantee.

Lemma 6.5 (Outer reduction and output). *Define*

$$C_{\text{LS}} := 24\sqrt{c_s} c_k, \quad \Lambda_{\text{out}} := C_{\text{LS}}/C = \frac{3}{\sqrt{2}}\sqrt{c_k} \geq 1.$$

After all radius constants, including c_R, A_E, A_L , have been fixed, choose

$$0 < c_{\text{sw}} \leq \min\{2^{-22}, c_R(\log 4)^2/(32A_E A_L)\} \quad (6)$$

and define

$$\begin{aligned} \eta_{\text{sw}} &= c_{\text{sw}} L_*^{-3}, & \eta_{\text{out}} &= \min\{c_{\text{sw}} L_*^{-3}, \epsilon/(32mU^2)\}, \\ \tau_{\text{sw}} &= \eta_{\text{sw}}/\Lambda_{\text{out}}, & \tau_{\text{out}} &= \eta_{\text{out}}/\Lambda_{\text{out}}, \end{aligned}$$

and, for a sufficiently large absolute A_{sw} ,

$$t_{\text{sw}} = (A_{\text{sw}} \Lambda_{\text{out}} m^{3/2} U^2 L_*^4)^{-1}, \quad t_{\text{out}} = 6m/\epsilon.$$

Whenever the two calls in Algorithm 9 satisfy their stated invariants, its output is feasible and obeys $c^\top x \leq \text{OPT} + \epsilon$.

Proof. For every vector v , the mixed norm with coefficient C_{LS} satisfies

$$\|v\|_\infty + C_{\text{LS}}\|v\|_w \leq \Lambda_{\text{out}}(\|v\|_\infty + C\|v\|_w).$$

Applying this inequality to every candidate in the minimization defining centrality gives $\delta_t^{\text{LS}}(x, w) \leq \Lambda_{\text{out}} \delta_t(x, w)$. Thus the stopping thresholds τ_{sw} and τ_{out} certify the corresponding Lee–Sidford centrality bounds η_{sw} and η_{out} .

The artificial cost $d = -w \odot \phi'(x_0)$ makes x_0 exactly central at $t = 1$. The first path-following call therefore starts with zero centrality; the zero-centrality branch of Algorithm 6 first changes t and avoids the vacuous ratio d^2/δ at $\delta = 0$. The signed path-change estimate used in Lemma 6.3 gives centrality at most $\alpha\beta < R$ after this exceptional first update.

We write a superscript for the cost vector and use the Lee–Sidford norm only in this paragraph. At the end of the first call,

$$\delta_{t_{\text{sw}}}^{d, \text{LS}}(x, w) \leq \Lambda_{\text{out}} \tau_{\text{sw}} = \eta_{\text{sw}}.$$

The global tracking invariant gives $e^{-K}n/m \leq w_i \leq 2e^K \leq 3$ at the switch. At initialization, $w_{0,i} \leq 3$, so $d = -w_0 \odot \phi'(x_0)$ and the barrier bounds in Section 1 imply $\|c - d\|_\infty \leq A_{\text{cost}}U$ for an absolute A_{cost} .

We derive the switch estimate directly. Let η_d minimize the Lee–Sidford mixed norm for cost d at the current (x, w) . Reusing η_d for cost c and applying the triangle inequality gives

$$\delta_{t_{\text{sw}}}^{c, \text{LS}}(x, w) \leq \delta_{t_{\text{sw}}}^{d, \text{LS}}(x, w) + t_{\text{sw}} \left\| \frac{c - d}{w \odot \sqrt{\phi''(x)}} \right\|_{w+\infty, \text{LS}}.$$

The lower bounds $w_i \geq e^{-K}n/m$ and $\sqrt{\phi''_i(x_i)} \geq 1/U$, together with $C_{\text{LS}} = 24\sqrt{c_s}c_k = O(L_*)$, imply

$$\left\| \frac{c - d}{w \odot \sqrt{\phi''(x)}} \right\|_{w+\infty, \text{LS}} \leq A_0 m^{3/2} L_* U^2$$

for an absolute A_0 . Hence, by the definition of t_{sw} ,

$$\delta_{t_{\text{sw}}}^{c, \text{LS}}(x, w) \leq \eta_{\text{sw}} + \frac{A_0}{A_{\text{sw}} \Lambda_{\text{out}} L_*^3} \leq 2\eta_{\text{sw}}$$

when A_{sw} is sufficiently large relative to A_0/c_{sw} . The Lee–Sidford norm dominates the maintained norm. Since $CK = 1/8$, $L_{\text{ch}} = A_L L_*$, and $L_* \geq \log 4$, the choice in Eq. (6) gives $2\eta_{\text{sw}} \leq R/2 < R$, so the second call is admissible. The tracking invariant, auxiliary-weight accuracy, and stochastic potential are unchanged at the switch because $x, w, \hat{g}$ do not change and the ideal weight depends on x , not on the cost vector. We regard the two calls as consecutive segments of one stopped residual game, so no initial-potential hypothesis is restarted at the switch.

At the end, Algorithm 6 gives $\delta_{t_{\text{out}}}^c(x, w) \leq \tau_{\text{out}} \leq \eta_{\text{out}}$. Let $x_{t_{\text{out}}}(w)$ be the exact minimizer for the final t, w . Lemma 6.4 gives

$$\left\| \sqrt{\phi''(x_{t_{\text{out}}}(w))} (x - x_{t_{\text{out}}}(w)) \right\|_\infty \leq 8\eta_{\text{out}}.$$

Consequently $\|x - x_{t_{\text{out}}}(w)\|_\infty \leq 8U\eta_{\text{out}}$. Also $\|w\|_1 \leq e^K \|g(x)\|_1 \leq 3n \leq 3m$. The weighted duality-gap lemma and $\|c\|_1 \leq mU$ now give

$$c^\top x - \text{OPT} \leq \frac{3m}{t_{\text{out}}} + 8mU^2\eta_{\text{out}} \leq \frac{3}{4}\epsilon.$$

Every Newton direction lies in $\ker(A^\top)$ and remains in the barrier domain, so feasibility is preserved. The remaining quarter of the error budget absorbs absolute-constant slack in the terminal thresholds. $\square$

6.3 Defect estimation and certification

The defect-adaptive construction chooses its residual radius from coarse data revealed before the fresh randomness of the step. The next lemma records the measurable-radius extension of the residual game needed for that ordering.

Lemma 6.6 (Adaptive stochastic residual game). *Lemma 5.1 remains valid when s_k is any $\mathcal{G}_k$ -measurable positive radius satisfying $s_k \leq E/100$, provided its three bias and fluctuation hypotheses hold conditionally on $\mathcal{G}_k$ and the movement bound is replaced by*

$$N_k(\hat{u}_k) + 1.1s_k \leq (1 - \frac{1}{c_k})\delta_k,$$

where $N_k(z) = \|z\|_\infty + C\|z\|_{w_{k-1}}$ is the predictable pre-step norm from that lemma. Assume also the same initial-potential condition $\Phi(d_0) \leq M_{\text{ch}}$.

Proof. Identify the pre-noise sigma-field of Lemma 5.1 with $\mathcal{G}_k$ and condition on it before drawing the fresh weighted sketches. In the proof of Lemma 5.1, the movement set enters only through

$$V_k = \{v : N_k(v) \leq s_k\}, \quad h_k(a) = s_k N_k^*(a),$$

and through $s_k \leq E/100$. Every one-step potential inequality is therefore unchanged. The coarse event is $\mathcal{G}_k$ -measurable before the independent weighted predictor and greedy move are drawn. Hence the analysis may freeze immediately before the first coarse failure using the external stop permitted by Lemma 5.1. On the event that no coarse failure occurs, this stopped process agrees with the implemented process through all path steps. The calls receive summable conditional failure budgets, so a union bound places the external stopping time after all path steps with the claimed constant probability. If the projection body is empty, the algorithm returns FAIL. Every other outcome outside the coarse event is counted as failure, and no claim is made about subsequent iterates. Doob's excursion argument is also unchanged because it already permits a different symmetric convex V_k at each step. Finally the deterministic movement paragraph uses only the displayed replacement inequality. These observations prove the claim. $\square$

The predicted warm start also survives. Indeed, $\|u_k - \hat{u}_k\|_\infty \leq s_k/32 \leq E/3200$, so advancing the old $E/2$ -accurate auxiliary vector by $\hat{u}_k$ leaves it strictly within E of the new target. Lemma 4.4 restores accuracy $E/2$ in an absolute number of rounds.

In this subsection all matrices are the certified surrogate of Lemma 7.1; superscripts are omitted. Recall

$$T = 2(I + aL)^{-1}L, \quad D = 2(I + aS)^{-1}S, \quad K := T - D.$$

Define $A_0 := (I + aS)^{-1}$ and, for an input x , define

$$y := (I + aL)^{-1}x, \quad z_* := A_0^{-1/2}y, \quad H := aA_0^{1/2}QA_0^{1/2}.$$

The resolvent identity gives

$$z_* = A_0^{1/2}x + Hz_*, \quad Kx = -\frac{2}{a}A_0^{1/2}Hz_*.$$

Since $Q \preceq S$,

$$0 \preceq H \preceq \rho I, \quad \rho := \frac{a}{1+a} < 1.$$

For the present p , ρ is bounded away from one by an absolute constant.

To exploit the retained defect, we need randomized estimators whose variance shrinks with that defect rather than with the full Newton motion. The next lemma constructs separate estimators for the weighted and coordinatewise norms.

Lemma 6.7 (Defect-scaled stochastic fixed point). *Let $x = W^{1/2}h$ and $d^2 := p^2\|x\|_2^2 - \|Tx\|_2^2$. There are two estimators $\widehat{Kx}_w$ and $\widehat{Kx}_\infty$ with the following properties. The first uses $O(c_k)$ systems in $O(\log c_k)$ sequential rounds and obeys*

$$\mathbb{E}[\|\widehat{Kx}_w - Kx\|_2^2] \leq \frac{a_4 d^2}{c_k}.$$

For every $\zeta \in (0, 1)$, the second uses $O(\log(m/\zeta))$ systems in an absolute number of sequential rounds and, with probability at least $1 - \zeta$, obeys

$$\max_{i \in [m]} \frac{|(\widehat{Kx}_\infty - Kx)_i|}{\sqrt{W_{ii}}} \leq \epsilon_\infty d,$$

where the absolute constant $\epsilon_\infty > 0$ may be made arbitrarily small by increasing the absolute sample constant and the absolute truncation depth $T = T(\epsilon_\infty)$.

Proof. The factorization in the proof of Lemma 4.6-(b) gives

$$d^2 = p^2 y^\top (I - L)(I + \frac{4-p}{p}L)y.$$

Since $Q \preceq I - L$,

$$\mathcal{D}^2 := z_*^\top H z_* = a y^\top Q y \leq \frac{a}{p^2} d^2.$$

To multiply a vector z by H , apply Lemma 7.2 to $A_0^{1/2}z$ and multiply the result by $aA_0^{1/2}$. With s independent samples this gives an unbiased estimate $\widehat{H}_s(z)$ satisfying

$$\mathbb{E}[\|\widehat{H}_s(z) - Hz\|_2^2 \mid z] \leq \frac{2a}{s} z^\top H z.$$

Run

$$\widehat{z}_{j+1} = A_0^{1/2}x + \widehat{H}_{s_j}(\widehat{z}_j), \quad \widehat{z}_0 = 0,$$

with fresh sketches. If $\bar{z}_j$ is the deterministic iterate and $\epsilon_j := \mathbb{E}[\|\widehat{z}_j - \bar{z}_j\|_2^2]$, conditional centering and $H \preceq \rho I$ give

$$\epsilon_{j+1} \leq (\rho^2 + \frac{4a\rho}{s_j})\epsilon_j + \frac{4a\mathcal{D}^2}{s_j}.$$

Choose an absolute s_0 so that the coefficient is at most a fixed $\lambda < 1$ whenever $s_j \geq s_0$. For a target η , take

$$s_j := \max\{s_0, \lceil c\eta^{-2}\lambda^{(T-1-j)/2} \rceil\}.$$

Then

$$\sum_{j=0}^{T-1} s_j = O(\eta^{-2} + T), \quad \sum_{j=0}^{T-1} \frac{\lambda^{T-1-j}}{s_j} = O(\eta^2).$$

The deterministic tail is bounded by $\|H^{T+1}z_*\|_2^2 \leq \rho^{2T+1}\mathcal{D}^2$. Taking $T = O(\log(1/\eta))$, followed by one fresh multiplication, proves

$$\mathbb{E}[\|\widehat{Kx} - Kx\|_2^2] = O(\eta^2 d^2).$$

The fresh multiplication uses $\Theta(\eta^{-2})$ samples; its conditional variance is $O(\eta^2 \mathcal{D}^2)$, while the propagated error of $\widehat{z}_T$ is $O(\eta^2 \mathcal{D}^2)$. Thus it does not change either the accuracy or the total $O(\eta^{-2} + T)$ sample count. The choice $\eta = \Theta(c_k^{-1/2})$ proves the first assertion.

For the coordinate assertion, take $T = T(\epsilon_\infty)$ to be an absolute constant and run independent copies of the whole stochastic chain, using an absolute number of samples in each multiplication. Let Y be the output of one copy, including its final fresh multiplication. The recurrence above, with constant sample sizes, gives

$$\mathbb{E}[\|\hat{z}_j - \bar{z}_j\|_2^2] = O(\mathcal{D}^2) \quad \text{and} \quad \mathbb{E}[\hat{z}_j^\top H \hat{z}_j] = O(\mathcal{D}^2)$$

for every $j \leq T$. The coordinate variance conclusion of Lemma 7.2, applied conditionally at the last multiplication and then averaged, therefore gives

$$Y - Kx = \underbrace{Y - \mathbb{E}[Y \mid \hat{z}_T]}_{\text{last multiplication}} + \underbrace{\mathbb{E}[Y \mid \hat{z}_T] - \mathbb{E}[Y]}_{\text{propagated chain noise}} + \underbrace{\mathbb{E}[Y] - Kx}_{\text{truncation bias}}.$$

For the first term, the conditional coordinate-variance conclusion of Lemma 7.2 and $\mathbb{E}[\hat{z}_T^\top H \hat{z}_T] = O(\mathcal{D}^2)$ give

$$\mathbb{E}[\text{Var}[Y_i / \sqrt{W_{ii}} \mid \hat{z}_T]] \leq c_5 \mathcal{D}^2 \leq c'_5 a^{-1} d^2.$$

For the second term, define $e := \hat{z}_T - \mathbb{E}[\hat{z}_T]$. Positive-semidefinite Cauchy–Schwarz gives

$$\frac{|(QA_0^{1/2}e)_i|^2}{W_{ii}} \leq e^\top A_0^{1/2} Q A_0^{1/2} e.$$

Its expectation is $O(\mathcal{D}^2) = O(d^2)$ by the Euclidean second-moment recurrence. The law of total variance therefore proves the fully normalized bound

$$\text{Var}\left[\frac{Y_i}{\sqrt{W_{ii}}}\right] \leq a_5 d^2 \quad \text{for every } i,$$

where deterministic truncation bias is absent from the variance and is handled next. Here the deterministic iterates cause no loss: since $\bar{z}_j = (I - H^j)z_*$, one has $\bar{z}_j^\top H \bar{z}_j \leq \mathcal{D}^2$.

It remains to bound the truncation bias in the same coordinate scale. For every $r \geq 0$, Cauchy–Schwarz for the positive semidefinite matrix Q and $Q_{ii} \leq W_{ii}$ give

$$\frac{|(QA_0^{1/2}H^r z_*)_i|}{\sqrt{W_{ii}}} \leq \sqrt{z_*^\top H^{2r+1} z_* / a} \leq \rho^r \sqrt{\mathcal{D}^2 / a}.$$

Thus the bias in $Y_i / \sqrt{W_{ii}}$ is at most $a_6 \rho^T d \leq \epsilon_\infty d / 4$ after increasing the absolute constant T . Form independent block means, each averaging $O(\epsilon_\infty^{-2})$ copies of Y . Chebyshev’s inequality makes one block mean accurate to $\epsilon_\infty d / 2$ with a fixed probability above $1/2$. The coordinatewise median of $O(\log(m/\zeta))$ block means, followed by a union bound, proves the simultaneous coordinate estimate. All copies are parallel and T is absolute, proving both the work and depth claims. $\square$

The mean-square estimators above must be amplified to a finite, simultaneous high-probability guarantee without using a nonconstructive selector. The next lemma gives the required medoid aggregation rule.

Lemma 6.8 (Finite medoid aggregation). *Let $Y^{(1)}, \dots, Y^{(B)} \in \mathbb{R}^m$ be independent and suppose*

$$\Pr[\|Y^{(j)} - y\|_2 \leq r] \geq 3/4 \quad \text{for every } j.$$

Take an odd $B \geq c_{\text{med}} \log(3/\zeta)$. For each j , let R_j be the distance from $Y^{(j)}$ to its $(B+1)/2$ -th closest candidate, and return $Y^{(j_)}$ for an index minimizing R_j . Except with probability $\zeta/3$, the returned candidate is within $3r$ of y . The rule uses $O(mB^2)$ exact real-RAM scalar operations and $O(\log m + \log B)$ arithmetic depth.*

Proof. A Chernoff bound and the choice of c_{med} show that, except with probability $\zeta/3$, at least $2B/3$ candidates lie in the radius- r ball about y . On this event, every good candidate has more than $B/2$ candidates within distance $2r$, so the selected median radius is at most $2r$. The selected ball contains more candidates than there are bad ones and therefore contains a good candidate. The triangle inequality gives distance at most $2r + r = 3r$ from the selected candidate to y . All pairwise squared distances are parallel reductions, after which each row of distances is selected and the minimum median radius is chosen. $\square$

The predictor also needs a one-sided numerical certificate for the retained defect. The next lemma constructs such a certificate using the same coarse chain and only logarithmically many additional systems.

Lemma 6.9 (Defect certificate). *For every $\zeta \in (0, 1)$ and every algorithmically supplied $\Delta \geq \delta$, the coarse chain in Lemma 6.7, followed by $O(\log(1/\zeta))$ additional systems, outputs a scalar $\bar{d}^2$ such that, with probability at least $1 - \zeta$,*

$$d_{\text{exact}}^2 \leq \bar{d}^2 \leq A_d(d_{\text{exact}}^2 + \frac{\Delta^2}{C^2 c_k}),$$

where d_{exact} is the defect in the active large-radius Lemma 5.6 and A_d is absolute. Throughout the algorithms and all later lemmas, $\bar{d}$ denotes the nonnegative square root $\sqrt{\bar{d}^2}$.

Proof. For the surrogate, let $y = (I + aL)^{-1}x$ and define $e_0 := y^\top(I - L)y$. The factorization above and $0 \preceq L \preceq I$ give

$$p^2 e_0 \leq d^2 \leq 4p e_0.$$

Also

$$e_0 = y^\top(I - S)y + y^\top Qy.$$

The first term is diagonal. For the second, draw a fresh Gaussian r , independently of all sketches used to construct the approximation to y , define $u = Pr$ and $v = P \text{Diag}(W^{-1/2}y)u$. In the notation of the proof of Lemma 7.2, $v = VZg$, and hence

$$\mathbb{E}[\|v\|_2^2] = \text{tr}[Z^2] = y^\top Qy, \quad \text{Var}[\|v\|_2^2] = 2 \text{tr}[Z^4] \leq 2(y^\top Qy)^2.$$

Using failure budget $\zeta/3$, a median of $B_Q = 2\lceil c_Q \log(6/\zeta) \rceil + 1$ constant-size block means gives a constant-factor estimate with $O(\log(1/\zeta))$ samples.

This approximation does not require prior knowledge of e_0 . Run the fixed-point chain for an absolute number of rounds with a sufficiently large absolute sample constant. The recurrence in Lemma 6.7 and $\mathcal{D}^2 \leq a e_0$ show that one run produces $\tilde{y}$ with

$$\Pr[\|\tilde{y} - y\|_2 \leq \epsilon_0 \sqrt{e_0}] \geq 3/4.$$

Take the odd integer $B = 2\lceil c_{\text{med}} \log(3/\zeta) \rceil + 1$ and apply the finite medoid rule of Lemma 6.8 to B independent runs. Except with probability $\zeta/3$, the selected candidate satisfies the same bound with $3\epsilon_0$ in place of ϵ_0 . This finite construction does not use e_0 and costs $O(m \log^2(1/\zeta))$ scalar work.

Both $I - S$ and Q are bounded above by $I - L$. Consequently replacing y by $\tilde{y}$ changes each quadratic form by at most $O(\epsilon_0)e_0$: use Cauchy–Schwarz in the corresponding positive semidefinite seminorm and $\|\tilde{y} - y\|_2^2 = O(\epsilon_0^2 e_0)$. Compute the diagonal form exactly and estimate the Q -form by the preceding median of block means, using samples independent of $\tilde{y}$. Denote the resulting

nonnegative scalar by $\tilde{e}_0$. The seminorm comparison in the previous paragraph shows that, for a sufficiently small fixed ϵ_0 and the stated block sizes, the joint medoid and scalar median event implies

$$\frac{3}{4}e_0 \leq \tilde{e}_0 \leq \frac{5}{4}e_0.$$

Define $\bar{e}_0 := (4/3)\tilde{e}_0$. This is the required one-sided inflation and it gives, without an unspecified confidence factor,

$$e_0 \leq \bar{e}_0 \leq \frac{5}{3}e_0 \leq 2e_0.$$

Define $\bar{d}_+^2 := 4p\bar{e}_0$. The inequalities in the first display of this proof then give explicitly

$$d_+^2 \leq \bar{d}_+^2 \leq \frac{8}{p}d_+^2 =: a_7 d_+^2.$$

Finally output $\bar{d}^2 := 2\bar{d}_+^2 + 2F$, where $F := a_3\Delta^2/(C^2c_k)$. Lemma 7.1 gives $|d_+^2 - d_{\text{exact}}^2| \leq F$. Therefore the required one-sided direction is explicit:

$$d_{\text{exact}}^2 \leq d_+^2 + F \leq \bar{d}_+^2 + F \leq \bar{d}^2.$$

Conversely,

$$\bar{d}^2 \leq 2a_7 d_+^2 + 2F \leq 2a_7 d_{\text{exact}}^2 + (2a_7 + 2)F \leq A_d(d_{\text{exact}}^2 + \frac{\Delta^2}{C^2c_k})$$

with $A_d := (2a_7 + 2)\max\{1, a_3\}$. This proves both certificate inequalities with their correct directions. The medoid and scalar block-median events each use budget $\zeta/3$; their union is below ζ , and every batch size is fixed before its samples are drawn. $\square$

Algorithm 4 Coarse defect certificate and split predictor

- 1: **procedure** COARSEDEFECTDATA($W^+, S^+, P^+, \psi, \Delta, \zeta$) $\triangleright$ Lemmas 6.7 and 6.9
 - 2: Fix the absolute constants $c_\infty, c_{\text{med}}, c_Q, c_w, c_{\text{fp}}, c_T, s_0$ and $\lambda < 1$ as in those lemmas.
 - 3: $\zeta_\infty \leftarrow \zeta/2$, $\zeta_{\text{cert}} \leftarrow \zeta/2$, $D^+ \leftarrow 2(I + aS^+)^{-1}S^+$, and $x \leftarrow (W^+)^{1/2}\psi$.
 - 4: $B_\infty \leftarrow 2\lceil c_\infty \log(4m/\zeta_\infty) \rceil + 1$ and $s_\infty \leftarrow \lceil c_\infty \epsilon_1^{-2} \rceil$.
 - 5: Form B_∞ independent block means, each from s_∞ independent constant-depth stochastic fixed-point chains, and set $\widehat{K}x_\infty$ to their coordinatewise median.
 - 6: $B_{\text{med}} \leftarrow 2\lceil c_{\text{med}} \log(6/\zeta_{\text{cert}}) \rceil + 1$.
 - 7: Form B_{med} independent constant-depth candidates for $y = (I + aL^+)^{-1}x$ and aggregate them by the finite medoid rule of Lemma 6.8.
 - 8: $B_Q \leftarrow 2\lceil c_Q \log(6/\zeta_{\text{cert}}) \rceil + 1$; estimate the Q^+ -form from B_Q independent constant-size polarization block means, take their scalar median, add the exact diagonal form, and apply the one-sided inflation of Lemma 6.9 to obtain $\bar{d}^2$.
 - 9: $p^\infty \leftarrow (W^+)^{-1/2}(D^+x + \widehat{K}x_\infty)$ and $\bar{d} \leftarrow \sqrt{\bar{d}^2}$.
 - 10: $\mathcal{G} \leftarrow \sigma(\text{past and all random variables drawn above})$.
 - 11: After $\mathcal{G}$ is fixed, set $\eta_w \leftarrow c_w/\sqrt{c_k}$ and $T_w \leftarrow \lceil c_T \log(2/\eta_w) \rceil$.
 - 12: Draw an independent weighted stochastic fixed-point chain as in Lemma 6.7.
 - 13: At round j , use $s_j \leftarrow \max\{s_0, \lceil c_{\text{fp}}\eta_w^{-2}\lambda^{(T_w-1-j)/2} \rceil\}$ for $j = 0, \dots, T_w - 1$.
 - 14: $p^w \leftarrow (W^+)^{-1/2}(D^+x + \widehat{K}x_w)$.
 - 15: **return** $(\bar{d}^2, p^\infty, p^w)$.
 - 16: **end procedure**
-

7 Certified prediction and path-following assembly

Section 7.1 builds the certified surrogate and the defect-adaptive predictor from observable quantities. Section 7.2 supplies the fresh Gaussian increment primitive for adaptive score queries. Section 7.3 fixes the information ordering within one centering step and verifies its conditional estimates. Finally, Section 7.4 assembles the centering calls into checkpointed path phases with explicit resource caps.

7.1 Certified predictor construction

The estimator must use algorithmically available quantities. The next lemma records a positive-semidefinite surrogate that preserves the defect up to the accuracy already present in the weight oracle.

Lemma 7.1 (Certified surrogate). *Assume the hypotheses of Lemma 5.6. In particular, $\delta := \delta_t(x, w) \leq R$, the tracking band holds along its Newton segment, and*

$$\psi := \log s(x^+) - \log s(x), \quad d^2 := p^2 \|\psi\|_g^2 - \|B\psi\|_g^2, \quad N(\psi) \leq 2\delta.$$

Suppose $\hat{g}$ satisfies $\|\log \hat{g} - \log g(x)\|_\infty \leq E/2$. Construct the diagonal W^+ and the positive query scaling D_{qry}^+ deterministically as in the proof, and define $P^+ := P(D_{\text{qry}}^+)$ as an implicit projection operator. Suppose a score provider, queried only after these objects are fixed, returns $\tilde{\sigma}$ satisfying

$$|\tilde{\sigma}_i - \sigma_i(P^+)| \leq c_0 E W_{ii}^+ \quad \text{for every } i \in [m].$$

A fresh Gaussian sketch is one admissible provider: conditionally on the fixed query it succeeds with probability at least $1 - \zeta$ using $O(E^{-2} \log(m/\zeta))$ permitted systems in one parallel round. From this provider output one can define diagonal S^+ and symmetric operators Q^+, L^+ such that, on the simultaneous event,

$$0 \preceq Q^+ \preceq S^+ \preceq I, \quad L^+ := S^+ - Q^+, \quad 0 \preceq L^+ \preceq I.$$

Define

$$T^+ := 2(I + aL^+)^{-1}L^+, \quad x^+ := (W^+)^{1/2}\psi, \quad d_+^2 := p^2 \|x^+\|_2^2 - \|T^+x^+\|_2^2.$$

Then

$$|d_+^2 - d^2| \leq a_2 E \frac{\delta^2}{C^2} \leq a_3 \frac{\delta^2}{C^2 c_k}$$

for absolute constants a_2, a_3 . The associated diagonal part $D^+ := 2(I + aS^+)^{-1}S^+$ and the operator $(W^+)^{-1/2}T^+(W^+)^{1/2}$ differ from their exact counterparts by $O(E)$ in both the ℓ_∞ operator norm and the Euclidean operator norm after conjugation to the exact $g(x)$ -normalized coordinates. The latter is equivalently the operator norm induced by $\|\cdot\|_{g(x)}$ on physical coordinates.

Proof. Let $G := \text{Diag}(g)$ and let P be the exact projection. Choose a sufficiently large absolute c_+ and define $W^+ := e^{c_+E} \text{Diag}(\hat{g})$. A global scalar does not change a projection, so define $D_{\text{qry}}^+ := (W^+)^{1/2} \text{Diag}(s(x))$ and let $P^+ := P(D_{\text{qry}}^+)$. The relative row scaling between the matrices defining P and P^+ is $e^{\pm O(E)}$. Hence Lemma 4.1, the fixed-point identity $\sigma_i(P) \leq g_i$, and the choice of c_+ give $\sigma_i(P^+) \leq W_{ii}^+$.

If the fresh provider is selected, condition on the fixed W^+ and P^+ . Standard simultaneous Gaussian leverage-score concentration gives relative error $O(E)$ using the stated number of systems.

Since $\sigma_i(P^+) \leq W_{ii}^+$, this gives the required additive guarantee. For any provider satisfying the displayed additive score interface, define

$$\sigma_i^+ := \min\{W_{ii}^+, \tilde{\sigma}_i + c_0 E W_{ii}^+\}.$$

Then

$$0 \leq \sigma_i^+ - \sigma_i(P^+) \leq 2c_0 E W_{ii}^+.$$

With

$$S^+ := (W^+)^{-1/2} \text{Diag}(\sigma^+)(W^+)^{-1/2}, \quad Q^+ := (W^+)^{-1/2}(P^+ \circ P^+)(W^+)^{-1/2},$$

the standard identity $\text{Diag}(\sigma(P^+)) - P^+ \circ P^+ \succeq 0$ proves the displayed semidefinite relations.

The implementation stores only the diagonal S^+ and an operator handle for P^+ . The matrices Q^+, L^+, T^+ are analysis notation and are never materialized: every required sample from Q^+ uses the two- P^+ polarization routine, and the stochastic fixed-point routines access L^+ only through those projection applications. One application of P^+ uses one permitted system and $O(\text{nnz}(A))$ scalar work.

We record the perturbation bounds explicitly. Write $W^+ = R_g G$, where R_g is diagonal and $\|R_g - I\|_{2 \rightarrow 2} = O(E)$. Define

$$\mathcal{L} := G^{-1}(\text{Diag}(\sigma(P)) - P \circ P), \quad \mathcal{L}^+ := (W^+)^{-1}(\text{Diag}(\sigma^+) - P^+ \circ P^+),$$

and distinguish these row-normalized operators from the symmetric ones

$$L := G^{1/2} \mathcal{L} G^{-1/2}, \quad L^+ := (W^+)^{1/2} \mathcal{L}^+ (W^+)^{-1/2}.$$

The row-sum calculation in Lemma 5.3, with row drift $O(E)$, bounds the contribution of $P^+ - P$ and $W^+ - G$ by $O(E)$. The remaining score excess has row norm at most $2c_0 E$. Thus

$$\|\mathcal{L}^+ - \mathcal{L}\|_{\infty \rightarrow \infty} = O(E).$$

This row estimate also gives the normalized spectral estimate without any dependence on the smallest weight. Both $G\mathcal{L}$ and $W^+\mathcal{L}^+$ are symmetric. Therefore, exactly as in the weighted column-sum paragraph of Lemma 5.3, the weighted column sum of $\mathcal{L}^+ - \mathcal{L}$ is at most its row sum plus

$$(e^{O(E)} - 1) \|\mathcal{L}^+\|_{\infty \rightarrow \infty} = O(E).$$

Here $\|\mathcal{L}^+\|_{\infty \rightarrow \infty} \leq 2$, because $\sigma_i(P^+) \leq \sigma_i^+ \leq W_{ii}^+$. The Schur test in $\ell_2(G)$ now gives

$$\|G^{1/2}(\mathcal{L}^+ - \mathcal{L})G^{-1/2}\|_{2 \rightarrow 2} = O(E).$$

Conjugating the first term by $R_g^{1/2} = I + O(E)$ proves

$$\|L^+ - L\|_{2 \rightarrow 2} = O(E).$$

The score perturbation and $W^+ = R_g G$ give $\|S^+ - S\|_{2 \rightarrow 2} = O(E)$ directly; hence $\|Q^+ - Q\|_{2 \rightarrow 2} = O(E)$ as well.

Although $\mathcal{L}^+ \mathbf{1}$ need not vanish, this does not affect the argument. The exact inverse $(I + a\mathcal{L})^{-1}$ has infinity norm one, and the resolvent identity shows $\|(I + a\mathcal{L}^+)^{-1}\|_{\infty \rightarrow \infty} \leq 2$ once the absolute accuracy constant is small. A second use of the resolvent identity gives

$$\|(W^+)^{-1/2} T^+ (W^+)^{1/2} - G^{-1/2} T G^{1/2}\|_{\infty \rightarrow \infty} = O(E).$$

The spectral resolvent identity and the preceding 2-norm bounds give the corresponding weighted Euclidean estimate. The same statements for D^+ follow directly from its diagonal formula.

Finally, $x^+ = R_g^{1/2}x$, so $\|x^+ - x\|_2 = O(E)\|x\|_2$. Since $0 \preceq T, T^+ \preceq pI$, the preceding spectral bound yields

$$|d_+^2 - d^2| = O(E)\|x\|_2^2 = O(E\delta^2/C^2).$$

Here metric equivalence and $C\|\psi\|_g \leq N(\psi) = O(\delta)$ were used. Finally $E = \Theta(q^{-1/2}\ell^{-1}) = O(1/c_k)$ because $q \leq \ell$. $\square$

The certified surrogate still requires an observable estimate of its quadratic defect energy. The following Gaussian construction supplies an unbiased estimator with the variance bound needed by the certification step.

Lemma 7.2 (Defect-energy estimator). *Let $P = P^\top = P^2$, let $W > 0$ be diagonal, assume $P_{ii} \leq W_{ii}$ for every i , and define $Q := W^{-1/2}(P \circ P)W^{-1/2}$. For $y \in \mathbb{R}^m$, set $t := W^{-1/2}y$. Draw $r \sim N(0, I_m)$ and define*

$$u_r := Pr, \quad v_r := P \text{Diag}(t)u_r, \quad \Xi(y; r) := W^{-1/2}(u_r \odot v_r).$$

Then

$$\mathbb{E}[\Xi(y; r)] = Qy, \quad \mathbb{E}[\|\Xi(y; r) - Qy\|_2^2] \leq 2y^\top Qy,$$

and, coordinatewise,

$$\text{Var}\left[\frac{\Xi_i(y; r)}{\sqrt{W_{ii}}}\right] \leq 2y^\top Qy.$$

Each sample uses two applications of P and hence two permitted linear-system solves.

Proof. Write $P = VV^\top$, where V has orthonormal columns, and denote the i -th row of V by $v_i^\top$. Define $Z := V^\top \text{Diag}(t)V$. Since $V^\top r \sim N(0, I_n)$, Gaussian polarization gives

$$\mathbb{E}[(Pr)_i(P \text{Diag}(t)Pr)_i] = v_i^\top Z v_i = \sum_{j=1}^m P_{ij}^2 t_j.$$

This proves unbiasedness. Moreover

$$\text{Var}[\Xi_i] = \frac{\sigma_i \|Z v_i\|_2^2 + (v_i^\top Z v_i)^2}{W_{ii}}.$$

Now

$$\|Z\|_F^2 = t^\top (P \circ P)t = y^\top Qy,$$

$\|Z v_i\|_2^2 \leq \sigma_i \|Z\|_F^2$, and $\sigma_i \leq W_{ii}$. These facts prove the coordinate estimate. For the total variance, define $H_V := V^\top \text{Diag}(\sigma_i/W_{ii})V \preceq I$. Summing the first variance term gives $\text{tr}[ZH_V Z] \leq \text{tr}[Z^2] = y^\top Qy$. The sum of the second terms is $\|Qy\|_2^2 \leq y^\top Qy$, because $0 \preceq Q \preceq I$. Adding proves the claim. $\square$

7.2 Gaussian increment primitive

We replace per-query score sketches by fresh shared-sketch increments whose endpoints are fixed before their randomness is drawn. The following lemma proves the conditional martingale and variance bounds needed for an adaptive query sequence; write $L_{\text{conf}} := \log(40mM_{\text{ep}})$, where M_{ep} bounds the number of randomized calls in one path epoch.

Lemma 7.3 (Adaptive shared-sketch increment). *Let D_- , D_+ be measurable with respect to a sigma-field $\mathcal{F}$, with $\|\log(D_+D_-^{-1})\|_\infty \leq \rho \leq 1/8$, and let $a_i \geq \max\{\sigma_i(D_-), \sigma_i(D_+)\}$ be $\mathcal{F}$ -measurable. Draw, after conditioning on $\mathcal{F}$, a fresh Gaussian sketch $S \in \mathbb{R}^{s \times m}$ with independent $N(0, 1/s)$ entries and set $\hat{\Delta}_i := \|SP(D_+)e_i\|_2^2 - \|SP(D_-)e_i\|_2^2$ and $\Delta_i := \sigma_i(D_+) - \sigma_i(D_-)$. Then $\mathbb{E}[\hat{\Delta}_i - \Delta_i \mid \mathcal{F}] = 0$ and there are absolute $c_2, C_2 > 0$ with*

$$\mathbb{E}[e^{\lambda(\hat{\Delta}_i - \Delta_i)} \mid \mathcal{F}] \leq \exp(C_2 \lambda^2 \rho^2 a_i^2 / s) \quad \text{whenever } |\lambda| \leq c_2 s / (\rho a_i).$$

Each sample of S costs two permitted solves.

Proof. Define $u := P(D_+)e_i$ and $v := P(D_-)e_i$. By polarization, $\|Su\|_2^2 - \|Sv\|_2^2 = \langle S(u-v), S(u+v) \rangle$, a sum of s independent products of jointly Gaussian variables with mean $\langle u-v, u+v \rangle = \|u\|_2^2 - \|v\|_2^2 = \Delta_i$; each summand is sub-exponential with scale $O(\|u-v\|_2 \|u+v\|_2 / s)$, so the sum has the displayed conditional moment generating function by the standard Bernstein computation, where the first step uses that S is independent of $\mathcal{F}$ and the second uses Lemma 4.1: $\|u-v\|_2 \leq 10\rho\sqrt{\sigma_i(D_-)} \leq 10\rho\sqrt{a_i}$ and $\|u+v\|_2 \leq 3\sqrt{a_i}$. Applying a row of S through $P(D_\pm)$ is one solve each. $\square$

7.3 Within-step hybrid filtration

Within-step filtration. For each centering call k , let $\mathcal{F}_{k,0}$ contain the entire past, the current primal, maintained weight, auxiliary weight, and score state, before the surrogate-score response is requested. The surrogate query matrices are $\mathcal{F}_{k,0}$ -measurable and their score randomness is drawn afterward. Let $\mathcal{F}_{k,1}$ contain that response together with the subsequently drawn coordinate predictor and defect certificate, and define $\mathcal{G}_k := \mathcal{F}_{k,1}$. Conditional on $\mathcal{G}_k$, draw the independent weighted predictor and let $\mathcal{F}_{k,2}$ include its output. After its endpoint is thereby fixed, draw the fixed-point score responses and let $\mathcal{F}_{k,3}$ include the returned auxiliary weight. The greedy correction is a deterministic function of $\mathcal{F}_{k,3}$, and the resulting state generates $\mathcal{F}_{k+1,0}$.

Thus the conditional bias and second-moment statements in Lemmas 5.7 and 5.9 are conditioned exactly on $\mathcal{G}_k$, while each score sketch is conditionally fresh given the sigma-field that fixes its query endpoints. Fixed-point responses drawn after $\mathcal{F}_{k,2}$ need not be independent of the weighted predictor; only the raw-relative-or-exact guarantee of Lemma 8.12 is used. This filtration is also the one used by the adaptive residual game, with its pre-noise sigma-field identified as $\mathcal{G}_k$.

We record the two routines that assemble the preceding ingredients into a centering step and then a path-following phase. The active cost vector is an explicit argument, so the two phases of the outer reduction are algorithmically distinct.

Algorithm 5 One hybrid-score predicted centering step

```

1: procedure PREDICTEDCENTER( $x, w, \hat{g}, t, c_{\text{path}}, \zeta, L_{\text{conf}}, \mathcal{S}$ ) ▷ Lemma 7.4
2:    $x^+ \leftarrow$  the Lee–Sidford projected Newton step at fixed  $t, w$  and cost  $c_{\text{path}}$ .
3:   if  $A^\top x^+ \neq b$  or  $x^+ \notin \Omega^\circ$  then
4:     return FAIL.
5:   end if
6:    $\psi \leftarrow \log s(x^+) - \log s(x)$  and  $\Delta \leftarrow \bar{\delta}_t^{c_{\text{path}}}(x, w)$  from the projected Newton residual.
7:    $(W^+, D_{\text{qry}}^+) \leftarrow$  the deterministic query of Lemma 7.1 at the pre-Newton base point  $x$ , and
   let  $P^+$  denote its implicit projection operator.
8:    $(\tilde{\sigma}^+, \mathcal{S}) \leftarrow \text{HYBRIDSCORE}(D_{\text{qry}}^+, \text{diag}(W^+), L_{\text{conf}}, \mathcal{S})$ . ▷ Algorithm 10
9:    $S^+ \leftarrow$  the certified diagonal from  $\tilde{\sigma}^+$ ; record only implicit operator access to  $P^+$  as in
   Lemma 7.1.
10:   $(\bar{d}^2, p^\infty, p^w) \leftarrow \text{COARSEDEFECTDATA}(W^+, S^+, P^+, \psi, \Delta, \zeta)$ . ▷ Algorithm 4
11:   $\bar{d} \leftarrow \sqrt{\bar{d}^2}$ .
12:   $s \leftarrow \eta_0(\bar{d} + \Delta/c_k)$  and  $r \leftarrow (1 - 59/(48c_k))\Delta - BC^2\bar{d}^2/\Delta$ .
13:   $\mathcal{C} \leftarrow \{u : N(u) \leq r, \|u - p^\infty\|_\infty \leq s/64\}$ .
14:  if  $\mathcal{C}$  is empty or  $r + 1.1s > (1 - 1/c_k)\Delta/\bar{c}_\gamma$  then
15:    return FAIL.
16:  end if
17:   $\hat{u} \leftarrow \text{DEFECTPROJECTION}(w, C, r, s, p^\infty, p^w)$ . ▷ Algorithm 2
18:   $z \leftarrow \exp(\log \hat{g} + \hat{u})$  and  $(\hat{g}^+, \mathcal{S}) \leftarrow \text{FIXEDPOINT}(x^+, z, E, E/2, \zeta, L_{\text{conf}}, \mathcal{S}, \text{auto})$ . ▷
   Algorithm 3
19:   $d_{\text{obs}} \leftarrow \log w + \hat{u} - \log \hat{g}^+$ .
20:   $\chi \leftarrow \text{MIXEDBALLLINEARORACLE}(w, C, 1.1s, \nabla \Phi(d_{\text{obs}}), \epsilon_g)$ . ▷ Algorithm 1
21:   $w^+ \leftarrow \exp(\log w + \hat{u} + \chi)$ .
22:  return  $(x^+, w^+, \hat{g}^+, \mathcal{S})$ .
23: end procedure

```

With the surrogate, defect certificate, and fresh score increments available, we can state a complete predicted centering call. The next lemma verifies the pseudocode above and accounts for every stopping event and resource charge.

Lemma 7.4 (Hybrid-score predicted centering step). *Suppose*

$$\begin{aligned}
0 < \delta &:= \delta_t^{c_{\text{path}}}(x, w) \leq R, & \Delta &:= \bar{\delta}_t^{c_{\text{path}}}(x, w), \\
\|\log w - \log g(x)\|_\infty &\leq K, & \|\log \hat{g} - \log g(x)\|_\infty &\leq E/2.
\end{aligned}$$

and let $\zeta \in (0, 1/10)$. Suppose the raw-relative initialization event holds, and use the coupled process of Lemma 8.12 up to the existing auxiliary stop. Lemma 8.13 supplies both score consumers without conditioning on future score success: a response before the first violation is raw-relative, and every replacement response is exact. Given that past, the simultaneous coarse good event has conditional failure probability at most ζ . On that event, Algorithm 5 does not return FAIL and returns $(x^+, w^+, \hat{g}^+, \mathcal{S})$ such that

$$\|\log \hat{g}^+ - \log g(x^+)\|_\infty \leq E/2, \quad \|\log w^+ - \log w\|_{w+\infty} \leq (1 - \frac{1}{c_k})\delta_t(x, w).$$

Conditionally on the coarse sigma-field, its residual error satisfies the three bias and fluctuation hypotheses of Lemma 6.6. It contributes $O(E)$ hybrid effective drift and $O(1)$ transitions to the score process. Outside that process, one call uses $O(L_* + q)$ permitted systems and $O(\log q)$ linear-system depth.

Proof. Lemma 7.1 makes the query matrices measurable before their fresh sketches are drawn. Lemmas 5.7 and 5.9, including the computable projection of Lemma 5.8, imply that $\mathcal{C}$ is nonempty and prove the literal proxy inequality tested by the algorithm,

$$r + 1.1s \leq (1 - \frac{1}{c_k})\Delta/\bar{c}_\gamma \leq (1 - \frac{1}{c_k})\delta.$$

They also give the conditional bias and fluctuation bounds required by Lemma 6.6. To connect the actual observation to that lemma, write

$$d_{k-1} := \log w - \log g(x), \quad e_k := u - \hat{u} = b_k + Z_k, \quad q_k := d_{k-1} - Z_k.$$

The line defining d_{obs} in the algorithm then gives the exact identity

$$d_{\text{obs}} - q_k = \log g(x^+) - \log \hat{g}^+ - b_k.$$

Since $\|\log \hat{g}^+ - \log g(x^+)\|_\infty \leq E/2$ and $N(b_k) \leq s/8$, its infinity norm is at most $E/2 + s/8 < E$. Thus the observed-softmax hypothesis and the greedy correction in Lemma 6.6 apply to the value actually passed by Algorithm 5. Finally, $N(\hat{u}) \leq r$ and $N(\chi) \leq 1.1s$, so the displayed movement bound follows from the triangle inequality.

Lemma 6.1 gives $\|\log z - \log g(x^+)\|_\infty \leq E$. Thus Algorithm 3, invoked in hybrid maintained mode with $D = E$ and desired accuracy $E/2$, uses an absolute number of rounds and proves the auxiliary-weight claim. Lemma 8.10 accounts for all score transitions and their effective drift globally. The coordinate predictor, certificate, and weighted predictor cost $O(L_* + q)$ systems outside that process, and the weighted predictor has $O(\log q)$ sequential rounds. This proves the cost and depth claims. $\square$

7.4 Path following and resource budgets

Algorithm 6 Hybrid-score path-following phase

```

1: procedure PATHFOLLOW( $x, w, \hat{g}, c_{\text{path}}, t_0, t_1, \tau, \zeta_{\text{step}}, L_{\text{conf}}, \mathcal{S}$ ) ▷ Lemma 7.5
2:    $\beta \leftarrow 1 + C\sqrt{2c_1}$ ,  $\alpha \leftarrow R/(16c_k\beta)$ , and  $t \leftarrow t_0$ .
3:    $M_{\text{path}} \leftarrow \lceil A_{\text{path}}(c_k\beta|\log(t_1/t_0)|/R + c_k\log(R/\tau) + 1) \rceil$ , including terminal-centering calls,
    $J \leftarrow \lceil 8c_k\log(R/\tau) \rceil$ , and  $j_{\text{path}} \leftarrow 0$ .
4:   while  $t \neq t_1$  do
5:      $j_{\text{path}} \leftarrow j_{\text{path}} + 1$ .
6:     if  $j_{\text{path}} > M_{\text{path}} - J$  then
7:       return FAIL.
8:     end if
9:      $\Delta \leftarrow \bar{\delta}_t^{c_{\text{path}}}(x, w)$  from one projected-residual solve.
10:    if  $\Delta = 0$  then
11:       $t \leftarrow \text{median}\{(1 - \alpha)t, t_1, (1 + \alpha)t\}$ .
12:    else
13:       $Y \leftarrow \text{PREDICTEDCENTER}(x, w, \hat{g}, t, c_{\text{path}}, \zeta_{\text{step}}, L_{\text{conf}}, \mathcal{S})$ . ▷ Algorithm 5
14:      if  $Y = \text{FAIL}$  then
15:        return FAIL.
16:      end if
17:       $(x, w, \hat{g}, \mathcal{S}) \leftarrow Y$ .
18:       $t \leftarrow \text{median}\{(1 - \alpha)t, t_1, (1 + \alpha)t\}$ .
19:    end if
20:  end while
21:  for  $j = 1, \dots, J$  do
22:     $\Delta \leftarrow \bar{\delta}_{t_1}^{c_{\text{path}}}(x, w)$  from one projected-residual solve.
23:    if  $\Delta > 0$  then
24:       $Y \leftarrow \text{PREDICTEDCENTER}(x, w, \hat{g}, t_1, c_{\text{path}}, \zeta_{\text{step}}, L_{\text{conf}}, \mathcal{S})$ . ▷ Algorithm 5
25:      if  $Y = \text{FAIL}$  then
26:        return FAIL.
27:      end if
28:       $(x, w, \hat{g}, \mathcal{S}) \leftarrow Y$ .
29:    end if
30:  end for
31:  return  $(x, w, \hat{g}, \mathcal{S})$ .
32: end procedure

```

The centering call is iterated while the path parameter moves toward its target. The following lemma proves that the phase routine preserves all invariants and gives its total number of calls.

Lemma 7.5 (Fresh path-following phase). *Suppose $t_0, t_1 > 0$, $0 < \tau \leq R$, and the input satisfies*

$$\delta_{t_0}^{c_{\text{path}}}(x, w) \leq R, \quad \|\log w - \log g(x)\|_\infty \leq K, \quad \|\log \hat{g} - \log g(x)\|_\infty \leq E/2,$$

together with the initial-potential hypothesis of Lemma 6.6; for a segment immediately continuing a preceding call inside the same attempt, it is enough that both segments use the same stopped residual game. Assume that the executed prefix belongs to an epoch of at most B_{ep} calls, with $L_{\text{ch}} \geq \log(A_0 B_{\text{ep}} m \beta)$ and $\zeta_{\text{step}} \leq (100 B_{\text{ep}})^{-1}$. On the raw-relative initialization event, analyze the

coupled hybrid score process of Lemma 8.12. Let $\mathcal{E}_{\text{sc}}$ be the event that no exact score replacement occurs before completion of this phase, and let $\mathcal{E}_{\text{coarse}}$ be the intersection of the per-call coarse events and $\mathcal{E}_{\text{ch}}$ the stopped tracking event of Lemma 6.6, both defined without conditioning on future score success. On $\mathcal{E}_{\text{coarse}} \cap \mathcal{E}_{\text{ch}} \cap \mathcal{E}_{\text{sc}}$, Algorithm 6 does not return FAIL and returns an iterate at t_1 with

$$\delta_{t_1}(x, w) \leq \tau,$$

while preserving the tracking and auxiliary-weight invariants. It makes

$$O\left(\frac{c_k \beta}{R} |\log(t_1/t_0)| + c_k \log(R/\tau) + 1\right)$$

calls to Algorithm 5.

Proof. On the intersection of the coarse, chasing, and score good events, Lemma 7.4 supplies the movement and tracking hypotheses of Lemma 6.3. Thus every nonzero-centrality iteration contracts at fixed t and permits the signed median update of magnitude at most α . If centrality is zero, the proof of the signed path-update part of Lemma 6.3, without its contraction term, gives centrality at most $\alpha\beta \leq R$ after the same median update. A full update changes $|\log t|$ by $\Omega(\alpha)$, and the last median update reaches t_1 exactly. This proves the first term in the call count.

At t_1 , each nonvacuous centering call contracts centrality by $1 - 1/(4c_k)$. Therefore the chosen value of J reduces centrality from at most R to at most τ . This contraction remains valid even when $\tau < E$: the hypotheses of Lemmas 5.6, 5.9, and 6.6 impose no lower bound on centrality relative to E , and Lemma 5.9 keeps the residual radius below the fixed observation scale E . Lemma 7.4 also preserves the auxiliary invariant, while the stochastic game preserves tracking simultaneously. Taking A_{path} sufficiently large makes the a priori call bound dominate both loops. Every zero test in the pseudocode is therefore a test of the computable projected-residual proxy, not of the mixed-norm minimization defining exact centrality. The pseudocode explicitly charges one projected-residual solve for that test. There is at most one such solve per loop iteration, so these solves are absorbed by the displayed call count and the $O(L_* + q)$ non-score work per iteration. This proves the lemma. $\square$

Algorithm 7 Deterministic normalization at a path-epoch boundary

- 1: **procedure** EPOCHNORMALIZE(x, w, t, c_{path}) $\triangleright$ Lemma 7.6
 - 2: Using exact score diagonals, run fixed-weight projected Newton steps until the computable proxy satisfies $\bar{\delta}_t^{c_{\text{path}}}(x, w) \leq \epsilon_{\text{rec}}$.
 - 3: $(\tilde{g}, \mathcal{S}_{\text{tmp}}) \leftarrow \text{FIXEDPOINT}(x, \mathbf{1}, q, \tau_{\text{norm}}, 1/2, L_{\text{conf}}, \emptyset, \text{exact})$.
 - 4: **while** $e^{\mu\tau_{\text{norm}}} \Phi(\log w - \log \tilde{g}) > M_{\text{ch}}/2$ **do**
 - 5: $\tilde{d} \leftarrow \log w - \log \tilde{g}$ and $\chi \leftarrow \text{MIXEDBALLLINEARORACLE}(w, C, s_{\text{norm}}, \nabla \Phi(\tilde{d}), \epsilon_{\text{norm}})$.
 - 6: $w \leftarrow w \odot e^\chi$ and run fixed-weight projected Newton steps until $\bar{\delta}_t^{c_{\text{path}}}(x, w) \leq \epsilon_{\text{rec}}$ at the same t, c_{path} .
 - 7: $(\tilde{g}, \mathcal{S}_{\text{tmp}}) \leftarrow \text{FIXEDPOINT}(x, \mathbf{1}, q, \tau_{\text{norm}}, 1/2, L_{\text{conf}}, \emptyset, \text{exact})$.
 - 8: **end while**
 - 9: $(\hat{g}, \mathcal{S}) \leftarrow \text{FIXEDPOINT}(x, \mathbf{1}, q, E/8, 1/2, L_{\text{conf}}, \emptyset, \text{exact})$; retain the exact score vector of its actual final query in $\mathcal{S}$.
 - 10: **return** $(x, w, \hat{g}, \mathcal{S})$.
 - 11: **end procedure**
-

Checkpointing requires all numerical scales to remain polynomially bounded between exact restarts. The next lemma shows that the normalization routine achieves this without changing the asymptotic path length.

Lemma 7.6 (Deterministic epoch normalization). *There are absolute constants $c_{\text{norm}}, A_{\text{norm}} > 0$ with the following property. Define*

$$\begin{aligned} s_{\text{norm}} &:= c_{\text{norm}} E / c_k, & \epsilon_{\text{norm}} &:= 1 / (16 c_k), \\ \tau_{\text{norm}} &:= \frac{s_{\text{norm}}}{2^{14} c_k \beta}, & \epsilon_{\text{rec}} &:= \min\{2^{-22} \ell^{-3}, \tau_{\text{norm}} / 2^{10}\}. \end{aligned}$$

Suppose that x is feasible and interior,

$$\delta_t^{\text{cpath}}(x, w) \leq R, \quad \|\log w - \log g(x)\|_\infty \leq K/2.$$

Then Algorithm 7 is deterministic, preserves feasibility and the path parameter, and returns a state satisfying

$$\delta_t^{\text{cpath}}(x, w) \leq \epsilon_{\text{rec}}, \quad \Phi(\log w - \log g(x)) \leq M_{\text{ch}}/2, \quad \|\log w - \log g(x)\|_\infty \leq K/4,$$

together with the stated auxiliary-weight and exact raw-score base invariants. It uses at most $(4m)^{A_{\text{norm}}}$ permitted systems, sequential system rounds, and exact scalar operations.

Proof. We first analyze exact fixed-weight centers; the finite stopping tolerance is inserted at the end. For a positive weight vector $\tilde{w}$, let $x(\tilde{w})$ be the minimizer of the fixed-weight barrier objective at t, c_{path} . Its existence and the convergence of projected Newton iteration follow as in Lemma 6.4; from an input of centrality at most R the total normalized displacement is $O(R)$. Target stability and the choice of c_R therefore show that the initial recentering changes $\log g$ by at most $K/16$. In particular the recentered state remains strictly inside the tracking band.

Fix a recentered state and a direction χ with $N_w(\chi) \leq s_{\text{norm}}$. Along $w_\theta = w \odot e^{\theta\chi}$ and $x_\theta = x(w_\theta)$, differentiate the fixed-weight stationarity condition. With

$$a_i(x) := \frac{\phi'_i(x_i)}{\sqrt{\phi''_i(x_i)}}, \quad |a_i(x)| \leq 1,$$

the normalized central response is

$$h_\theta := \sqrt{\Phi''(x_\theta)} \dot{x}_\theta = -P_{x_\theta, w_\theta}(\chi \odot a(x_\theta)).$$

Lemma 4.5, applied along this segment, gives $N_{w_\theta}(h_\theta) \leq c_\gamma N_{w_\theta}(\chi)$. At an exact center the fixed-weight centrality is zero, so Lemma 4.9 applies pointwise as a degenerate Newton segment whenever the strict tracking band holds. On any such prefix it gives

$$N_{w_\theta}(\frac{d}{d\theta} \log g(x_\theta)) \leq (1 - \frac{15}{8c_k}) c_\gamma N_{w_\theta}(\chi).$$

Define $\lambda := (1 - 15/(8c_k))c_\gamma$. Since the weighted component of N_{w_θ} differs from that of N_w by at most $e^{s_{\text{norm}}/2}$, integration gives, for

$$u := \log g(x(w \odot e^\chi)) - \log g(x(w))$$

the bound $N_w(u) \leq \lambda e^{s_{\text{norm}}} N_w(\chi)$. Because $c_\gamma \leq 1 + 1/(128c_k)$, decreasing c_{norm} makes $\lambda e^{s_{\text{norm}}} \leq 1 - 3/(2c_k)$, and hence

$$N_w(u) \leq (1 - \frac{3}{2c_k}) N_w(\chi). \tag{7}$$

Thus the exact centered update changes $d := \log w - \log g(x)$ by $d^+ = d + \chi - u$.

Let $\tilde{d} = \log w - \log \tilde{g}$, so that $\|\tilde{d} - d\|_\infty \leq \tau_{\text{norm}}$. The support guarantee of Algorithm 1, Eq. (7), duality, and the coordinate estimate

$$|\phi'(\tilde{d}_i) - \phi'(d_i)| \leq (e^{\mu\tau_{\text{norm}}} - 1)(|\phi'(d_i)| + 2\mu)$$

show, for $a = \nabla\Phi(d)$, that evaluating the oracle at the certified residual changes the exact support calculation by at most

$$O(\mu\tau_{\text{norm}})s_{\text{norm}}N_w^*(a) + O(\mu^2m\tau_{\text{norm}}s_{\text{norm}}).$$

To pass from exact centers to the finite stopping rule, let $x^\circ(w)$ denote the exact fixed-weight center. The unconditional lower side of Eq. (2) gives $\delta \leq \bar{\delta} \leq \epsilon_{\text{rec}}$. The strengthened fixed-weight distance bound of Lemma 6.4, self-concordant transport, and the target-Jacobian estimate give, at each endpoint,

$$N_w(\log g(x) - \log g(x^\circ(w))) \leq 32c_\gamma\epsilon_{\text{rec}}.$$

Consequently

$$N_w([\log g(x^+) - \log g(x)] - [\log g(x^\circ(we^X)) - \log g(x^\circ(w))]) \leq 64c_\gamma\epsilon_{\text{rec}}.$$

Its contribution to the support calculation is at most $64c_\gamma\epsilon_{\text{rec}}N_w^*(a)$ and is absorbed because $\epsilon_{\text{rec}} \leq \tau_{\text{norm}}/2^{10}$. Hence

$$\langle a, \chi - u \rangle \leq -\frac{3s_{\text{norm}}}{4c_k}N_w^*(a) + O(\mu^2m\tau_{\text{norm}}s_{\text{norm}}).$$

Taylor's theorem, $\phi'' = \mu^2\phi$, and the same mixed-norm quadratic estimate used in Lemma 5.1 give

$$\Phi(d^+) \leq \Phi(d) - \frac{s_{\text{norm}}}{2c_k}N_w^*(a) + C_{\text{norm}}\mu^2ms_{\text{norm}}^2.$$

The gradient-dependent Taylor remainder is absorbed by the preceding linear term because

$$\mu s_{\text{norm}} = c_{\text{norm}}/(100c_k).$$

This is the reason for using the smaller normalization radius. The additive term $C_{\text{norm}}\mu^2ms_{\text{norm}}^2$ is absorbed only above the threshold fixed next. Moreover

$$N_w^*(a) \geq \frac{1}{\beta}\|a\|_1, \quad \|a\|_1 \geq \mu(\Phi(d) - 2m).$$

The choice $\tau_{\text{norm}} = s_{\text{norm}}/(2^{14}c_k\beta)$ absorbs the certified-gradient and finite-centering perturbations as well. Taking A_{ch} large after the absolute Taylor constants are fixed, every iteration with $\Phi(d) > M_{\text{ch}}/4$, where $M_{\text{ch}} = A_{\text{ch}}m\beta$, therefore decreases Φ by a factor $1 - \Omega((c_k^2\beta)^{-1})$. Consequently at most

$$O(c_k^2\beta \log(2 + \Phi(d_{\text{start}})/M_{\text{ch}}))$$

normalization iterations are executed. If the outward loop test fails, then $\Phi(d) \leq e^{\mu\tau_{\text{norm}}}\Phi(\tilde{d}) \leq M_{\text{ch}}/2$. If it has not yet failed, the true potential is larger than $e^{-2\mu\tau_{\text{norm}}}M_{\text{ch}}/2 > M_{\text{ch}}/4$, so the descent estimate continues to apply.

At a certified epoch endpoint the starting tracking error is at most $K/2$. The initial fixed-weight recentering moves the target by $O(R) \leq K/16$, so $\log \Phi(d_{\text{start}}) = O(\ell)$. We close the tracking hypothesis used above by a first-exit argument. Suppose that the exact-center interpolation has a first point at which $\|d\|_\infty = K$. On the preceding prefix the pointwise invocation of Lemma 4.9 is valid, so Eq. (7) and the potential descent derived from it hold at every completed update on

that prefix. Hence Φ is nonincreasing there. More quantitatively, the initial margin and $\|d\|_\infty \leq \mu^{-1} \log \Phi(d)$ give, after increasing A_E , $\|d\|_\infty \leq 5K/8$ at every completed exact-center update. During one interpolation,

$$\|d_\theta - d_0\|_\infty \leq N_w(\theta\chi) + N_w(\log g(x_\theta) - \log g(x_0)) \leq (2 + o(1))s_{\text{norm}} < K/8.$$

The two finite recentering errors contribute at most $64c_\gamma\epsilon_{\text{rec}} < K/16$. The intervening fixed-weight Newton path has no larger total target displacement: apply the Newton-series estimate in Lemma 6.4 to each of its prefixes. Thus the putative first exit is still strictly below K , a contradiction. This justifies the operator estimate on the entire normalization path without circularity. Finally the stopping threshold, rather than the starting potential, gives $\|d\|_\infty \leq K/4$, because $\mu^{-1} = 100E$, $L_{\text{ch}} = A_L\ell$, and $E = K/(A_EL_{\text{ch}})$.

It remains to justify finite execution. After changing the weight by $s_{\text{norm}} < 1/10$, the weight-response inequality of Lemma 3.2 gives fixed-weight centrality $O(s_{\text{norm}})$. Projected Newton iteration reduces this quadratically, so $O(\log \log(2 + s_{\text{norm}}/\epsilon_{\text{rec}}))$ systems suffice to reach ϵ_{rec} . The exact-score branch of Algorithm 3 computes each required certified regularized Lewis vector in $O(n \log(2 + q/\epsilon_{\text{rec}}))$ permitted systems. Choosing A_{norm} above the resulting fixed polynomial bounds makes all finite recentering errors at most $\epsilon_{\text{rec}} \leq \tau_{\text{norm}}/2^{10}$ in the preceding inequalities and bounds the total work by $(4m)^{A_{\text{norm}}}$. The final exact-score response supplies a zero-error raw relative base, so it also satisfies every score-state convention. All Newton directions lie in $\ker(A^\top)$ and all iterates remain in the barrier domain, proving feasibility and the remaining claims. $\square$

The preceding lemmas introduced several absolute constants whose choices must be mutually compatible and independent of the input. The next lemma fixes the structural subset without circular dependence.

Lemma 7.7 (Compatible structural constants). *All structural absolute constants used by Algorithms 2–6 can be fixed once, independently of m, n, U, ϵ . The remaining scale, hybrid-provider, outer-switch, and resource-cap constants are then fixed, without changing these structural choices, in Lemma 8.9.*

Proof. First fix the numerical constants in the Lee–Sidford facts, the choices $\epsilon_0 = 1/10$ and $\nu = 1/5$, and the structural definitions of p, c_k, C, K . Next fix the constant-depth stochastic-chain, medoid, and scalar-median parameters; this fixes a_2, a_3, a_4, a_7, A_d and the absolute equivalence constants. Choose $B > 0$ so that $BA_d \leq \min\{1/12, \gamma/2\}$, then choose $\eta_0 > 0$ so that Eq. (8) below holds. Choose $\epsilon_1, \epsilon_P, \epsilon_g, c_0 > 0$ in that order, small enough that

$$1.1\eta_0 + \frac{(1.1\eta_0)^2}{512B} \leq \frac{1}{6}, \quad 16\epsilon_1/\eta_0 + \epsilon_P \leq 1/16. \quad (8)$$

and so that the fixed support-oracle and surrogate perturbation inequalities hold.

After those choices, decrease r_0 to satisfy every finite-segment estimate, including $r_0 \leq \epsilon_1/(8a_2)$, and fix A_q large enough to dominate the deterministic companion-error constants. Every remaining requirement encountered above is now a finite monotone upper bound on $c_R, E, R, c_{\text{anc}}$ or a lower bound on a sample, path, horizon, or budget constant. Lemma 8.9 resolves those bounds in an order that never changes a previously frozen radius. $\square$

8 The five-logarithm bound

Section 8.1 derives the exact coordinatewise transport identity and bounds the nonlocal endpoint error. Section 8.2 combines transported and ordinary transitions with block stopping to maintain

all scores uniformly. Finally, Section 8.3 verifies the companion bands, initialization, coupling, and deterministic budget needed by the path algorithm.

Proof boundary. Lemma 3.2 records the Lee–Sidford Newton contraction, normalized movement, response to a weight change, signed path-parameter change, and weighted duality-gap estimate used below. The outer reduction also uses their interval-barrier derivative bounds, which are built into the definition of U in Section 1; the cost-switch estimate and the terminal-distance conversion are derived explicitly in Lemma 6.5. Every other ingredient used for the solve bound—well-posed regularized weights, the computable centrality proxy, linear-defect projection, defect-adaptive predictor, observed softmax step, hybrid relative score maintenance, residual chasing, and unconditional initialization—is proved in the preceding sections and below. The hybrid provider transports the own-row response exactly and sketches only the nonlocal residual. Every random sketch is drawn after its query endpoints are fixed.

Algorithms 3–6 implement fixed-point computation, one hybrid-score centering step, and one path phase. The final block applies the two-phase outer reduction. In the budgeted execution, every descendant call is made through the same resource wrapper. Inside Algorithm 8, a descendant FAIL or an attempt-local-cap failure is caught and turns the current attempt into a rejection. A shared-global-cap failure is not caught: it propagates immediately to Algorithm 9. Outside this retry boundary every descendant failure likewise propagates before the enclosing procedure executes its next line, and tuple assignment is performed only on a successful return.

Regard the two path phases, including their terminal-centering loops and the cost-vector switch, as one resumable instruction stream $\mathfrak{P}$. Its state records the active phase, t , both deterministic loop counters, and $(x, w, \hat{g}, \mathcal{S})$. One *atomic path instruction* is a complete iteration of either loop of Algorithm 6: it includes the projected-residual test, the optional centering call, and, in the path loop, the ensuing median update of t . The cost-vector switch is atomic as well. Thus a checkpoint never splits a centering call from the path update licensed by it. We count every atomic loop iteration, including a zero-centrality iteration, against the epoch and global path caps. Enlarging the absolute constant A_N absorbs this equivalent count. Pausing and resuming only at these boundaries does not alter the iterates of the two displayed path loops.

For $1 \leq b \leq B_{\text{ep}}$ define $P_{\text{norm}} := (4m)^{A_{\text{norm}}}$ and fix a sufficiently large absolute A_{loc} . Every attempt uses the deterministic local caps

$$\begin{aligned} M_{\text{att}}(b) &:= A_{\text{loc}}(b + q\ell + 1 + \mathbf{1}_{\{b=B_{\text{ep}}\}}P_{\text{norm}}), \\ S_{\text{att}}(b) &:= A_{\text{loc}}(b(E^{-1}\ell + \ell + q + 1) + E^{-2}\ell + \mathbf{1}_{\{b=B_{\text{ep}}\}}P_{\text{norm}}), \\ D_{\text{att}}(b) &:= A_{\text{loc}}(b(1 + \log q) + 1 + \mathbf{1}_{\{b=B_{\text{ep}}\}}P_{\text{norm}}), \\ W_{\text{att}}(b) &:= A_{\text{loc}}(\text{nnz}(A) + m)(S_{\text{att}}(b) + b\ell^2 + \mathbf{1}_{\{b=B_{\text{ep}}\}}P_{\text{norm}}). \end{aligned} \tag{9}$$

The first cap counts score queries and exact-score queries; the other three count permitted systems, sequential system rounds, and scalar work. A routine announces its full charge before work or randomness, and an attempt is rejected before exceeding any local cap.

Algorithm 8 Checkpointed execution of the two path phases

```

1: procedure CHECKPOINTEDPATH( $\mathfrak{P}$ ,  $N_{\text{cap}}$ ) ▷ Lemma 8.1
2:    $B_{\text{ep}} \leftarrow (4m)^{A_{\text{ep}}}$ ,  $M_{\text{ep}} \leftarrow A_{\text{call}}(B_{\text{ep}} + q\ell + 1)$ ,  $L_{\text{conf}} \leftarrow \log(40mM_{\text{ep}})$ , and  $\zeta_{\text{step}} \leftarrow (100B_{\text{ep}})^{-1}$ ,
    $\epsilon_{\text{chk}} \leftarrow s_{\text{norm}}/(2^{14}c_k\beta)$ .
3:    $J_{\text{ep}} \leftarrow \lceil N_{\text{cap}}/B_{\text{ep}} \rceil + 1$ ,  $A_{\text{att}} \leftarrow 4J_{\text{ep}}$ ,  $a \leftarrow 0$ , and  $r \leftarrow 0$ .
4:   while  $\mathfrak{P}$  is not complete and  $a < A_{\text{att}}$  do
5:      $b \leftarrow \min\{B_{\text{ep}}, N_{\text{cap}} - r\}$ ; if  $b = 0$ , return FAIL.
6:     Announce the four attempt-local caps in Eq. (9).
7:     Charge every descendant operation to its local and shared global counters.
8:     Save the mathematical path and score state  $Q$  of  $\mathfrak{P}$ ; set  $j \leftarrow 0$ ,  $a \leftarrow a + 1$ , reject  $\leftarrow$  false,
     start fresh score and residual-game epochs, and begin a fresh attempt. Resource counters are
     never rolled back.
9:     while  $j < b$  and  $\mathfrak{P}$  is not complete and reject = false do
10:      Execute the next atomic instruction of  $\mathfrak{P}$ , using  $\zeta_{\text{step}}$  in Algorithm 5. On a descendant
      or attempt-local FAIL, set reject  $\leftarrow$  true; on a shared-global-cap FAIL, return FAIL.
11:      if the instruction completed then
12:         $j \leftarrow j + 1$ .
13:      end if
14:    end while
15:    if reject = true then
16:      Restore  $Q$  and continue to the next attempt.
17:    end if
18:    if  $\mathfrak{P}$  is complete then
19:      return its primal output.
20:    end if
21:    if  $r + j = N_{\text{cap}}$  then
22:      Restore  $Q$  and continue to the next attempt.
23:    end if
24:    if  $w \not\geq 0$ , the point violates  $A^\top x = b$ , is not interior, or has a path/phase descriptor
    inconsistent with  $\mathfrak{P}$  then
25:      Restore  $Q$  and continue to the next attempt.
26:    end if
27:     $(g_{\text{chk}}, \mathcal{S}_{\text{chk}}) \leftarrow \text{FIXEDPOINT}(x, \mathbf{1}, q, \epsilon_{\text{chk}}, 1/2, L_{\text{conf}}, \emptyset, \text{exact})$ , and compute  $\bar{\delta}_t^{c_{\text{path}}}(x, w)$ .
28:    if  $\bar{\delta}_t^{c_{\text{path}}}(x, w) > R - R/(32c_k)$  or  $\|\log w - \log g_{\text{chk}}\|_\infty + \epsilon_{\text{chk}} > K/2$  then
29:      Restore  $Q$  and continue to the next attempt.
30:    else
31:       $(x, w, \hat{g}, \mathcal{S}) \leftarrow \text{EPOCHNORMALIZE}(x, w, t, c_{\text{path}})$ .
32:       $(g_{\text{chk}}, \mathcal{S}_{\text{chk}}) \leftarrow \text{FIXEDPOINT}(x, \mathbf{1}, q, \epsilon_{\text{chk}}, 1/2, L_{\text{conf}}, \emptyset, \text{exact})$  and certify  $\bar{\delta}_t^{c_{\text{path}}}(x, w) \leq$ 
       $R/4$ ,  $\Phi(\log w - \log g_{\text{chk}}) \leq e^{-\mu\epsilon_{\text{chk}}} M_{\text{ch}}$ , and  $\|\log \hat{g} - \log g_{\text{chk}}\|_\infty \leq E/4$ ; on failure restore  $Q$  and
      continue; otherwise accept this epoch and start the next attempt from the normalized state.
33:       $r \leftarrow r + j$ .
34:    end if
35:  end while
36:  return FAIL.
37: end procedure

```

After normalization and the structural constants are fixed, the path can be partitioned into

bounded attempts with deterministic caps. The next lemma assembles these attempts and controls their total failure probability.

Lemma 8.1 (Checkpointed polynomial epochs). *Fix a sufficiently large absolute $A_{\text{ep}} > A_{\text{norm}} + 10$ and define $B_{\text{ep}} = (4m)^{A_{\text{ep}}}$. Run all stochastic routines inside one attempt with*

$$L_{\text{ch}} = A_L \ell, \quad M_{\text{ep}} = A_{\text{call}}(B_{\text{ep}} + q\ell + 1), \quad L_{\text{conf}} = \log(40mM_{\text{ep}}), \quad \zeta_{\text{step}} = (100B_{\text{ep}})^{-1}.$$

Suppose an attempted epoch starts either on the raw-relative initialization event or from an accepted output of Lemma 7.6. Conditional on its entire starting state, a full B_{ep} -instruction attempt that does not finish the stream is accepted with probability at least $4/5$; if it finishes the stream, the same good event implies a correct output. Algorithm 8, capped at $4J_{\text{ep}}$ attempts where $J_{\text{ep}} = \lceil N_{\text{cap}}/B_{\text{ep}} \rceil + 1$, halts on every outcome and completes the two path phases with constant probability. Its total systems, system depth, and scalar work are within an absolute constant of the uncheckpointed bounds with $L_ = \ell$.*

Proof. At an attempt boundary the algorithm snapshots only the mathematical state: the active phase, t , loop counters, $x, w, \hat{g}, \mathcal{S}$, and the local residual-game state. Shared resource counters are monotone and are never rolled back. A rejected attempt restores the snapshot, resets its local stochastic-game and score-block counters, and draws fresh randomness; all work already done remains charged. Immediately after forming a proposed Newton point, membership in Ω° is tested before evaluating any barrier quantity, and a failed test rejects the attempt.

On the local good event, the B_{ep} coarse events fail with total probability at most $1/100$, the stopped residual game fails with probability at most $1/20$, and raw score transport fails with probability at most $1/40$. Thus their intersection has conditional probability at least $4/5$. The enlarged choice of A_E makes the last display in the proof of Lemma 5.1 at most $K/4$. On this event every nonzero path instruction ends, after its path-parameter update, with $\delta \leq R - R/(8c_k)$. Therefore

$$\bar{\delta} \leq \bar{c}_\gamma(1 - \frac{1}{8c_k})R \leq R - \frac{R}{32c_k}.$$

The zero-centrality instruction has $\bar{\delta} \leq \bar{c}_\gamma R/(16c_k)$ and obeys the same test. An exact-score approximation $g_{\text{chk}} = e^{\pm \epsilon_{\text{chk}}} g(x)$ passes $\|\log w - \log g_{\text{chk}}\|_\infty + \epsilon_{\text{chk}} \leq K/2$. Affine feasibility, interiority, weight positivity, and the active phase and t descriptor are checked exactly.

For an attempt of length cap b , the same good event gives $O(b)$ score transitions and total effective drift $O(bE)$. Hence the hybrid provider uses

$$O(b + bE^{-1}\ell + E^{-2}\ell)$$

systems, while all non-score operations use $O(b(\ell + q))$ systems, $O(b(1 + \log q))$ system rounds, and $O((\text{nnz}(A) + m)b\ell^2)$ scalar work. If $b = B_{\text{ep}}$, the two cold checkpoint computations and epoch normalization are bounded by a constant multiple of P_{norm} . Thus, after fixing A_{loc} , a good attempt never reaches any cap in Eq. (9). On an arbitrary bad attempt, those same caps force rejection before a larger charge is incurred; no drift assertion is used for rejected work.

Lemma 7.6 then returns the complete fresh-start invariants, including computable centrality at most ϵ_{rec} , potential at most $M_{\text{ch}}/2$, an $E/8$ -accurate auxiliary vector, and an exact raw-score base. The algorithm recomputes outward certificates after normalization and rejects on any failure. Hence the local good event implies acceptance. An accepted state contains every hypothesis of a fresh invocation of the path lemmas: in particular the potential, not merely the ℓ_∞ tracking bound, is reset. Past sketches occur in no future hypothesis, and the phase and t encode all deterministic

progress. The two costs share one residual game only if the switch occurs inside the same attempt; every accepted checkpoint starts a new game.

Let T be the number of accepted checkpoints strictly preceding the final segment (which may itself contain B_{ep} instructions and complete without a checkpoint). Then $T \leq J_{\text{ep}} - 1$. The conditional acceptance probabilities stochastically dominate Bernoulli variables of mean $4/5$. A Chernoff bound shows that the first $3J_{\text{ep}}$ attempts contain T accepted full epochs except with probability $e^{-\Omega(J_{\text{ep}})}$; for bounded J_{ep} the same claim follows directly after decreasing the absolute success constant. This leaves at least one of the total $4J_{\text{ep}}$ attempts for the final segment. That segment has at most B_{ep} instructions and, conditionally on the state at which it starts, its local good event has probability at least $4/5$ and implies a correct output. If its good event fails, an uncertified output is counted as part of the theorem's constant failure probability. The final segment is not followed by a mandatory exact checkpoint, which preserves the cheaper initialization branch. Together with the one-time initialization event, this gives constant overall success.

Every attempt announces local deterministic caps before work. Let $A_{\text{att}} \leq 4J_{\text{ep}}$ be the number of attempts, and write b_a for the announced instruction cap of attempt a . Because $J_{\text{ep}} = \lceil N_{\text{cap}}/B_{\text{ep}} \rceil + 1$ and $b_a \leq \min\{B_{\text{ep}}, N_{\text{cap}}\}$,

$$\sum_{a=1}^{A_{\text{att}}} b_a = O(N_{\text{cap}}), \quad A_{\text{att}} = O(N_{\text{cap}}/B_{\text{ep}} + 1).$$

Summing Eq. (9) therefore gives

$$O(N_{\text{cap}}(E^{-1}\ell + \ell + q) + (N_{\text{cap}}/B_{\text{ep}} + 1)E^{-2}\ell)$$

systems before normalization. The choice of A_{ep} ensures

$$B_{\text{ep}} \geq E^{-2}\ell, \quad (N_{\text{cap}}/B_{\text{ep}})P_{\text{norm}} = O(N_{\text{cap}}).$$

When $N_{\text{cap}} < B_{\text{ep}}$ no normalization is invoked. The same summation applies to rounds and scalar work. Every confidence logarithm is $O(\ell)$ independently of κ , so the total is

$$O(\sqrt{n} q^2 \ell^3 \kappa + \min\{q \ell^3, n \log \ell\}).$$

The final partial segment pays no extra exact checkpoint. System depth and scalar work obey the same constant-factor accounting. $\square$

Algorithm 9 Two-phase linear-programming solver

```

1: procedure LPSOLVE( $x_0, U, \epsilon$ ) ▷ Theorem 8.16
2:    $\ell \leftarrow \log(4m)$ ,  $q \leftarrow \log(4m/n)$ , and  $\kappa \leftarrow A_\kappa(1 + \log(emU/\epsilon))$ .
3:   Set  $p, v, c_k, C, K$  as in Lemma 4.9,  $L_* \leftarrow \ell$ ,  $L_{\text{ch}} \leftarrow A_L L_*$ ,  $E \leftarrow K/(A_E L_{\text{ch}})$ , and  $R \leftarrow c_R C E$ .
4:    $(N_{\text{cap}}, M_{\text{calls}}, L_{\text{conf}}, B_{\text{sys}}, B_{\text{rnd}}, B_{\text{ar}}) \leftarrow$  the explicit quantities in Lemma 8.15, with  $B_{\text{sys}} \leftarrow B_{\text{sys}}^{(5)}$  and  $B_{\text{ar}} \leftarrow B_{\text{ar}}^{(5)}$ .
5:   Initialize the four shared counters—score queries, permitted systems, sequential system rounds, and exact scalar operations—at zero. Before each such operation, announce its deterministic charge and compare it respectively with  $M_{\text{calls}}$ ,  $B_{\text{sys}}$ ,  $B_{\text{rnd}}$ , or  $B_{\text{ar}}$ ; return FAIL before work or randomness if the corresponding cap would be exceeded.
6:    $\zeta \leftarrow 1/100$  and  $Y \leftarrow \text{FIXEDPOINT}(x_0, \mathbf{1}, q, E/4, \zeta/3, L_{\text{conf}}, \emptyset, \text{auto})$ . ▷ Algorithm 3
7:   if  $Y = \text{FAIL}$  then
8:     return FAIL.
9:   end if
10:   $(\hat{g}, \mathcal{S}) \leftarrow Y$ .
11:   $w \leftarrow \hat{g}$  and  $d \leftarrow -w \odot \phi'(x_0)$ .
12:   $(t_{\text{sw}}, t_{\text{out}}, \tau_{\text{sw}}, \tau_{\text{out}}) \leftarrow$  the parameters of Lemma 6.5.
13:  Form the resumable stream  $\mathfrak{P}$  from the two path-phase descriptors  $(d, 1, t_{\text{sw}}, \tau_{\text{sw}})$  and  $(c, t_{\text{sw}}, t_{\text{out}}, \tau_{\text{out}})$ . Algorithm 8 supplies  $\zeta_{\text{step}}$ ,  $L_{\text{conf}}$ , and the current score state to each atomic instruction. The cost switch shares the residual game only when it occurs inside one attempt; every accepted checkpoint starts a fresh game.
14:   $Y \leftarrow \text{CHECKPOINTEDPATH}(\mathfrak{P}, N_{\text{cap}})$ . ▷ Algorithm 8
15:  if  $Y = \text{FAIL}$  then
16:    return FAIL.
17:  end if
18:  return  $Y$ .
19: end procedure

```

We prove the hybrid score-maintenance bound by transporting relative leverage-score errors through diagonal rescalings. The provider transports the own-row response exactly and sketches only the nonlocal endpoint discrepancy; no path-geometry or residual-game lemma is altered. For a positive diagonal matrix D , write

$$P_D := P(D), \quad \sigma_i(D) := (P_D)_{ii} = \|P_D e_i\|_2^2.$$

All sigma-fields below are taken immediately before the fresh Gaussian sketch of the transition under discussion.

8.1 Exact diagonal self-transport

For $\gamma > 0$ and $s \in [0, 1]$, define

$$\mathcal{T}_\gamma(s) := \frac{\gamma^2 s}{1 + (\gamma^2 - 1)s}.$$

Diagonal self-transport starts from the exact effect of rescaling a single row, including its action on an imperfect incoming score. The next lemma records that identity and bounds the transported relative error.

Lemma 8.2 (One-row transport and approximate anchor). *Let D be positive diagonal, define $P = P_D$, $p_i = P e_i$, and $s_i = P_{ii}$. Let $D^{(i,\gamma)}$ be obtained from D by multiplying its i -th diagonal entry by γ . Then*

$$P_{D^{(i,\gamma)}} e_i = \frac{\gamma}{1 + (\gamma^2 - 1)s_i} (p_i + (\gamma - 1)s_i e_i)$$

and

$$\sigma_i(D^{(i,\gamma)}) = \mathcal{T}_\gamma(s_i).$$

For $t \in [0, 1]$, define

$$q_i(t) := \frac{\gamma}{1 + (\gamma^2 - 1)t} (p_i + (\gamma - 1)t e_i).$$

Then, for every $s, t \in [0, 1]$ with $s = s_i$,

$$\mathcal{T}_\gamma(t) - \|q_i(t)\|_2^2 = A_\gamma(t)(t - s), \quad A_\gamma(t) := \frac{\gamma^2(1 + 2(\gamma - 1)t)}{(1 + (\gamma^2 - 1)t)^2}. \quad (10)$$

If $s > 0$, $|\log \gamma| \leq 1/8$, $|t - s| \leq \epsilon s$, and $0 \leq \epsilon \leq 1/4$, then $A_\gamma(t) \geq 0$ and

$$\frac{A_\gamma(t)s}{\mathcal{T}_\gamma(s)} \leq \exp(12\epsilon |\log \gamma| s). \quad (11)$$

Proof. Write $B = DA$, $M = B^\top B$, and let $b_i^\top$ be the i -th row of B . Thus $p_i = BM^{-1}b_i$ and $s_i = b_i^\top M^{-1}b_i$. The new Gram matrix is $M + (\gamma^2 - 1)b_i b_i^\top$. Sherman–Morrison gives

$$(M + (\gamma^2 - 1)b_i b_i^\top)^{-1} b_i = \frac{M^{-1}b_i}{1 + (\gamma^2 - 1)s_i}.$$

Multiplying the i -th row of B by γ now gives the first display. The norm identity follows because $P_{D^{(i,\gamma)}}$ is an orthogonal projection.

For Eq. (10), define $b = \gamma - 1$ and $d(t) = 1 + (\gamma^2 - 1)t$. Since $\|p_i\|_2^2 = e_i^\top p_i = s$, direct expansion gives

$$\|q_i(t)\|_2^2 = \frac{\gamma^2}{d(t)^2} (s + 2bts + b^2 t^2).$$

Subtracting this quantity from $\gamma^2 t/d(t)$ yields $\gamma^2(t - s)(1 + 2bt)/d(t)^2$.

When $|\log \gamma| \leq 1/8$, one has $\gamma > 1/2$, so $1 + 2(\gamma - 1)t \geq 2\gamma - 1 > 0$. Define

$$R_\gamma(t) := \frac{A_\gamma(t)s}{\mathcal{T}_\gamma(s)}.$$

At $t = s$,

$$R_\gamma(s) = \frac{1 + 2(\gamma - 1)s}{1 + (\gamma^2 - 1)s} = 1 - \frac{(\gamma - 1)^2 s}{1 + (\gamma^2 - 1)s} \leq 1.$$

Moreover,

$$\frac{d}{dt} \log R_\gamma(t) = \frac{2(\gamma - 1)}{1 + 2(\gamma - 1)t} - \frac{2(\gamma^2 - 1)}{1 + (\gamma^2 - 1)t}.$$

Define $a := |\log \gamma| \leq 1/8$. Then $7/8 \leq \gamma \leq 8/7$, both denominators are at least $3/4$, and $|\gamma - 1| \leq 8a/7$ and $|\gamma^2 - 1| \leq 120a/49$. Consequently,

$$\left| \frac{d}{dt} \log R_\gamma(t) \right| \leq \left(\frac{64}{21} + \frac{320}{49} \right) a < 10a < 12a.$$

Integrating from s to t and using $|t - s| \leq \epsilon s$ proves Eq. (11). $\square$

The one-row identity must be combined with the simultaneous motion of all other rows between two score queries. The following endpoint estimate shows that the nonlocal remainder is governed by the companion-weighted drift.

Lemma 8.3 (Finite nonlocal endpoint bound). *Let D_-, D_+ and a positive vector $\tilde{w}$ be measurable with respect to a sigma-field $\mathcal{F}$. Define*

$$d := \log \text{diag}(D_+ D_-^{-1}), \quad \lambda_w := \frac{\sum_{j=1}^m \tilde{w}_j d_j}{\sum_{j=1}^m \tilde{w}_j},$$

$$\delta := d - \lambda_w \mathbf{1}, \quad \rho_w := \left(\sum_{j=1}^m \tilde{w}_j \delta_j^2 \right)^{1/2}.$$

Suppose

$$\|\delta\|_\infty \leq 1/8, \quad \sigma_j(D_-) \leq Q \tilde{w}_j \quad (j \in [m]), \quad (12)$$

where Q is an absolute constant. For a fixed coordinate i , define $s_i = \sigma_i(D_-)$, $\gamma_i = e^{\delta_i}$, and let $q_i(t)$ be the vector in Lemma 8.2, formed from $P_{D_-} e_i$. If $|t_i - s_i| \leq \epsilon s_i$ with $0 \leq \epsilon \leq 1/4$, then

$$\|P_{D_+} e_i - q_i(t_i)\|_2 \|P_{D_+} e_i + q_i(t_i)\|_2 \leq K_Q \rho_w \sigma_i(D_+) \quad (13)$$

for an absolute constant K_Q depending only on Q , provided $\rho_w \leq \rho_Q$ for a sufficiently small absolute $\rho_Q > 0$. Furthermore,

$$e^{-K_Q \rho_w} \mathcal{T}_{\gamma_i}(s_i) \leq \sigma_i(D_+) \leq e^{K_Q \rho_w} \mathcal{T}_{\gamma_i}(s_i). \quad (14)$$

Proof. If $s_i = 0$, then row i of A is zero because D_- is positive. Positive diagonal rescaling preserves this property, so both projection columns in the statement are zero and all conclusions are immediate. Assume below that $s_i > 0$. Multiplying a diagonal query by a scalar does not change its projection, so replace D_+ by $e^{-\lambda_w} D_+$. First multiply only row i of D_- by γ_i . By Lemma 8.2, the resulting i -th projection column is $q_i(s_i)$ and its squared norm is $\mathcal{T}_{\gamma_i}(s_i)$.

Connect this intermediate query to D_+ by multiplying every row $j \neq i$ by $e^{t \delta_j}$, $t \in [0, 1]$. Along this path write $P(t)$ for the projection, $\sigma_j(t) = P(t)_{jj}$, and $\Delta = \text{Diag}(\delta)$ with $\delta_i = 0$. Differentiation of the projection gives

$$\dot{P} = (I - P)\Delta P + P\Delta(I - P).$$

Since $\Delta e_i = 0$,

$$\dot{P} e_i = (I - 2P)\Delta P e_i.$$

The matrix $I - 2P$ is an orthogonal reflection. Consequently

$$\|\dot{P} e_i\|_2^2 = \sum_{j=1}^m \delta_j^2 P_{ji}^2 \leq \sigma_i(t) \sum_{j=1}^m \sigma_j(t) \delta_j^2, \quad (15)$$

where the last step uses $P_{ji}^2 \leq \sigma_j(t) \sigma_i(t)$. Similarly,

$$\dot{\sigma}_i(t) = -2 \sum_{j=1}^m \delta_j P_{ij}(t)^2.$$

Weighted Cauchy–Schwarz and $\sum_{j=1}^m P_{ij}^4 / \sigma_j(t) \leq \sigma_i(t) \sum_{j=1}^m P_{ij}^2 = \sigma_i(t)^2$ show that

$$\left| \frac{d}{dt} \log \sigma_i(t) \right| \leq 2 \left(\sum_{j=1}^m \sigma_j(t) \delta_j^2 \right)^{1/2}. \quad (16)$$

Every intermediate query is within logarithmic ℓ_∞ distance $1/8$ of D_- . Lemma 4.1 and Eq. (12) therefore give

$$\sum_{j=1}^m \sigma_j(t) \delta_j^2 \leq e^{1/2} Q \rho_w^2.$$

Integrating Eq. (16) proves Eq. (14), after enlarging K_Q . Integrating Eq. (15) and using the same score comparison gives

$$\|P_{D_+} e_i - q_i(s_i)\|_2 \leq K_Q \rho_w \sqrt{\sigma_i(D_+)}. \quad (17)$$

It remains to replace s_i by t_i . Direct differentiation gives

$$q'_i(t) = \frac{\gamma_i(\gamma_i - 1)}{(1 + (\gamma_i^2 - 1)t)^2} (e_i - (\gamma_i + 1)P_{D_-} e_i),$$

and

$$\|e_i - (\gamma_i + 1)P_{D_-} e_i\|_2^2 = 1 + (\gamma_i^2 - 1)s_i.$$

Thus

$$\|q_i(t_i) - q_i(s_i)\|_2 \leq 3\epsilon |\delta_i| s_i \leq 3\epsilon \sqrt{Q} \rho_w \sqrt{s_i}.$$

Since $|\log \gamma_i| \leq 1/8$, one has $\mathcal{T}_{\gamma_i}(s_i) = e^{\pm 1/4} s_i$. Eq. (14) therefore converts the last factor to $O(\sqrt{\sigma_i(D_+)})$. Combining this estimate with Eq. (17) proves the first-factor bound $O(\rho_w \sqrt{\sigma_i(D_+)})$. Both columns have norm $O(\sqrt{\sigma_i(D_+)})$ when ρ_Q is sufficiently small. Their product proves Eq. (13). $\square$

The endpoint estimate now lets us transport an already approximate score and use fresh randomness only for the residual discrepancy. The next lemma gives the conditional recurrence that drives the maintained raw-relative error.

Lemma 8.4 (Conditional self-transport transition). *Assume the hypotheses of Lemma 8.3. Suppose an $\mathcal{F}$ -measurable incoming state $h \in \mathbb{R}^m$ satisfies*

$$|h_i - \sigma_i(D_-)| \leq \epsilon \sigma_i(D_-), \quad 0 \leq \epsilon \leq 1/4.$$

Define $t_i = \min\{\max\{h_i, 0\}, 1\}$ and

$$a_i := \frac{\gamma_i}{1 + (\gamma_i^2 - 1)t_i}, \quad c_i := a_i(\gamma_i - 1)t_i,$$

$$Q_t := P_{D_-} \text{Diag}(a) + \text{Diag}(c).$$

After all these quantities and both endpoints are fixed, draw a fresh $S \in \mathbb{R}^{k \times m}$ with independent $N(0, 1/k)$ entries and set

$$h_i^+ := \mathcal{T}_{\gamma_i}(t_i) + \|SP_{D_+} e_i\|_2^2 - \|SQ_t e_i\|_2^2.$$

Then

$$\frac{h_i^+ - \sigma_i(D_+)}{\sigma_i(D_+)} = b_i \frac{t_i - \sigma_i(D_-)}{\sigma_i(D_-)} + \xi_i, \quad 0 \leq b_i \leq e^{K_Q \rho_w}, \quad (18)$$

and, conditionally on $\mathcal{F}$,

$$\mathbb{E}[e^{\lambda \xi_i} \mid \mathcal{F}] \leq \exp(K_Q^2 \lambda^2 \rho_w^2 / k), \quad |\lambda| \leq k / (4K_Q \rho_w). \quad (19)$$

The update for all coordinates uses one application of each endpoint projection per sketch row because

$$SQ_t = (SP_{D_-}) \text{Diag}(a) + S \text{Diag}(c).$$

If $\rho_w = 0$, the two endpoints differ only by a scalar and the algorithm returns t without drawing a sketch. If $\sigma_i(D_-) = 0$, use the exact convention $h_i^+ = 0$, $b_i = 0$, and $\xi_i = 0$ in Eq. (18).

Proof. Clipping to $[0, 1]$ cannot increase distance from the true score, so $|t_i - \sigma_i(D_-)| \leq \epsilon \sigma_i(D_-)$. The i -th column of Q_t is exactly $q_i(t_i)$. Conditional on $\mathcal{F}$, Gaussian isotropy and Eq. (10) give

$$\mathbb{E}[h_i^+ - \sigma_i(D_+) \mid \mathcal{F}] = A_{\gamma_i}(t_i)(t_i - \sigma_i(D_-)).$$

Thus Eq. (18) holds with

$$b_i := \frac{A_{\gamma_i}(t_i)\sigma_i(D_-)}{\sigma_i(D_+)}.$$

It is nonnegative by Lemma 8.2. Eq. (11) and Eq. (14), together with

$$|\delta_i| \sqrt{\sigma_i(D_-)} \leq \sqrt{Q} \rho_w,$$

give $b_i \leq e^{K_Q \rho_w}$ after enlarging K_Q .

For the centered innovation, use the identity

$$\|Su\|_2^2 - \|Sv\|_2^2 = \langle S(u-v), S(u+v) \rangle$$

with $u = P_{D_+} e_i$ and $v = q_i(t_i)$. The centered product of two jointly Gaussian variables is sub-exponential. Equation Eq. (13) therefore gives

$$\mathbb{E}[e^{\lambda X_i} \mid \mathcal{F}] \leq \exp(K_Q^2 \lambda^2 \rho_w^2 \sigma_i(D_+)^2 / k)$$

for $|\lambda| \leq k/(4K_Q \rho_w \sigma_i(D_+))$, where X_i is the centered unnormalized sketch error. Substitute $\xi_i = X_i/\sigma_i(D_+)$ to obtain Eq. (19).

If $\rho_w = 0$, positivity of $\tilde{w}$ gives $\delta = 0$. Hence the endpoints differ only by a scalar, $P_{D_+} = P_{D_-}$, and the exact transported value is t . $\square$

8.2 Hybrid maintenance and stopping

For a transition define the ordinary centered drift

$$\lambda_\infty := \frac{\max_{i \in [m]} d_i + \min_{i \in [m]} d_i}{2}, \quad \rho_\infty := \|d - \lambda_\infty \mathbf{1}\|_\infty.$$

Not every score transition satisfies the weighted eligibility condition for self-transport. The following ordinary transition provides a uniform fallback whose cost is charged to centered ℓ_∞ drift.

Lemma 8.5 (Ordinary relative transition). *Let D_-, D_+ be fixed by a sigma-field $\mathcal{F}$, define $d = \log \text{diag}(D_+ D_-^{-1})$, and suppose $\rho_\infty \leq 1/8$. If an incoming raw state h obeys $|h_i - \sigma_i(D_-)| \leq \epsilon \sigma_i(D_-)$ with $0 \leq \epsilon \leq 1/4$, define $t_i = \min\{\max\{h_i, 0\}, 1\}$. After the endpoints and t are fixed, draw a fresh $S \in \mathbb{R}^{k \times m}$ with independent $N(0, 1/k)$ entries and set*

$$h_i^+ := t_i + \|SP_{D_+} e_i\|_2^2 - \|SP_{D_-} e_i\|_2^2.$$

For every coordinate with nonzero score,

$$\frac{h_i^+ - \sigma_i(D_+)}{\sigma_i(D_+)} = b_i \frac{t_i - \sigma_i(D_-)}{\sigma_i(D_-)} + \xi_i, \quad b_i := \frac{\sigma_i(D_-)}{\sigma_i(D_+)},$$

where $0 \leq b_i \leq e^{4\rho_\infty}$ and, for absolute constants $c, C > 0$,

$$\mathbb{E}[e^{\lambda \xi_i} \mid \mathcal{F}] \leq \exp(C\lambda^2 \rho_\infty^2 / k), \quad |\lambda| \leq ck / \rho_\infty.$$

If the score is zero, then the corresponding row of every projection query is zero and the transition returns zero exactly. If $\rho_\infty = 0$, the two projections coincide and the procedure returns t without a sketch.

Proof. Projection matrices are invariant under a common scalar rescaling of D , so remove $\lambda_\infty \mathbf{1}$ from d . The relative-drift lemma gives $\sigma_i(D_+) = e^{\pm 4\rho_\infty} \sigma_i(D_-)$. Conditional Gaussian isotropy shows that the transition increment is unbiased. Apply Lemma 7.3 with $a_i = \max\{\sigma_i(D_-), \sigma_i(D_+)\}$, which is $\mathcal{F}$ -measurable. After division by $\sigma_i(D_+)$, score comparison turns its conditional moment-generating-function bound into the displayed relative bound. The formula for b_i is the remaining deterministic part. Finally, clipping to $[0, 1]$, an interval containing the true score, multiplies the incoming relative error by a number in $[0, 1]$. $\square$

For the same transition, and for the positive companion $\tilde{w}^-$ stored with the old query, define

$$\lambda_w := \frac{\sum_{i=1}^m \tilde{w}_i^- d_i}{\sum_{i=1}^m \tilde{w}_i^-}, \quad \rho_w := \left(\sum_{i=1}^m \tilde{w}_i^- (d_i - \lambda_w)^2 \right)^{1/2}.$$

Fix absolute constants C_∞, C_w large enough to dominate all constants in the ordinary and weighted transition bounds, and then choose

$$h_0 \leq \min\{\log(4/3), C_\infty/8, C_w \rho_w\}.$$

The ordinary branch has effective drift $C_\infty \rho_\infty$ and is eligible when this quantity is at most h_0 . The weighted branch is eligible only if

$$\|d - \lambda_w \mathbf{1}\|_\infty \leq 1/8 \quad \text{and} \quad C_w \rho_w \leq h_0;$$

when eligible, its effective drift is $C_w \rho_w$. Define η as the smaller eligible effective drift. If no branch is eligible, charge the transition the finite amount h_0 and force a fresh base at the new endpoint; no transition recurrence is then used. On every nonrestart transition, both branches have a common recurrence

$$z_{t+1,i} = b_{t,i} r_{t,i} z_{t,i} + \xi_{t,i}, \quad 0 \leq r_{t,i} \leq 1, \quad 0 \leq b_{t,i} \leq e^{\eta_t}, \quad (20)$$

and

$$\mathbb{E}[e^{\lambda \xi_{t,i}} \mid \mathcal{F}_t] \leq e^{\lambda^2 \eta_t^2 / k_t}, \quad |\lambda| \leq k_t / (4\eta_t). \quad (21)$$

where $z_{t,i} := (h_{t,i} - \sigma_i(D_t)) / \sigma_i(D_t)$ for a nonzero score. The zero-score coordinates are returned exactly and omitted from this normalization. Here the factor $r_{t,i}$ is the contraction caused by clipping the previous raw state to $[0, 1]$. The ordinary branch follows from Lemma 8.5; the weighted branch follows from Lemma 8.4. The constants C_∞, C_w are fixed once so that Eq. (20) and Eq. (21) hold literally in both branches.

Algorithm 10 Hybrid relative score maintenance

1: **procedure** HYBRIDSCORE($D, \tilde{w}, L_{\text{conf}}, \mathcal{S}$) ▷ Lemma 8.7
2: **if** $\mathcal{S} = \emptyset$ **then**
3: Go to FRESHBASE.
4: **else**
5: $(D_-, \tilde{w}^-, h, H) \leftarrow \mathcal{S}$ and $d \leftarrow \log \text{diag}(DD_-^{-1})$.
6: Compute ρ_∞ and ρ_w from the two displays above.
7: $\eta_\infty \leftarrow C_\infty \rho_\infty$ if $C_\infty \rho_\infty \leq h_0$, and $\eta_\infty \leftarrow +\infty$ otherwise.
8: $\eta_w \leftarrow C_w \rho_w$ if $\|d - \lambda_w \mathbf{1}\|_\infty \leq 1/8$ and $C_w \rho_w \leq h_0$, and $\eta_w \leftarrow +\infty$ otherwise.
9: $\eta \leftarrow \min\{\eta_\infty, \eta_w\}$, breaking ties deterministically.
10: **if** $\eta = +\infty$ **then**
11: Charge effective drift h_0 and go to FRESHBASE.
12: **else if** $H + \eta > h_0$ **then**
13: Charge effective drift η and go to FRESHBASE.
14: **end if**
15: $t_i \leftarrow \min\{\max\{h_i, 0\}, 1\}$ for every $i \in [m]$.
16: **if** $\eta = 0$ **then**
17: $h \leftarrow t$.
18: **else**
19: $k \leftarrow \max\{1, \lceil C_1 \eta E^{-2} L_{\text{conf}} \rceil\}$ and draw a fresh $S \in \mathbb{R}^{k \times m}$ with independent $N(0, 1/k)$ entries after all preceding quantities are fixed.
20: **if** $\eta = \eta_\infty$ **then**
21: Update h by Lemma 8.5, using the already drawn matrix S .
22: **else**
23: Update h by Lemma 8.4, using the already drawn matrix S and $\tilde{w}^-$ to define λ_w and ρ_w .
24: **end if**
25: **end if**
26: $H \leftarrow H + \eta$ and go to RETURNSTATE.
27: **end if**
 FRESHBASE:
28: $k_0 \leftarrow \lceil C_0 E^{-2} L_{\text{conf}} \rceil$, draw a fresh $S \in \mathbb{R}^{k_0 \times m}$ with independent $N(0, 1/k_0)$ entries after $D, \tilde{w}$ are fixed, set $h_i \leftarrow \|SP_D e_i\|_2^2$ for every i , and set $H \leftarrow 0$.
29: Go to RETURNSTATE.
 RETURNSTATE:
30: $h_i^{\text{out}} \leftarrow \min\{\max\{h_i, 0\}, 1\}$ for every i .
31: $\mathcal{S} \leftarrow (D, \tilde{w}, h, H)$.
32: **return** $(h^{\text{out}}, \mathcal{S})$.
33: **end procedure**

The block analysis needs a maximal inequality that allows inherited error to be predictably contracted before each fresh innovation is added. The next lemma supplies exactly this stopped, time-uniform concentration bound.

Lemma 8.6 (Contractive maximal Bernstein). *Let $T \geq 1$ and*

$$Y_t = a_t Y_{t-1} + X_t, \quad 0 \leq a_t \leq 1, \quad t = 1, \dots, T,$$

where a_t is predictable and

$$\mathbb{E}[e^{\lambda X_t} \mid \mathcal{F}_{t-1}] \leq e^{\lambda^2 v_t/2} \quad \text{for } |\lambda| \leq b_t^{-1}.$$

If $\sum_{t=1}^T v_t \leq V_0$, $\max_{t \in [T]} b_t \leq b_0$, and $|Y_0| \leq u_0$, then, for $u > u_0$,

$$\Pr[\max_{t \in [T]} |Y_t| \geq u] \leq 2 \exp(-c \min\{\frac{(u - u_0)^2}{V_0}, \frac{u - u_0}{b_0}\}).$$

The conclusion remains valid at a first-exit stopping time, with the crossing update included and the process frozen thereafter; equivalently, its pre-exit activity indicator is predictable.

Proof. For every fixed admissible $\lambda > 0$,

$$\mathbb{E}[\cosh(\lambda Y_t) \mid \mathcal{F}_{t-1}] \leq e^{\lambda^2 v_t/2} \cosh(\lambda a_t Y_{t-1}) \leq e^{\lambda^2 v_t/2} \cosh(\lambda Y_{t-1}).$$

Thus

$$\cosh(\lambda Y_t) \exp(-\frac{\lambda^2}{2} \sum_{j=1}^t v_j)$$

is a nonnegative supermartingale. Stop at the first crossing of u and apply Ville's inequality. Since $\cosh(\lambda u) \geq e^{\lambda u}/2$ and $\cosh(\lambda Y_0) \leq e^{\lambda u_0}$, the crossing probability is at most

$$2 \exp(-\lambda(u - u_0) + \lambda^2 V_0/2).$$

Take $\lambda = \min\{(u - u_0)/V_0, (2b_0)^{-1}\}$. For a stopped process, multiply every innovation by the predictable indicator of not having stopped, and after stopping take coefficient one and innovation zero. $\square$

The two transition recurrences and the maximal inequality can now be combined into one adaptive score provider. The following lemma states its uniform accuracy event and prices bases, transitions, and restarts in a single ledger.

Lemma 8.7 (Hybrid self-transport score maintenance). *Consider an adaptive sequence of at most M_{ep} score queries. Each query D_t has a positive companion $\tilde{w}^{(t)}$ fixed before its score sketch and satisfying, before the companion stopping time,*

$$Q^{-1}(\sigma_i(D_t) + v_i) \leq \tilde{w}_i^{(t)} \leq Q(\sigma_i(D_t) + v_i) \quad (i \in [m]) \quad (22)$$

for an absolute Q . Assume the block input satisfies the raw-relative condition

$$|h_{b,i} - \sigma_i(D_b)| \leq (c_{\text{anc}} E/8) \sigma_i(D_b).$$

Restart immediately before cumulative effective drift would exceed a fixed small h_0 , and on a non-restart transition use

$$k_t := \max\{1, \lceil C_1 \eta_t E^{-2} L_{\text{conf}} \rceil\}, \quad L_{\text{conf}} := \log(40mM_{\text{ep}}).$$

Every transition sketch is fresh and is drawn only after both endpoints, the old companion, both normalizations, the branch, η_t , and k_t are fixed. Run Algorithm 10. In the accounting below, η_t is the selected finite drift whenever a branch is eligible, including when that transition starts a fresh

block, and is h_0 only when no branch is eligible. Then, for sufficiently large absolute base and transition constants, the raw states satisfy

$$|h_{t,i} - \sigma_i(D_t)| \leq c_{\text{anc}} E \sigma_i(D_t) \quad (23)$$

simultaneously before the companion stopping time, except with probability at most $1/40$. The same inequality then holds for the clipped outputs. If M is the actual number of transitions and $\Omega := \sum_{t=1}^M \eta_t$, the total number of permitted systems is

$$O(M + E^{-2} L_{\text{conf}}(1 + \Omega)). \quad (24)$$

Proof. At every fresh base, condition on the complete pre-base sigma-field and draw

$$k_0 = \lceil C_0 E^{-2} L_{\text{conf}} \rceil$$

Gaussian rows. For $\sigma_i(D_b) > 0$,

$$\frac{\|SP_{D_b} e_i\|_2^2}{\sigma_i(D_b)} \stackrel{d}{=} \frac{\chi_{k_0}^2}{k_0}.$$

Under the standing no-zero-row assumption, every score is positive. If zero rows are allowed, a zero score produces the exact zero column and is returned exactly. The chi-square tail and a union bound give the displayed base condition with conditional failure at most $e^{-3L_{\text{conf}}}$.

Fix a block with transition indices $b, \dots, e-1$ and a coordinate, and stop immediately before the first incoming raw-relative violation or companion violation. Let $H_t = \sum_{j=b}^{t-1} \eta_j$. All η_j, H_t, k_t are predictable. By Eq. (20),

$$Y_t := e^{-H_t} z_t$$

satisfies

$$Y_{t+1} = a_t Y_t + \tilde{\xi}_t, \quad 0 \leq a_t \leq 1.$$

Eq. (21) remains valid after multiplication of the innovation by $e^{-H_{t+1}} \leq 1$. The sample rule gives

$$\sum_{t=b}^{e-1} \frac{\eta_t^2}{k_t} \leq \frac{h_0 E^2}{C_1 L_{\text{conf}}}, \quad \max_{b \leq t \leq e-1} \frac{\eta_t}{k_t} \leq \frac{E^2}{C_1 L_{\text{conf}}}.$$

Choose h_0 so that $e^{-h_0} \geq 3/4$. A crossing $|z_t| \geq c_{\text{anc}} E$ implies $|Y_t| \geq 3c_{\text{anc}} E/4$, whereas the base has magnitude at most $c_{\text{anc}} E/8$. Lemma 8.6, with C_1 sufficiently large, makes the coordinate-block crossing probability at most $2e^{-3L_{\text{conf}}}$. The first violating update is included by freezing the stopped process after it. Thus there is no conditioning on a future success event.

The restart decision is predictable. If adding a transition would exceed h_0 , then either the completed block has accumulated at least $h_0/2$ or the triggering transition has effective drift at least $h_0/2$. These charges are disjoint. A transition with no eligible branch contributes h_0 by definition. Thus the number of fresh bases is at most $1 + 2\Omega/h_0$. Including a possible externally supplied initial base, there are at most $M_{\text{ep}} + 1$ blocks. The base failure is at most $e^{-3L_{\text{conf}}}$ per coordinate and block, and the crossing failure is at most $2e^{-3L_{\text{conf}}}$. Hence their union is at most

$$3m(M_{\text{ep}} + 1)e^{-3L_{\text{conf}}} < 1/40.$$

Finally,

$$\sum_{t=1}^M k_t = O(M + E^{-2} L_{\text{conf}} \Omega),$$

and the fresh bases cost $O(E^{-2} L_{\text{conf}}(1 + \Omega))$. Each row uses two endpoint systems for a transition and one system for a base; constant factors are absorbed in Eq. (24). $\square$

8.3 Compatibility with the path algorithm

The maintenance theorem assumes a two-sided companion band at every query. The next lemma checks that all fixed-point and surrogate queries generated by the path algorithm satisfy this hypothesis.

Lemma 8.8 (Uniform query companions). *Every maintained-mode fixed-point and inflated-surrogate score query made after the raw-relative state handoff and before the existing auxiliary stopping time has a companion satisfying Eq. (22) for an absolute Q .*

Proof. For an exact fixed-point query $D_*(x)$, $g_i(x) = \sigma_i(D_*(x)) + v_i$. Every fixed-point companion z used by the algorithm satisfies $z = e^{\pm A_q E} g(x)$. The query formed from z is within logarithmic row drift $O(E)$ of $D_*(x)$, so Lemma 4.1 gives $\sigma_i(D_z) = e^{\pm O(E)} \sigma_i(D_*(x))$. Hence $z_i = e^{\pm O(E)} (\sigma_i(D_z) + v_i)$.

For the surrogate query, $W^+ = e^{c+E} \text{Diag}(\hat{g})$ with $\hat{g} = e^{\pm E/2} g(x)$. Its query is again within $O(E)$ row drift of $D_*(x)$, and the same argument yields $W_{ii}^+ = e^{\pm O(E)} (\sigma_i(P^+) + v_i)$. This gives both directions of Eq. (22), not only the already used upper bound $\sigma_i(P^+) \leq W_{ii}^+$. $\square$

The score provider's thresholds must coexist with the earlier geometric and path-following constants without a circular choice. The next lemma gives an explicit admissible ordering for all remaining constants.

Lemma 8.9 (Compatible constants for the hybrid score provider). *The constants in the hybrid provider can be fixed compatibly with Lemma 7.7. In particular, they can be chosen in the noncircular order*

$$\begin{aligned} & Q, \quad (K_Q, \rho_Q), \quad (C_\infty, C_w), \quad h_0, \quad c_R, \quad A_E, \\ & c_{\text{anc}}, \quad (C_0, C_1, C_{\text{fp}}), \quad A_{\text{path}}, \quad c_{\text{norm}}, \quad A_{\text{norm}}, \quad A_{\text{ep}}, \quad A_L, \quad c_{\text{sw}}, \quad A_{\text{sw}}, \\ & A_\kappa, \quad A_N, \quad A_{\text{call}}, \quad A_{\text{loc}}, \quad A_B. \end{aligned}$$

With these choices, before the auxiliary stopping time every query has the companion band in Eq. (22), every successful transition has an eligible branch, every transition lemma is invoked inside its stated radius and incoming-error range, and the base and crossing probabilities in Lemma 8.7 have the claimed bounds.

Proof. Retain first all structural and deterministic constants already fixed in Lemma 7.7. The proof of Lemma 8.8 gives an absolute $E_{\text{cmp}} > 0$ and an absolute Q for every $E \leq E_{\text{cmp}}$. Fix this Q . Lemma 8.3 then fixes absolute constants K_Q and ρ_Q .

Let $c_{\text{ord}}, C_{\text{ord}}$ be the constants in the moment-generating function bound of Lemma 8.5. Choose

$$C_\infty \geq \max\{4, \sqrt{C_{\text{ord}}}, 1/(4c_{\text{ord}})\}, \quad C_w \geq \max\{K_Q, 1\}.$$

These choices make Eq. (20) and Eq. (21) literal for both transition branches. Next choose

$$h_0 \leq \min\{\log(4/3), C_\infty/8, C_w \rho_Q, 1/16\}.$$

Thus eligibility implies $\rho_\infty \leq 1/8$ in the ordinary branch and $\rho_w \leq \rho_Q$ in the weighted branch, while $e^{-h_0} \geq 3/4$.

The already proved query and movement bounds give absolute constants $A_{\text{ord}}, A_w, A_\infty$ such that all old-query-to-surrogate and same-point refinement transitions satisfy $\rho_\infty \leq A_{\text{ord}} E$, and the principal transition satisfies

$$\rho_w \leq A_w E, \quad \|d^\circ\|_\infty \leq A_\infty R.$$

Decrease the absolute radius multiplier c_R once, and then increase A_E once, so that

$$E \leq \min\{E_{\text{cmp}}, 1, h_0/(C_\infty A_{\text{ord}}), h_0/(C_w A_w)\}, \quad A_E \geq 400, \quad 2A_\infty R \leq 1/8.$$

This is compatible with all old upper-bound requirements because $E = K/(A_E L_{\text{ch}})$ and $R = c_R C E$. The ordinary transitions are now eligible. For the principal transition, its weighted mean lies between the smallest and largest coordinates, and hence

$$\|d^\circ - \lambda_w \mathbf{1}\|_\infty \leq 2\|d^\circ\|_\infty \leq 1/8, \quad C_w \rho_w \leq h_0.$$

Thus its weighted branch is eligible as well.

Choose c_{anc} so that $c_{\text{anc}} \leq \min\{1/8, c_0/2\}$ and all old consumer tolerances hold. This only strengthens the old score requirements. Since $E \leq 1$, the incoming relative error is then below $1/4$. Finally choose $C_0 = \Omega(c_{\text{anc}}^{-2})$ for the raw chi-square base and $C_1 = \Omega(c_{\text{anc}}^{-2})$ for the maximal Bernstein bound, and choose C_{fp} large enough for the fixed-point chi-square bound. Increasing these sample multipliers changes no earlier radius or eligibility choice.

Next choose A_{path} to dominate both deterministic path loops, choose c_{norm} small enough for Lemma 7.6, and choose A_{norm} to dominate its fixed polynomial work. Fix $A_{\text{ep}} > A_{\text{norm}} + 10$. Now choose A_L large enough that $L_{\text{ch}} = A_L L_*$ dominates every epoch-local tracking and confidence horizon, including $\log(A_0 B_{\text{ep}} m \beta)$ for $B_{\text{ep}} = (4m)^{A_{\text{ep}}}$. This is noncircular because A_{ep} and every preceding constant are already fixed, while $\log B_{\text{ep}} = O(A_{\text{ep}} \ell)$.

With c_R, A_E, A_L now frozen, choose c_{sw} according to Eq. (6), and then choose A_{sw} large enough for the switch estimate. Choose A_κ to dominate both phase log-ratios and the terminal logarithms, including the now-fixed switch constants. Finally choose A_N to dominate the frozen analytic atomic-path instruction count, then A_{call} to dominate all score transitions and initializer rounds, then A_{loc} to dominate the four good-attempt charges in Eq. (9), and then A_B to dominate the global system, round, and scalar work constants, including C_0, C_1, C_{fp} and $1/h_0$.

After A_B no absolute constant is changed. In particular, no radius is shrunk after c_{sw} is selected and no analytic work constant is increased after the caps are selected. This proves the claimed order and fixes one algorithm rather than a separate choice for each call.

Under the standing no-zero-row assumption, every score is positive. If zero rows are retained instead, positive diagonal rescaling preserves them, their projection columns and sketches are identically zero, and the exact zero conventions in the three transition lemmas apply. $\square$

The score complexity is determined by the total effective drift accumulated across all queries in one centering call. The following lemma charges every such transition and proves the required $O(E)$ bound.

Lemma 8.10 (Effective drift of one centering call). *Before the minimum of the existing coarse, tracking, companion, and raw-score stopping times, all score transitions in one call to Algorithm 5 have total hybrid effective drift $O(E)$ and use $O(1)$ transitions.*

Proof. There are exactly three types of transition.

First, the last old fixed-point query at x moves to the inflated surrogate query at the same x . Its old companion and W^+ are both $e^{\pm O(E)} g(x)$, so the ordinary centered drift is $O(E)$. After the absolute constants are fixed, the existing choice of sufficiently large A_E makes its effective drift at most h_0 .

Second, the surrogate query at x moves to the first predicted fixed-point query at x^+ . Modulo the scalar $c_+ E$, its row-log drift is

$$d^\circ = \psi + \frac{\vartheta}{2} \hat{u}.$$

The existing movement lemmas give $N_w(\psi) = O(R)$ and $N_w(\hat{u}) \leq r \leq \Delta = O(R)$. Therefore

$$\|d^\circ\|_w = O(R/C) = O(E), \quad \|d^\circ\|_\infty = O(R) \leq 1/8.$$

The path weight and W^+ are within an absolute coordinatewise factor before the tracking stop. Thus the computable weighted centered drift is at most $\|d^\circ\|_{W^+} = O(E)$. Since a weighted mean lies between the minimum and maximum coordinates,

$$\|d^\circ - \lambda_w \mathbf{1}\|_\infty \leq 2\|d^\circ\|_\infty \leq 1/8.$$

These are precisely the inequalities enforced by the already frozen c_R, A_E in Lemma 8.9; in particular $C_w \rho_w \leq h_0$, so the weighted branch is eligible.

Third, the remaining constant number of fixed-point refinements are all at x^+ . Lemma 6.1 gives total ordinary row drift $O(E)$, and again every such branch is eligible. At an ordinary block boundary the algorithm charges the selected drift, not h_0 ; hence adding the three contributions proves the stated $O(E)$ effective charge even when the call starts a fresh block. No score query is made by the predictors, defect certificate, projection, greedy correction, residual solve, path-parameter update, or cost-vector switch. $\square$

Raw-relative maintenance also needs an exact specification of its initial state and of the companion associated with the last actual query. The next lemma establishes this handoff from the fresh fixed-point routine.

Lemma 8.11 (Raw-relative initialization and state handoff). *In the fresh branch of Algorithm 3, the final Gaussian score sketch satisfies, on the same final-round event used in the fixed-point proof,*

$$|\tilde{\sigma}_i^{(J-1)} - \sigma_i(D_{J-1})| \leq (c_{\text{anc}}E/8)\sigma_i(D_{J-1}) \quad (i \in [m]).$$

The exact branch satisfies the same statement with zero error. The score state handed to Algorithm 10 is

$$\mathcal{S} = (D_{J-1}, z^{(J-1)}, \tilde{\sigma}^{(J-1)}, 0),$$

where $z^{(J-1)}$ is the companion of the actual last query D_{J-1} ; the returned fixed-point vector $z^{(J)}$ is not substituted for this stored companion. In general the final Gaussian row count is $O((\epsilon_{\text{fp}}^{-2} + E^{-2}) \log(2mJ/\zeta))$; in the outer solver $\epsilon_{\text{fp}} = E/4$, so this is $O(E^{-2} \log(2mJ/\zeta))$.

Proof. Conditional on D_{J-1} , every nonzero coordinate of a k -row Gaussian score sketch obeys

$$\frac{\tilde{\sigma}_i^{(J-1)}}{\sigma_i(D_{J-1})} \stackrel{d}{=} \frac{\chi_k^2}{k}.$$

The explicit schedule has $\theta_{J-1} \leq c_{\text{anc}}E/16$ and

$$k_{J-1} = \lceil C_{\text{fp}} \theta_{J-1}^{-2} \log(2mJ/\zeta) \rceil = O((\epsilon_{\text{fp}}^{-2} + E^{-2}) \log(2mJ/\zeta)).$$

Here minimality of J gives $e_{J-1} \geq \epsilon_{\text{fp}}$; equality can occur in the forced one-round case because the input assumption has $D \geq \epsilon_{\text{fp}}$. Hence the unclamped last-round tolerance is $\Omega(\epsilon_{\text{fp}})$, which justifies the displayed upper bound after taking the minimum with $c_{\text{anc}}E/16$. The chi-square event of the fixed-point proof therefore gives the displayed simultaneous raw-relative event with failure probability at most ζ/J ; it is not a new event and requires no extra sketch. For the call made by Algorithm 9, $\epsilon_{\text{fp}} = E/4$, giving the claimed E^{-2} specialization. Zero-score coordinates and the exact branch are exact. By construction, D_{J-1} was formed from $z^{(J-1)}$; $z^{(J)}$ is computed only after that query response. Storing the former therefore gives precisely the old-query companion needed to compute the next weighted drift. $\square$

The accuracy proof must stop at the first violation without conditioning the past on a future success event. The following exact-continuation coupling provides that filtration-safe comparison process.

Lemma 8.12 (Raw-relative score coupling with exact continuation). *Let τ_{aux} be the existing first coarse, tracking, algorithmic, or companion failure, taken immediately before the next score query. Let τ_{sc} be the first query before τ_{aux} whose stored raw post-update state violates Eq. (23). Couple the implemented process to an analysis process that agrees with it strictly before τ_{sc} . At τ_{sc} and at every later score query, the analysis process supplies the exact score vector instead of the maintained response. Thus every response consumed by the analysis process either satisfies the raw-relative guarantee or is exact, and a score violation never stops a centering call after its predictor noise has been drawn. Moreover,*

$$\Pr[\tau_{\text{sc}} < \tau_{\text{aux}} \mid \text{raw-relative initialization event}] \leq 1/40.$$

The predictor bias and fluctuation identities retain their original conditional sigma-fields, and the coupled process may be stopped only at the existing pre-noise auxiliary stops.

Proof. Strictly before τ_{aux} , Lemma 8.8 supplies the companion band of Lemma 8.7. That lemma therefore bounds the first raw-relative score violation by $1/40$. All transition endpoints, companions, branch choices, effective drifts, and row counts are functions of the past and are fixed before the transition's fresh sketch. At the first bad response, switch the analysis process permanently to exact scores. All later exact responses are deterministic functions of their queries, while predictor randomness continues to be drawn with its original conditional law. Hence no current centering call is censored according to a score event revealed after its weighted predictor. The implemented and analysis processes agree until the first violation and agree throughout on the event that no violation occurs. No argument conditions on that event, so the existing predictor martingale identities are unchanged. If an independent auxiliary stop is used, it is taken only at its already specified pre-noise boundary. $\square$

The raw-relative guarantee must finally be translated into the additive score accuracy expected by the fixed-point, surrogate, and predictor interfaces. The next lemma performs this transfer, including clipping and zero scores.

Lemma 8.13 (Transfer from raw-relative scores to the path interfaces). *On the raw-relative initialization event, consider the coupled process of Lemma 8.12. At an executed query D , let $\sigma_i = \sigma_i(D)$ and let $\hat{\sigma}_i = \min\{\max\{h_i, 0\}, 1\}$. At the first violating query and at every later query replaced by the coupling, define $\hat{\sigma} = \sigma$. If $c_{\text{anc}}E \leq 1/2$ and $c_{\text{anc}} \leq \min\{1/8, c_0\}$, then*

$$|\hat{\sigma}_i - \sigma_i| \leq c_{\text{anc}}E\sigma_i \leq c_{\text{anc}}E(\sigma_i + v_i). \quad (25)$$

Thus every fixed-point score satisfies the hypothesis of Lemma 4.4, and every inflated-surrogate score satisfies the additive hypothesis of Lemma 7.1. These are the only hybrid-score consumers in Algorithm 5. Every centering call completed before the auxiliary stop incurs $O(E)$ total hybrid effective drift and $O(1)$ score transitions, while its non-score work and depth are $O(L_ + q)$ systems and $O(\log q)$ system rounds.*

Proof. Every projection score lies in $[0, 1]$. Projection onto this interval is nonexpansive and fixes σ_i , so the raw guarantee gives

$$|\hat{\sigma}_i - \sigma_i| \leq |h_i - \sigma_i| \leq c_{\text{anc}}E\sigma_i.$$

At every replaced query the error is zero. This proves Eq. (25).

At a fixed-point query, define $\tau_i = \sigma_i + v_i$ and $\hat{\tau}_i = \hat{\sigma}_i + v_i$. Then

$$(1 - c_{\text{anc}}E)\tau_i \leq \hat{\tau}_i \leq (1 + c_{\text{anc}}E)\tau_i,$$

and hence

$$|\log \hat{\tau}_i - \log \tau_i| \leq \frac{c_{\text{anc}}E}{1 - c_{\text{anc}}E} \leq 2c_{\text{anc}}E.$$

After multiplication by the fixed-point exponent $p/2$, this is at most $pE/8$ once $c_{\text{anc}} \leq 1/8$. At the surrogate, $\sigma_i(P^+) \leq W_{ii}^+$, and therefore

$$|\hat{\sigma}_i - \sigma_i(P^+)| \leq c_{\text{anc}}E\sigma_i(P^+) \leq c_{\text{anc}}EW_{ii}^+,$$

which is the certified-surrogate hypothesis when $c_{\text{anc}} \leq c_0$.

Lemma 8.12 neither conditions on future success nor changes any coarse conditional kernel. Hence the bias and second-moment statements of Lemmas 5.7 and 5.9 retain their stated sigma-fields. Lemma 8.10 accounts for every score transition in the call. The coordinate predictor, certificate, projection, Newton step, and fixed-point refinement have the non-score work and depth claimed in Lemmas 5.7, 5.8, and 6.1. This proves the direct one-call interface. $\square$

Corollary 8.14 (Five-logarithm ledger). *Use the raw-relative final initialization response, store the companion of the actual final score query, and run the hybrid maintenance process of Lemma 8.7 inside the checkpointed epochs of Lemma 8.1. On accepted good attempts both score consumers receive their required accuracy. After charging all checkpointed attempts, including rejected ones, the total system count is*

$$O(\sqrt{n}q^2\ell^3\kappa + \min\{q\ell^3, n\log\ell\}).$$

For arbitrary finite U and ϵ , this yields

$$O(\sqrt{n}\log^5(4m)\log\frac{emU}{\epsilon}\mathcal{T}_w)$$

work. The linear-system depth is

$$O((\sqrt{n}q^{3/2}\ell\log q\kappa + \log\ell)\mathcal{T}_d).$$

Proof. Lemma 8.11 gives the required external base event and stores the companion belonging to the actual last query. On the no-replacement event, Lemma 8.13 directly supplies both score interfaces in every centering call before the auxiliary stop.

Lemmas 8.1, 7.5, and 6.5, together with Lemma 4.8's bound $c_1 \leq 2n$, $C = \Theta(\sqrt{q})$, $c_k = \Theta(q)$, and $R = \Theta(\ell^{-1})$, give

$$N_{\text{step}} = O(\sqrt{n}q^{3/2}\ell\kappa), \quad E = \Theta((\sqrt{q}\ell)^{-1}), \quad L_{\text{conf}} = O(\ell).$$

Lemma 8.10 gives

$$\Omega = O(N_{\text{step}}E) = O(\sqrt{n}q\kappa).$$

Consequently Eq. (24) is

$$O(\sqrt{n}q^2\ell^3\kappa + q\ell^3).$$

The transition-count term $M = O(N_{\text{step}})$ and the non-score cost $O(N_{\text{step}}(\ell + q))$ are lower order. Initialization remains $O(\min\{q\ell^3, n\log\ell\})$.

Every transition sketch consists of parallel systems at two endpoints and introduces no additional sequential query. Combining the $O(\log q)$ predictor depth per centering call with the initialization depth gives the displayed depth bound. Finally $q \leq \ell$ gives $q^2\ell^3 = O(\ell^5)$. $\square$

It remains to convert the high-probability score analysis into a deterministic cap on the work of every attempted epoch. The final lemma supplies this wrapper and closes the five-logarithm resource ledger.

Lemma 8.15 (Deterministic hybrid-score budget). *Let $A_\kappa, A_N, A_{\text{call}}, A_{\text{loc}}, A_B$ be the constants fixed, in this order, by Lemma 8.9, and define*

$$\begin{aligned}\kappa &:= A_\kappa(1 + \log(emU/\epsilon)), & N_{\text{cap}} &:= \lceil A_N \sqrt{n} q^{3/2} L_* \kappa \rceil, \\ B_{\text{ep}} &:= (4m)^{A_{\text{ep}}}, & M_{\text{ep}} &:= \lceil A_{\text{call}}(B_{\text{ep}} + qL_* + 1) \rceil, & L_{\text{conf}} &:= \log(40mM_{\text{ep}}), \\ M_{\text{calls}} &:= \lceil 32A_{\text{call}}(N_{\text{cap}} + q\kappa + \log(q/E) + 1) \rceil.\end{aligned}$$

For the hybrid provider set

$$\begin{aligned}B_{\text{sys}}^{(5)} &:= \lceil A_B(\sqrt{n} q^2 L_*^3 \kappa + qL_*^3 + \min\{qL_*^3, n \log L_*\}) \rceil, \\ B_{\text{rnd}} &:= \lceil A_B(\sqrt{n} q^{3/2} L_* \log q \kappa + \log(qL_*) + 1) \rceil, & B_{\text{ar}}^{(5)} &:= \lceil A_B(\text{nnz}(A) + m)B_{\text{sys}}^{(5)} \rceil.\end{aligned}$$

Wrap every score query, requested batch of permitted systems, sequential system round, and exact real-RAM scalar-work block in four shared counters capped respectively by M_{calls} , B_{sys} , B_{rnd} , and B_{ar} . Each routine announces its complete deterministic charge before drawing randomness or doing the work. If a cap would be exceeded, propagate FAIL immediately to Algorithm 9.

The wrapped five-logarithm algorithm halts on every random outcome within these caps. On the initialization event and the accepted epoch events of Lemma 8.1, no cap is reached, so the wrapped and analytic checkpointed algorithms coincide.

Proof. Algorithm 6 has the explicit call cap M_{path} and terminal-loop length J . Algorithms 1 and 2 have the certified iteration caps in their correctness lemmas, every fixed-point loop has its displayed deterministic schedule, and Algorithm 4 announces all batch sizes before drawing randomness. Thus all non-score loops terminate deterministically. The wrapper additionally checks the local M_{ep} cap before another score transition and the monotone global M_{calls} cap before another attempted transition.

On the analytic good event, the two path phases use at most N_{cap} atomic instructions after the prior choice of A_N . A good attempt of announced length b has $O(b)$ score transitions and effective drift $O(bE)$ by Lemma 8.10; hence it stays strictly within the four local caps in Eq. (9). On a rejected attempt no such drift claim is made: the local counters stop it before the same deterministic caps are exceeded.

Writing $A_{\text{att}} \leq 4J_{\text{ep}}$ for the number of attempts, the summation in the proof of Lemma 8.1 gives

$$\sum_{a=1}^{A_{\text{att}}} b_a = O(N_{\text{cap}}), \quad A_{\text{att}} = O(N_{\text{cap}}/B_{\text{ep}} + 1).$$

Consequently all attempted path work, including rejected attempts, uses

$$O(N_{\text{cap}}(E^{-1}L_* + L_* + q) + (N_{\text{cap}}/B_{\text{ep}} + 1)E^{-2}L_*)$$

systems. Since B_{ep} dominates $E^{-2}L_*$ and $A_{\text{ep}} > A_{\text{norm}}$, checkpoint computations and normalizations add only $O(N_{\text{cap}})$. Substituting

$$E = \Theta((\sqrt{q}L_*)^{-1}), \quad L_{\text{conf}} = O(L_*)$$

gives $O(\sqrt{n} q^2 L_*^3 \kappa + q L_*^3)$. The cold initializer adds $O(\min\{q L_*^3, n \log L_*\})$, so the global system cap is not reached.

All systems within one score sketch are parallel, and each non-score atomic instruction has $O(\log q)$ system depth. Summing the local round caps gives the stated B_{rnd} bound, including normalization. Likewise, summing $W_{\text{att}}(b)$ and adding the initializer gives $O((\text{nnz}(A) + m) B_{\text{sys}}^{(5)})$ scalar work. The shared score-query cap is valid by the same sum, since exact checkpoint queries contribute at most $O((N_{\text{cap}}/B_{\text{ep}}) P_{\text{norm}}) = O(N_{\text{cap}})$.

On an arbitrary trajectory the local and global call, batch, round, and scalar counters reject the next operation before its randomness or work. Thus the algorithm halts on every trajectory within the stated caps; on the success event of Lemma 8.1, none is reached. $\square$

Theorem 8.16 (Five-logarithm bound; formal version of Theorem 1.1). *Let the linear program in Eq. (1), its interior point x_0 , and a finite certified magnitude bound U supplied to the algorithm satisfy the assumptions in Section 1, and let $\epsilon \in (0, 1)$. Define*

$$\ell = \log(4m), \quad q = \log(4m/n), \quad \kappa = A_\kappa(1 + \log(emU/\epsilon)), \quad L_* = \ell,$$

take $L_{\text{ch}} = A_L L_*$ for a sufficiently large absolute A_L , and set

$$E = \frac{K}{A_E L_{\text{ch}}}, \quad R = c_R C E.$$

Run the budgeted Algorithm 9, using the raw-relative state handoff of Lemma 8.11 and the resource caps of Lemma 8.15; execute its path through the checkpointed epochs of Lemma 8.1 and analyze each epoch's score process through the exact-continuation coupling of Lemma 8.12. The algorithm halts on every random outcome and, with constant probability, returns a feasible x satisfying $c^\top x \leq \text{OPT} + \epsilon$. Its permitted-system count is

$$O(\sqrt{n} q^2 \ell^3 \kappa + \min\{q \ell^3, n \log \ell\}).$$

For arbitrary finite U and ϵ , its work is

$$O(\sqrt{n} \log^5(4m) \log \frac{emU}{\epsilon} \mathcal{T}_w),$$

and its linear-system depth is

$$O((\sqrt{n} q^{3/2} \ell \log q \kappa + \log \ell) \mathcal{T}_d).$$

Proof. Fix the structural constants according to Lemma 7.7 and complete the global hierarchy in the order of Lemma 8.9. The latter order already includes the constants of Lemma 8.15 and ensures $E, R \leq r_0/c_k$, $R \leq 1/(160c_k)$, and that L_{ch} dominates the logarithmic horizon required by Lemma 6.6. No absolute constant is changed in this proof.

For the floor $v = (n/m)\mathbf{1}$, the ideal initial weight satisfies $n/m \leq g_i(x_0) \leq 2$. Hence the cold start $\mathbf{1}$ has logarithmic error at most q . Lemma 6.2, with $D = q$ and target accuracy $E/4$, gives

$$\|\log \hat{g} - \log g(x_0)\|_\infty \leq E/4, \quad \Phi(\log \hat{g} - \log g(x_0)) \leq 2me^{1/400} \leq 3m \leq M_{\text{ch}}.$$

Lemma 8.11 makes the final initializer response raw-relative and stores the companion of its actual query. This event is measurable before either path phase.

Within each attempt run the coupled process of Lemma 8.12. Before its first replacement, Lemma 8.8 verifies the companion band and Lemma 8.13 supplies both score interfaces; no predictor

identity is conditioned on future score success. Lemma 8.1 combines the local score, coarse, and stopped-tracking events, certifies every full-epoch endpoint, and restores the potential and raw score base before the next epoch. It completes both path phases with constant probability. On its final partial-epoch good event, Lemmas 7.4 and 7.5 finish the stream, and Lemma 6.5 yields a feasible ϵ -optimal output. Together with the one-time initialization event, their intersection has constant probability.

For the resource count, Lemma 4.8 gives $c_1 \leq 2n$; also $C = \Theta(\sqrt{q})$, $c_k = \Theta(q)$, and $R = \Theta(\ell^{-1})$, so the two path phases and their terminal calls use

$$N_{\text{step}} = O\left(\frac{c_k \sqrt{nq}}{R} \kappa\right) = O(\sqrt{n} q^{3/2} \ell \kappa)$$

centering calls. Lemma 8.10 gives $O(1)$ score transitions and $O(E)$ hybrid effective drift per call. Corollary 8.14 therefore proves the displayed permitted-system count and depth. Lemma 8.15 proves that no cap changes a good trajectory and that every other trajectory terminates before exceeding the same system, round, and scalar-work caps. Finally $q \leq \ell$ gives $q^2 \ell^3 = O(\ell^5)$, and $\mathcal{T}_w = \Omega(\text{nnz}(A) + m)$ absorbs the associated exact scalar work. $\square$

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